

Floods at Bengaluru City - A Technical Study

Dr. Sudhir Sajjan, B.E., M.B.A., Ph.D.
sajjan.sudhir@gmail.com

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Preface

Bengaluru city experienced devastating floods during September, 2022. Its large portions were under water after two days of torrential rains uprooted trees, caused crippling traffic and forced offices to issue work-from-home orders. Many sparkling villas and apartment buildings were submerged in chest-deep water. **Inflated boats were employed** at many locations to rescue the residents. The east and Mahadevapur zones of the city were the most affected. Life came to a standstill at its Outer Ring Road area, which houses more than 700 companies, including the offices of ones like Microsoft, Intel, Goldman Sachs and Morgan Stanley.

A close observation of the affected areas reveals that the floods were earlier **limited to the core city**, where the stormwater was just draining from the ridge. The city had large number of parks and lakes. The total run-off was limited. It was obstructed at the stormwater drains due to their poor maintenance or encroachments. It was causing waterlogging limited to certain areas of the streets.

But, the city has now grown manifold. The earlier towns and villages at its periphery are added to Bruhat Bengaluru Mahanagara Palike (BBMP). They are experiencing floods. The flooding pattern at them is entirely different. Earlier they were just street floodings, but **the entire valleys are affected** now. They are taking too significant time to recede.

A number of factors are cited as the reasons for the occurrence of the floods. The drains are encroached upon. They are narrowed down at many locations. Water retention capacity of the lakes has been reduced significantly. There is increase in the stormwater discharge due to un-checked concretization and acute loss of vegetation cover over the years. The pattern of rainfall has changed. There are increasing instances of heavy and incessant rainfall leading to overflowing of lakes. The city is not prepared for this sort of rainfall intensity, and so on.

However, a detailed study reveals that they are not the principal reasons for the floods at the city, particularly for those experienced at its peripheral area. It reveals that,

- i. At the core city, the stormwater was just draining from the ridge to the lakes. At the periphery, it is discharged from one lake to another.

- ii. The then entire outskirts of the city was agriculture land. A significant portion of the stormwater runoff was percolated into the ground. The lakes were virgin. They had significant storage capacity. They were spilling over occasionally. The discharge from them was thinly discharged over the agriculture land. It had **no defined stream path**.

- iii. The erstwhile towns and villages were administered by the local bodies. There was a rapid development in their outskirts, particularly in 1980s and 1990s. The local bodies had no expertise or resources to plan a comprehensive stormwater and sewage drainage system for their outskirts. The layouts were developed **right on the stream paths** between the lakes without knowledge of their existence.

iv. The stormwater and sewage drains planned at the layouts or at their boundaries were joined together to connect to the nearby lakes to discharge the sewage from the surrounding locality. They **also took the role of** discharging the stormwater from the locality to the lakes and from one lake to another in due course.

v. Most of the drains, principally those connecting one lake to another carry sewage. They always run to their maximum capacity. The flow from the lakes **is** invariably spilled over them and is discharged on their banks. The drains just provide the direction for it.

vi. Earlier, the roads were at a lower level. The spill-over from the lakes was freely discharged over them during the rains. It was not causing significant obstruction then. But they are raised and widened in due course as the city is developed. Huge ring roads and periphery roads are proposed. But, the bridges and culverts on them are built with **too small vents and inadequate waterway**.

vii. As a result, the spill-over from one lake to another is now stuck at the above bridges and culverts and is impounded at the valley behind their high embankments. A large area adjacent to the stream path is inundated in the above water.

Thus, what we see at the RBD layout, Ring Road near EPZ Ecospace, Yemalur, Panathur-Balagere road is just the impounded water at certain bridges/culverts on the drains/stream paths between the lakes. The bridges/culverts get invariably choked up by weed due to their too small vents. The impounded water is released only when the weed/debris at their mouth is cleared and an escape route is provided for it.

Similarly, the discharge from the lakes is also obstructed when huge buildings are constructed on the banks of the drains adjacent to one another leaving absolutely no overbank space. They act like bund holding back the spill-over from the lakes. The Wipro campus on Sarjapur road and Borewell road at Whitefield are inundated by such phenomena.

Thus, floods at the city have assumed new dimensions in the recent years. Earlier they were just **“Street floodings”** limited to certain areas of the streets and lasting for few hours after the rain. But they are now **“Valley floodings”**, engulfing the entire valleys between the lakes and taking days to get cleared. The discharge from one lake to another is obstructed at a number of places along its stream path. Artificial lakes are created at each of the obstructions. The spill-over from the lake is impounded at the above artificial lakes and is slowly released from one artificial lake to another.

For example, the discharge from Halanayakanahalli lake to Saul Kere is obstructed at Sarjapur road at the entrance of RBD layout and then at the Wipro Campus. Two artificial lakes are formed between them. The discharge from Saul Kere to main stream from Bellandur lake is obstructed at Outer Ring Road near EPZ Ecospace and an artificial lake is formed at it. Similarly, the discharge from Bellandur lake to Varthur lake is obstructed at Outer Ring Road in Kadubeesanahalli and at the newly built road just

at the upstream of Varthur lake. Two artificial lakes are formed at the two above roads. Epsilon, DivyaSree and other layouts are inundated in the backwater of the first artificial lake and Panathur-Balagere road in the backwater of the second artificial lake.

Another equally contributing factor for the phenomenal floods at the periphery of the city is the total concretization of the area with absolutely no park area. The entire rain water is turned into run-off and it is quickly discharged at the lakes. The capacities of the lakes on the other hand are significantly reduced over the years. While the core city was known as “the city of gardens and lakes”, we may call its periphery as **“the city of concrete jungle and artificial lakes”**.

The floods are a sort of boon for the city

The city with a population about ten million demands nearly about 60 TMC of water to meet its domestic and industrial needs. About 28.8 TMC or about 50% of its total demand is met from the Cauvery water supply schemes. The entire other 50% is met from the ground water extraction. There are lakhs of borewells. Thousands of new ones are drilled every year. The entire sewage and stormwater during rain is percolated into the ground and they are recharged. Thus, the city could fairly meet all its demand. It should vow this entirely to its stormwater drainage system. It may be facing floods for few days in a year during rainy season. But if there was really an effective stormwater drainage system, the near entire run-off would have drained away to the Cauvery and Pennar rivers.

The city lies in two river basins, viz., Pennar and Cauvery. All the flow from the city is finally drained to the above two rivers. The Varthur lake which is the last lake in its Pennar basin hardly overflows. The average total volume of flow at the gauging station at T. Bekuppe on Vrishabavathi River at 60 km downstream of the city in Cauvery basin, is only about 8 TMC during the entire year, including the flow from its intermittent catchment downstream of the city. (Source: Water Year Book- 2019, CWC)

To refer to a particular instance, the RBD layout on Sarjapur road is able to meet its entire drinking water need from groundwater, since it is waterlogged during every year in rainy season. The discharge from the four lakes on its upstream is obstructed at it and percolated into the ground. Had there been an effective stormwater drainage system at the reach, it would have invariably depended on the Cauvery water and tankers to meet its water demand.

Encroachment eviction drive

It is strongly believed that the floods at the city are mainly caused due to encroachment of the drains. Therefore, a major drive is initiated to clear them immediately after every flood. However, a study of the stormwater system at the cities in the Silicon Valley of US reveals that it has two distinct components, viz., (i) creeks and (ii) stormwater drains. The stormwater drains collect water from the streets and drain to creek. The creek would have a natural channel at the centre and a free overbank area on it for a certain width called floodway to carry the large stormwater discharge during flood, as shown in Figure 1. It is designed to discharge 100-year return flood. No development is entertained on it. Any development on it is treated as encroachment.

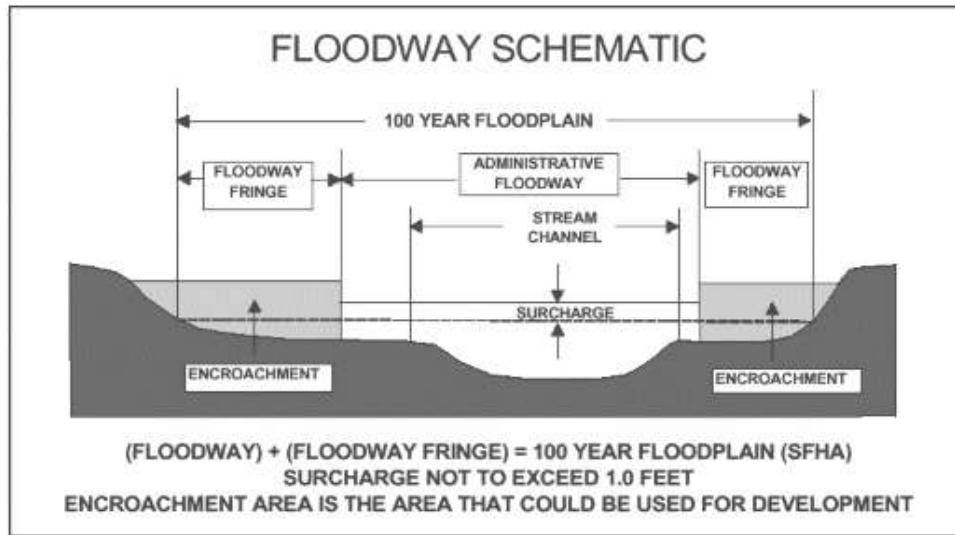


Figure 1: Section of the Creek at the cities in Silicon Valley of US
(Source: Santa Clara County Floodplain Ordinance)

But, both drain and creek are considered to be the same at the city. Only construction of any structure right on the drain or laying of slab on it is called as the encroachment. There is no provision of free overbank area. All drains carry sewage at the city and have the capacity just adequate to discharge it. The stormwater invariably spills over them during rain. Clearing the slab or structure on them **would not facilitate** its entire discharge through them. It is invariably held back if there is any obstruction on their bank.

According to the data submitted by BBMP to the High Court, 2,626 encroachments were identified on drains in the period 2016-17 within municipal limits, of which 627 are still left to be removed. (Indian Express, September 16, 2022) Even if all the encroachments are cleared, there would be no significant reduction in the total number of waterlogged areas or in the intensity of flood at any of them. Infact, if they are cleared without actually attending the major obstacles on the floodway or at the bridges/culverts on them, all the stormwater would drain at the above obstacles and cause more impounding at one location. For example,

- i. The dismantling of the slab on the drain at Wipro campus and Greenwood residency would not prevent the discharge from Halanayakanahalli lake entering through the main gate of the campus and impounding inside it, unless certain openings are provided in the dividers on the Sarjapur road.
- ii. The demolition of certain villas at RBD layout will not prevent waterlogging at it, unless the culvert on the Sarjapur road near Keerthi Regalia Apartment is cleaned and connected to the drain to Saul Kere.
- iii. Their demolition on the other hand may quickly drain off the impounded water at the layout and inside the campus to Saul Kere and which will then be directed at Outer Ring Road near Ecospace. It would create more impounding at Outer Ring Road.

- iv. The demolition of labour sheds at the downstream of the Outer Ring Road near Ecospace would not prevent the waterlogging on it. It may just facilitate quick discharge of the percolated water through the embankment of the road and impounded at the upstream of the labour sheds.
- vi. The demolition of certain villas at Epsilon and DivyaSree layouts would not prevent their flooding, as it is actually the impounded water at the bridge on Outer Ring Road in Kadubeesanahalli.
- vii. The demolition of old bridge on the upstream of Outer Ring Road in Kadubeesanahalli near Marathahalli Police Station would not prevent the weed choking up the small vent of its main bridge and impounding of the discharge of Bellandur lake on its upstream.
- viii. It may quickly drain off the impounded water at the above road, but it would get accumulated at the bridge on the newly constructed road on the upstream of Varthur lake. It would increase the intensity of waterlogging at Panathur-Balagere road on its upstream.

We often experience pain in our body at a location other than the site of its painful stimulus/origin. It is known as "Referred pain". The experience of dentally related pain during a heart attack is cited as a classical example of it. In the similar parlance, we may term the floods at the city as "**Referred floods**". They are not caused by the obstructions in the drains in their vicinity. They are caused by the obstructions on the stream paths between the lakes and far at their downstream. Therefore, the present eviction drive would be like treating teeth/gums for a referred pain of heart attack.

This study makes an attempt to understand the actual phenomena of the floods at the city and propose ideal measures for their control.

Case Study

In order to understand the phenomena of the floods at the city, five severely affected locations in its Mahadevapur zone are selected in this study and the actual reasons for their flooding/waterlogging are analyzed. Their locations are shown in the following figure.

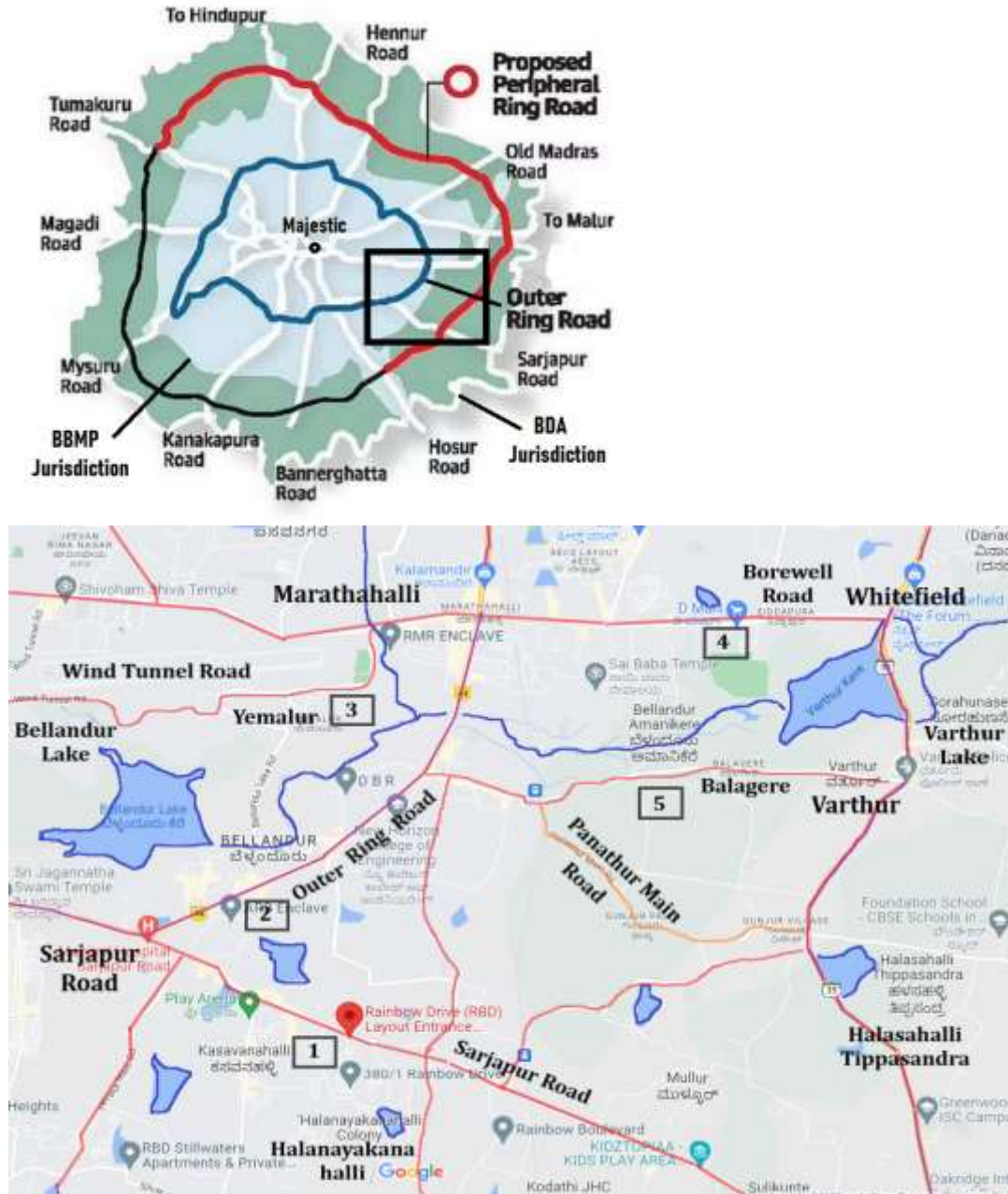


Figure 2: Index map showing the five locations which were severely affected at the city during the flood of September, 2022 and for which the case study has been carried out. They are (1) RBD layout and Wipro Corporate Office in Sarjapur Road, (2) Outer Ring Road near Ecospace, (3) DivyaSree and Epsilon layouts in Yemalur (4) Borewell Road in Whitefield and (5) Panathur-Balagere road. Their locations are shown by their number in the map.

Previous Studies of the author

This study is **8th in the series** of the studies carried out by the author on flood management at different river basins and cities. The previous studies are,

- i. Flood Control in Don River Valley - A Scientific Proposal (2010)
- ii. Flood Control and Utilization of Water in Bennihalla Basin (2015)
- iii. The Flood Phenomena in the Krishna Basin of Karnataka (2019)
- iv. The Flood Phenomena in the Krishna Basin of Maharashtra (2020)
- v. The Flood Phenomena at the Malaprabha Basin (2020)
- vi. Impact of 4-laning of NH-50 bypass on the stormwater drainage at Vijayapur city (2021)
- vii. The Actual Reasons for floods at Hampi World Heritage Site (2022)

It is the **3rd in the series** of studies on the water resources management at Bengaluru city. The previous studies are,

- i. Sharavathi Diversion – A proposal for augmentation of water supply to Bangalore city (2012). It was presented before the **Expert Committee of BWSSB** headed by Mr. Thyagaraj and constituted by GoK for “Identification of Sources for Sustainable Water Supply to Greater Bangalore” in 2011 and was recommended by it for acceptance in its Report submitted to the Government in August 2013.
- ii. Makedatu Balancing Reservoir cum Drinking Water Supply Project – Technical Study (2022)

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Chapter 1

The Bengaluru City

1.1 History

Bengaluru, also spelled **Bengalooru**, formerly **Bangalore**, city is the capital of Karnataka State, southern India. It is one of the India's largest cities. It lies 3,113 feet (949 metres) above the sea level, atop an east-west ridge in the Karnataka Plateau of the south-eastern part of the state.

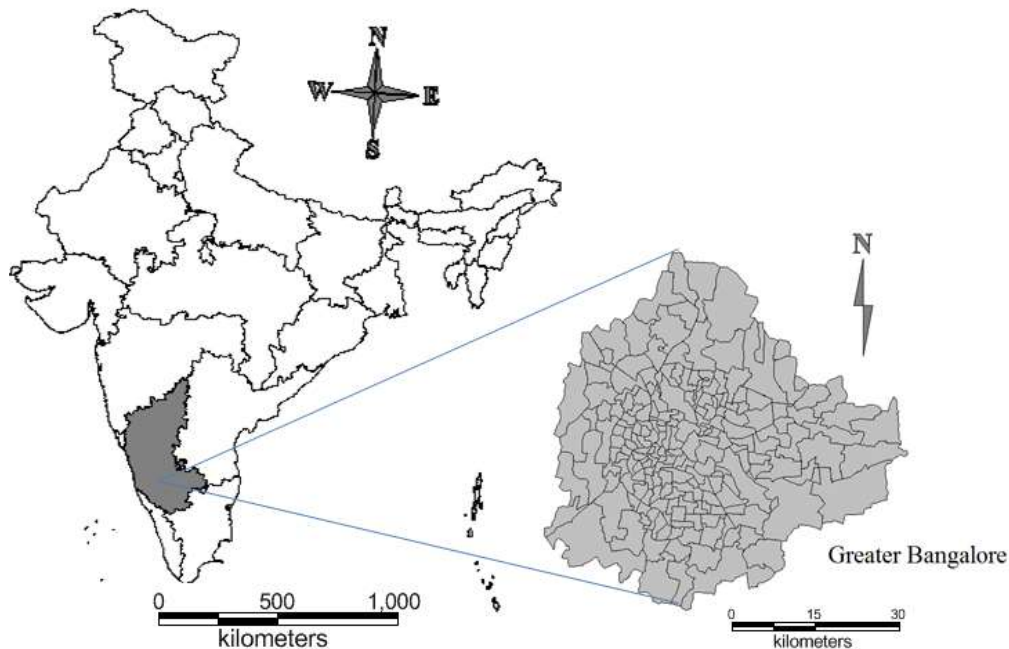


Figure 1.1: The location map of the Bengaluru City

The city's nucleus was a settlement around a mud fort, built in 1537 by a local chief, Kempe Gowda. The fort was reconstructed of stone in 1761. Bangalore was the headquarters of the British administration from 1831 to 1881, when the raja was restored. However, Britain retained an administrative and military presence in the city until Indian independence in 1947. It subsequently became the capital of the new state of Mysore and remained the capital during the reorganization of States in 1956; Mysore was renamed Karnataka in 1973.

Large-scale immigration to Bangalore began in the 1950s as the State invested heavily in the public sector and in education. Employment opportunities were created for tens of thousands of people in southern India, and Bangalore grew to become one of the largest cities in the country. For several decades the city's economic growth was based largely on manufacturing industries. **In the early 1990s**, however, a combination of new national economic liberalization policies and the emergence of a strong education system in the city encouraged the development of an information and communications technology (ICT) sector in Bangalore and paved the way for its rapid development into both a national and a global ICT centre.

Bengaluru consists of the closely built old town, together with a number of modern outlying areas (former suburbs) laid out in a gridiron pattern to the north and south, with many parks and wide streets. A large military area is situated just southeast of central Bengaluru. Outlying areas incorporated into the city in 2007 include large green spaces as well as farmland and are sites of population and ICT-related growth.

Bengaluru has **pleasant summers and mild winters**. Summer temperatures average in the low to mid-90s F (about 34 °C), and winter temperatures rarely drop below 60 °F (16 °C). The city receives an average annual rainfall of about 787 mm, which has been inadequate as a water supply for its increasing population and industry. Most of the city's water comes from the Kaveri (Cauvery) River, about 45 miles (70 km) to the south.

From the late 20th century, the city became a centre of high-technology industry (notably ICT), and a number of large multinational technology corporations opened offices there. In addition, major domestic firms such as Infosys and Wipro established headquarters in the city. In 1998 an ICT park opened in what was then the suburb of Whitefield, about 10 miles (16 km) east of central Bengaluru. As a self-contained city with hundreds of technology, software, and telecommunications companies, the park became known as **the Silicon Valley of India**. (Source: www.britannica.com, retrieved on 14.09.2022)

1.2 Bangalore Metropolitan Region (BMR)

The City of Bangalore is fast developing. Besides the Bangalore urban corridor, Bangalore rural district is also expected to experience population influx due to several developmental activities happening in Bangalore and its outer fringes. For the purpose of planning, co-ordinating and supervising the proper and orderly development of the areas around the City, the Government of Karnataka has created an autonomous body, Bangalore Metropolitan Region Development Authority (BMRDA) under the BMRDA Act 1985 comprising of Bangalore urban district, Bangalore rural district and Ramanagar district. The entire region is known as **Bangalore Metropolitan Region**.

As per the Structure Plan, apart from BMA (BDA jurisdiction), the rest of the Bangalore Metropolitan Region [BMR] is divided into five **Area Planning Zones** (APZ's) and six Interstitial Zones (IZ's). The APZ's are proposed along the corridors, which are,

- (1) Bangalore-Bidadi,
- (2) Bangalore-Nelamangala,
- (3) Bangalore-Devanahalli
- (4) Bangalore- Whitefield, Hoskote
- (5) Bangalore-Anekal, Sarjapur-Hosur.

The Area Planning Zones (APZ's) are areas where urban development is permitted subject to certain regulations. The Interstitial Zones are the areas lying between APZ's where urban activities are restricted giving more emphasis to environmental – issues like conservation of forest area, agriculture etc. For planned urban growth, local planning areas under Karnataka Town & Country Planning Act-1961 are declared in the five APZ's and IZ's 1 & 2.

Table 1.1 gives the details of the Planning / Development Authorities functioning in the Bangalore Metropolitan Region (BMR).

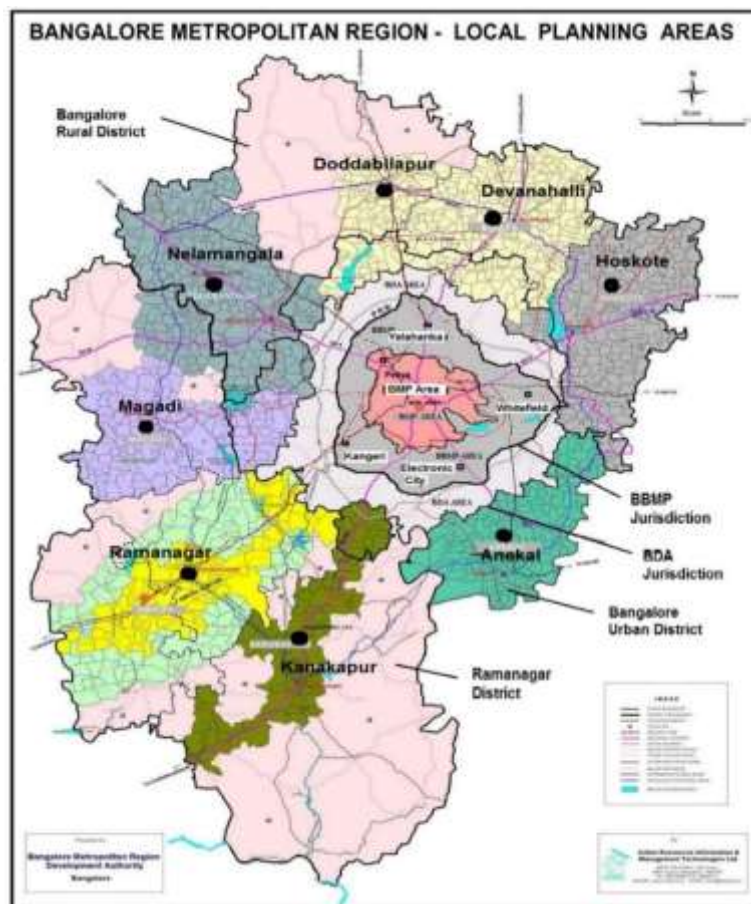


Figure 1.3: Bangalore Metropolitan Region and BBMP Jurisdiction

Table 1.1: Authorities functioning within Bangalore Metropolitan Region

Sl. No.	Authority	Area in Sq.Km.
1	Bangalore Development Authority [BDA]	1219.50
2	Bangalore – Mysore Infrastructure Corridor Area Planning Authority [BMICAPA]	426.24
3	Ramanagar-Channapatna Urban Development Authority [RCIDA]	63.06
4	Anekal Planning Authority [APA]	402.30
5	Nelamangala Planning Authority [NPA]	735.00
6	Magadi Planning Authority [MPA]	501.52
7	Hosakote Planning Authority [HPA]	535.00
8	Kanakapura Planning Authority [HPA]	412.78
9	Bangalore International Airport Area Planning Authority [BIAAPA]	792.00
10	Area Planning Zone-1 [APZ-1] (Excluding RDUDA & BMICAPA)	462.60
11	All Interstitial Zones in Bangalore Metropolitan Region [IZ's BMR]	2455.00
Total		8005.00

1.3 Bruhat Bengaluru Mahanagara Palike (BBMP)

The **Bruhat Bengaluru Mahanagara Palike (BBMP)** is the administrative body responsible for civic amenities and some infrastructural assets of the Greater Bangalore metropolitan area. It is the fourth largest Municipal Corporation in India and is responsible for a population of 8.4 million in an area of 741 km². Its roles and responsibilities include the "orderly development of the city" — zoning and building regulations, health, hygiene, licensing, trade and education, as well as quality of life issues such as public open space, water bodies, parks and greenery. It often works in conjunction with other civic bodies such as the Agenda for Bangalore Infrastructure Development Task Force (ABIDE) and the Bangalore Development Authority (BDA) to design and implement civic and infrastructural projects.

The BBMP represents **the third level of government** (the Central Government and State Government being the first two levels). BBMP is run by a city council composed of elected representatives, called "corporators", one from each of the wards (localities) of the city.

The history of municipal governance of Bangalore dates back to 27 March 1862, when nine leading citizens of the old city formed a Municipal Board under the Improvement of Towns Act of 1850 with a similar Municipal Board formed for the newer Cantonment area. The two boards were legalised in 1881, and functioned as two independent bodies called Bangalore City Municipality and Bangalore Civil and Military Station Municipality (Cantonment). The following year, half of the municipal councilors were permitted to be elected, property tax was introduced and greater powers given over police and local improvement.

In 1913, an honorary president was introduced, and seven years later made an elected position. An appointed Municipal Commissioner was introduced in 1926 on the Cantonment board as the executive authority. After Indian independence, the two Municipal Boards were merged to form the Corporation of the City of Bangalore in 1949, under the Bangalore City Corporation Act. The corporation then consisted of 70 elected representatives and 50 electoral divisions and the office of Mayor introduced for the first time. The first elections were held in 1950.

The name of the council changed - first to Bangalore City Corporation (BCC) and then to Bengaluru Mahanagara Palike (BMP). In 1989, the BMP expanded to include 87 wards and further increased to 100 wards in 1995, covering an extra area of 75 sq. km. In **January 2007**, the Karnataka Government issued a notification to merge 100 wards of the erstwhile Bengaluru Mahanagara Palike with seven City Municipal Councils (CMC)s, one Town Municipal Council (TMC) and 110 villages around the city to form a single administrative area (111 villages mentioned in initial Notification. Later 2 villages omitted from the list and another village added before final Notification). The process was completed by April 2007 and the body was renamed 'Bruhat Bengaluru Mahanagara Palike (Greater Bangalore Municipal Corporation).

In the absence of an elected body, the BBMP is at present run by an Administrator and a Commissioner, who is appointed by the State Government. The Additional Commissioner is a specially appointed IRS officer.

(Source: https://en.wikipedia.org/wiki/Bruhat_Bengaluru_Mahanagara_Palike, retrieved on 17.10.2022)

1.4 Administrative Zones

The BBMP is divided into 8 zones for ease of administration, each administered by a Zonal Commissioner. They are, i. Yelahanka, ii. Dasarahalli, iii. Rajarajeshwarinagar (RR Nagar), iv. Bommanahalli v. East, vi. West, vii. South and viii. Mahadevapura. Their jurisdictions are shown in Figure 1.6.



Figure 1.6: Bruhat Bengaluru Mahanagara Palike (BBMP) Zones

1.5 Role of various organizations

Bruhat Bengaluru Mahanagara Palike (BBMP): The Bruhat Bengaluru Mahanagara Palike is Bangalore's administrative body. It is the fourth largest municipal corporation in India after Mumbai, Delhi and Chennai. Commonly known as BBMP, it is administered by a city council that has elected representatives called 'corporators'. It looks after the civic and infrastructure requirements of Bangalore. It also includes the maintenance of the civic and infrastructure assets of the city. Some of the major responsibilities of the BBMP include:

- Undertaking zoning and building regulations
- Ensuring health and hygiene in the city
- Maintenance of open spaces and water bodies
- Building and maintaining sewage
- Maintenance of roads and bridges
- Waste management

Bangalore Development Authority (BDA): Falling under the jurisdiction of the Government of Karnataka, the Bangalore Development Authority is responsible for the entire city's planning. It has the jurisdiction of the BBMP area and certain surrounding

towns and villages. The jurisdiction of the BBMP is 716 sq km, while of the BDA is 1219.5 sq km. The primary role of the BDA involves zoning, planning, and regulation of the land in Bangalore. Some of the focus areas of the BDA include:

- Planned development of Bangalore
- Creation of quality infrastructure
- Maintenance of lakes
- Addressing the housing requirements of economically weaker sections

Bangalore Metropolitan Region Development Authority (BMRDA): It is an autonomous body in charge of planning, coordinating, and supervising the systematic development of areas falling under the Bangalore Metropolitan Region. This includes Bangalore Urban District, Bangalore Rural District, and Ramnagar District. The main role of the BMRDA is to bring changes in the policies related to urban development. It does not provide approval for the layouts directly. The layout approvals should come from the local planning authorities. Development in Anekal, Hoskote, Kanakapura, Magadi, Nelamangala, and Ramanagaram-Channapatna is permitted with planning authorities' approval. For Devanahalli, the Bangalore International Airport Area Planning Authority (BIAAPA) is the planning approval authority. The BMRDA only sets the rule for these planning authorities to help their master plans and does not approve a layout directly.

Department of Town and Country Planning (DTCP): The DTCP is involved in the task of preparing master plans and their enforcement. Also, it sanctions the layout plan in the state, where the planning bodies like BMRDA, BDA, BIAAPA, etc. do not exist. The main functions of the DTCP include:

- Preparation of master plans
- Assisting the Karnataka State Town and Country Planning Board and giving technical assistance to Urban Development Authorities/ Planning Authorities in plan preparation, implementation and enforcement under the Karnataka Town & Country Planning Act, 1961
- Offering assistance to developmental agencies like Karnataka Housing Board, Karnataka Industrial Areas Development Board, and other such agencies
- Preparation of systematic, efficient, and safe traffic management schemes.

1.6 Rainfall

The analysis of 115 years rainfall data from 1901 to 2015 reveals that rainfall in Bengaluru is spread across seven months (i.e., 87 per cent of rainfall occurs between the months of May and November, September being the highest with average rainfall of 156 mm). The average annual rainfall at the city is about 787 mm with 75 per cent dependability and return period of five years. The runoff yield is in the range of 600-700 mm in most of Bengaluru. Catchment wise water yield analysis indicates about 49.5 per cent of water yield in the Vrishabhavathi valley, followed by 35.2 per cent in K&C valley and 15.3 per cent in Hebbal valley. The average annual water yield in Bengaluru is **about 14.80 TMC**. (Ref: Ramachandra, et al., 2016)

Chapter 2

Drainage Valleys and Lake System

2.1 Drainage Valleys

Figure 2.1 shows the drainage valleys at the city. It has three major natural valleys and three minor ones. The major valleys are Vrishabhavathi Valley, Koramangala-Challaghatta Valley and Hebbal Valley. The minor valleys are Arkavathi, Kathriguppe and Tavarekere.

The topographic setting of the city has radial slopes towards east and west with a smooth ridge running north to south; rainfall over the ridge area gets divided and flows east or west into the three gentle slopes and valleys of Koramangala–Challagatta, Hebbal and Vrishabhavathi. Doddabettahalli 1,062 m (3,484.3 ft) is **the highest point** on this ridge. Both K&C valley and Hebbal valley join at Nagondanahalli village which further flows to Dakshina Pinakini river while Vrishabhavathi valley joins Arkavathi river which is a tributary of river Cauvery.

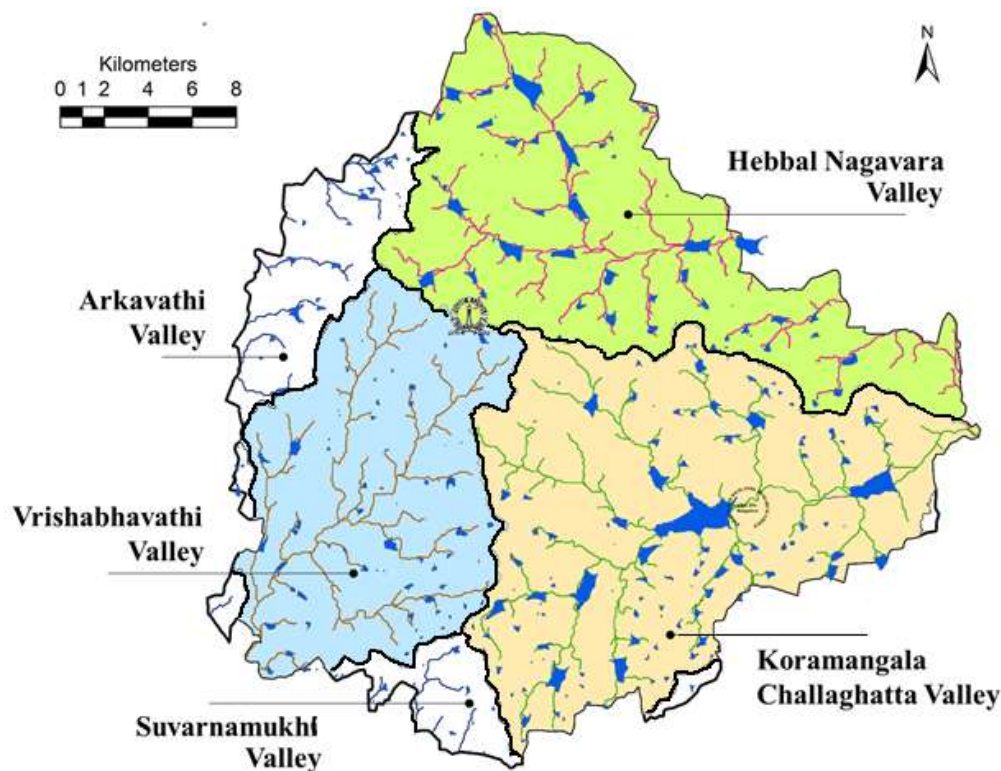


Figure 2.1: The Drainage Valleys of the city

2.2 Lakes and Lake Series

The naturally undulating terrain of the city, with its hills and valleys, lends itself perfectly to the development of lakes that impound rainwater, store it and ensure ground water recharge. They are designed in cascades from higher to lower elevations; as a lake

overflowed the excess water would flow into the next lake in the cascade. The flow of water is from North to South-east as well North to South-west along the natural gradient of the land. They form a chain of reservoirs in each of the three valley systems. Each valley at the ridge top gives birth to small streams. The valleys are thus the repository of all the lakes in Bangalore and these lakes themselves are interlinked to each other through a series of chains of lakes giving a cascading effect to the whole system.

The lakes at the city form into **6 series** in the three major valleys as listed below and shown in Figure 2.2,

- i. Yellamallappa Chetty Lake series
- ii. Varthur Lake series
- iii. Hulimavu Lake series
- iv. Byramangala Lake series
- v. Puttenahalli Lake series
- vi. Madavara Lake series

Figures 2.3 to Figure 2.8 show the individual lake series.



Figure 2.2: Important Lake series in Bengaluru region

As per the study of Ramachandra T V., et. al., the city had 1,452 water bodies with a total storage capacity of 35 TMC (thousand million cubic feet) (in the current spatial extent of 741 sq km) during early 1800s. By 2016, the number of water bodies in the same area reduced to 194 with a storage capacity of 5 TMC. Due to siltation, the current storage capacity further decreased to 1.2 TMC (2016).

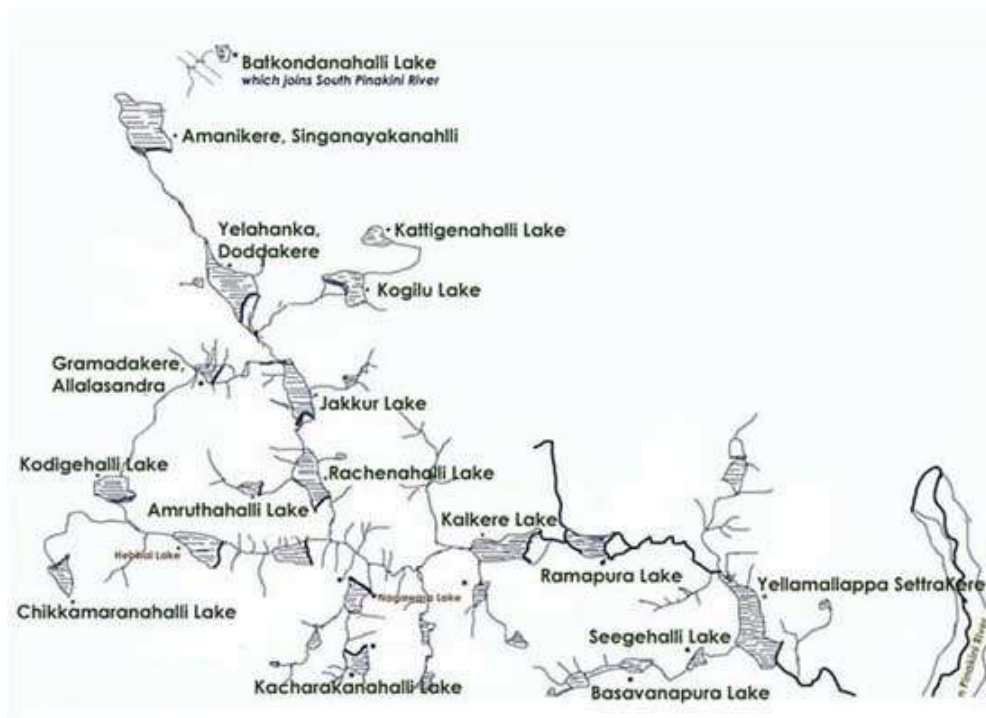


Figure 2.3: Yellamallappa Chetty Lake series

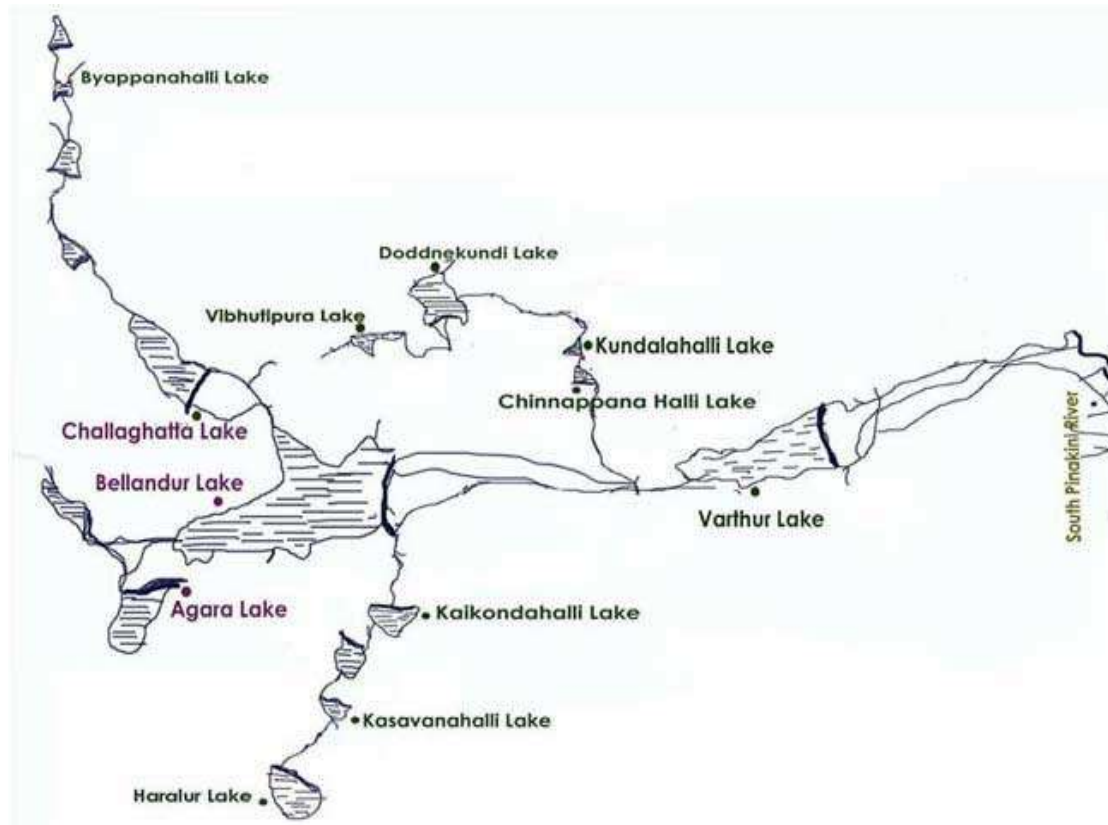


Figure 2.4: Varthur Lake series



Figure 2.5: Hulimavu Lake series

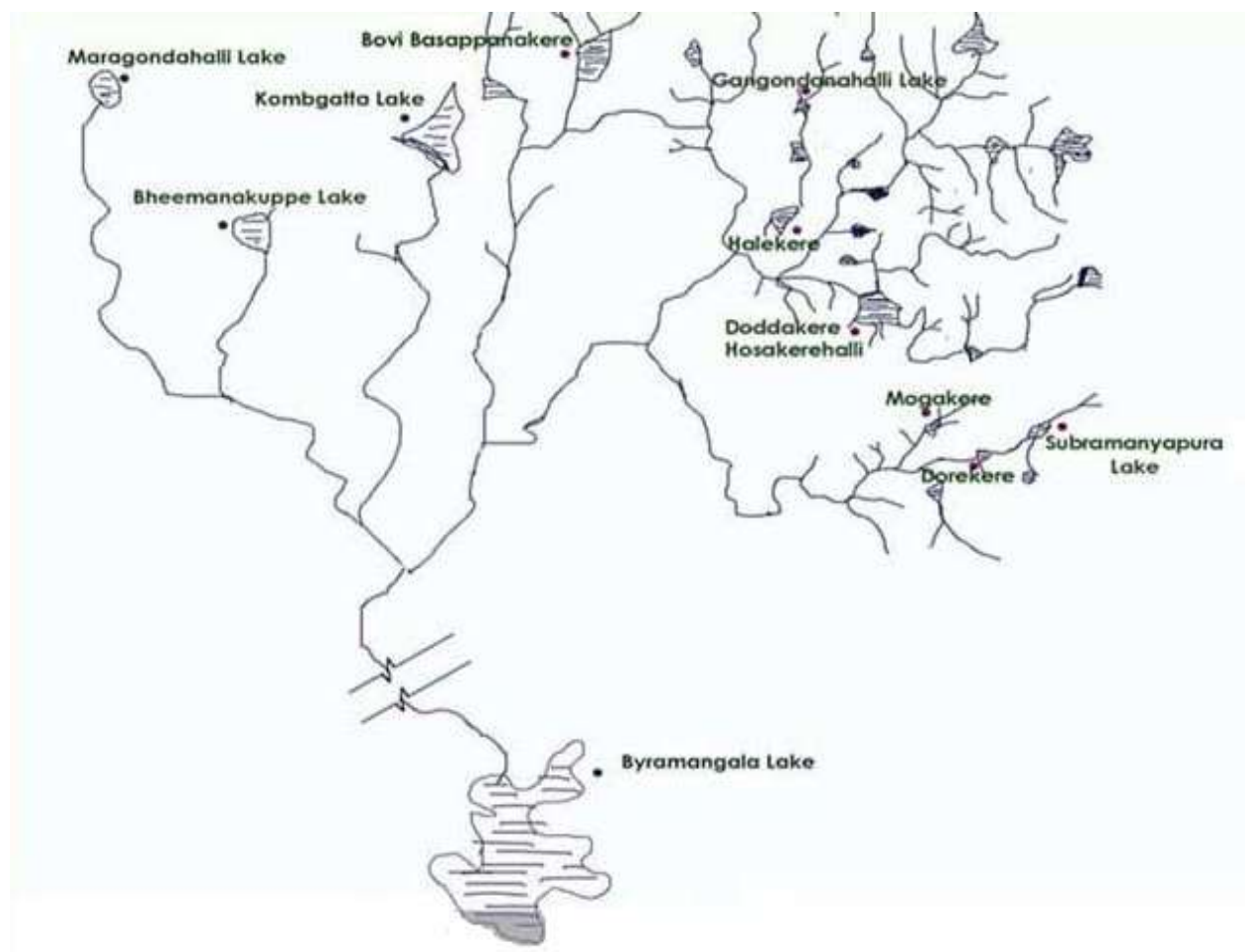


Figure 2.6: Byramangala Lake series



Figure 2.7: Puttenahalli Lake series



Figure 2.8: Madavara Lake series

2.3 Management of Lakes

A performance audit on the “Management of Storm Water in Bengaluru Urban Area” was conducted by the Controller and Auditor General of India in 2021 to examine the existence and adequacy of the mechanism for the management of storm water. It gives an account of management of lakes at the city; its sections are reproduced below.

3.3.1 Management of lakes

Bengaluru city was once aptly known as ‘city of lakes’ due to the presence of large number of lakes. As per the 2016 IISc report, it had around 1,452 water bodies during the early 1800s. These were gradually reduced due to conversions/encroachments as discussed in the subsequent paragraph. The lakes/water bodies in Bengaluru city were under the custody of different authorities such as BBMP, Forest Department, BDA, KLDCA, *etc.* The Government ordered (September 2016 and December 2019) the transfer of all lakes (except Bellandur and Varthur lakes which remained with BDA) to BBMP. The BBMP did not possess comprehensive data on the actual number of lakes, their status and custodians. As per data furnished (December 2020) by the CE, Lakes Division, BBMP, there were 210 lakes under its jurisdiction. Details are furnished in the **Table 3.1** below.

Table 3.1: Status of maintenance of lakes/water bodies in Bengaluru

Status of lakes	Number of lakes
Developed lakes	89
Work in progress	37
To be Developed	66
Disused lakes	18
Total	210

Source: CE, Lakes Division, BBMP

It could be observed that while 66 lakes were yet to be developed, 18 lakes with a total area of 254 acres and 17 guntas were reported as disused lakes. The current status of the disused lakes as per the Google Earth images is shown in **Exhibit 3.2** and the list of disused lakes is given in **Appendix 3.2**. There is a significant risk of these lakes being vulnerable to encroachments and future conversions. (page 16 of CAG Report)

3.3.2 Decrease in water bodies and length of drains

Audit conducted an independent study of long-term changes in land use patterns of Bengaluru city utilising geospatial inputs with technical support from Regional Remote Sensing Centre–South, Indian Space Research Organization, Bengaluru (RRSC). The methodology adopted for the study is already explained in Paragraph 2.4. The study corroborates the fact that urbanisation resulted in decrease in water bodies and length of drains as discussed below.

Exhibit 3.2: Current status of disused lakes

(Source: Google Earth images as on 19 January 2021)



Chennamma kere

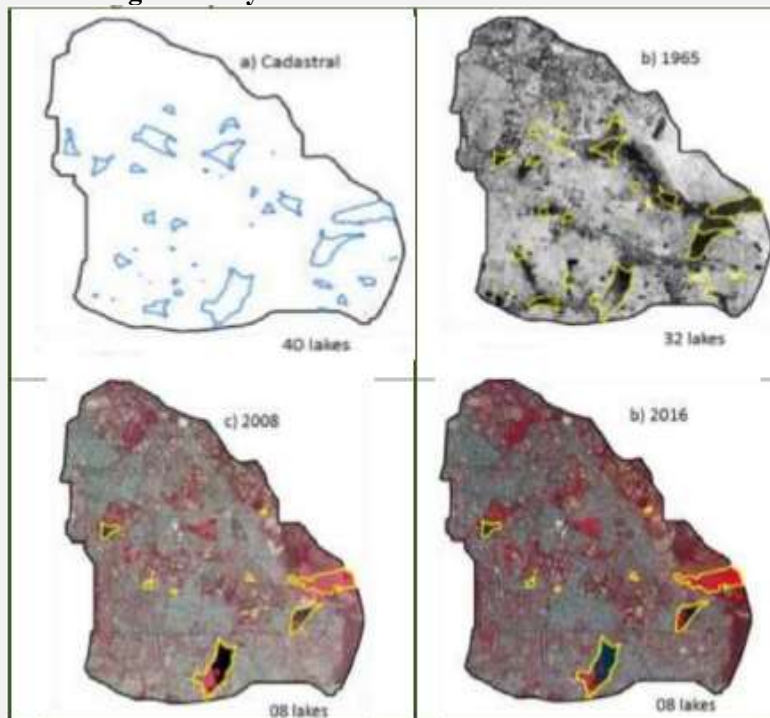


Doresanipalya kere

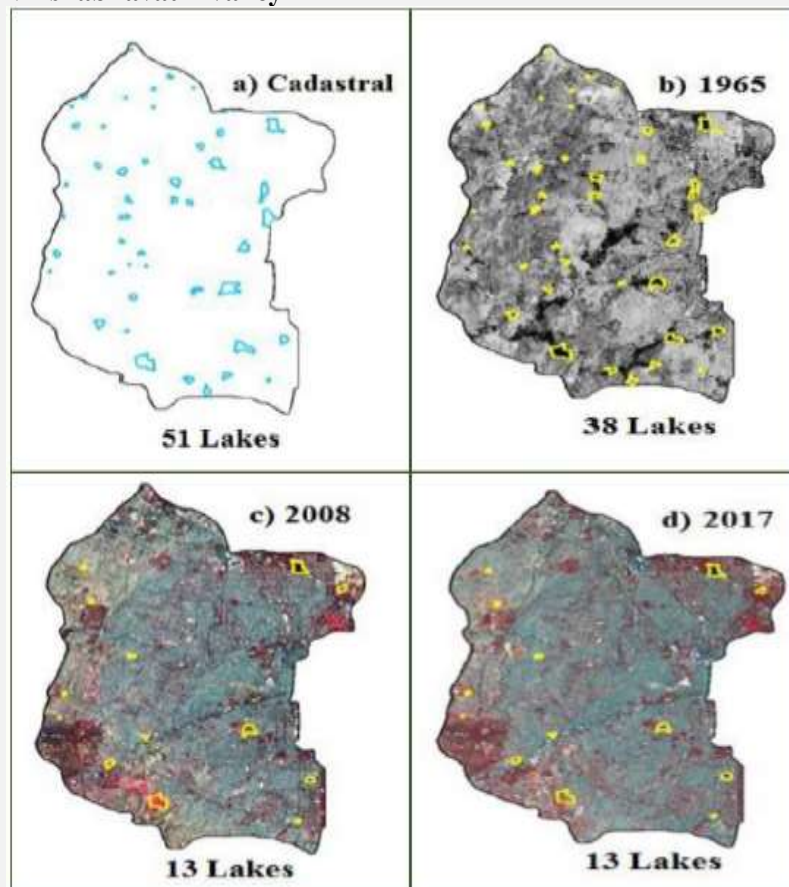
Decrease in water bodies

The study revealed that about 41 and 51 water bodies, which existed in Koramangala and Vrishabhavathi valleys respectively as per the cadastral map (prepared through field survey during early 1900s) were reduced to 8 and 13 by the year 2008 indicating the severity of lake conversion. Further, the wetland system (lakes, tanks, *kere* and *katte*), which contributed about 479.48 ha (0.75 per cent) and 215.46 ha (2.24 per cent) to the geographical area of the valleys as per the cadastral map decreased to 262.37 ha and 62.05 ha during 2016/2017 indicating erosion due to land use changes. The time series map showing the changes in lakes/tanks is indicated in **Chart 3.1**.

**Chart 3.1: Time series maps showing changes in lakes / tanks
Koramangala valley**



Vrishabhavathi valley



The change in size and shape of a few lakes in both the valleys which is indicative of significant reduction of lake foot print is indicated in **Appendix 3.3**. The reduction in the number of lakes can be linked to conversion of lakes for other purposes (**Exhibit 3.3**) such as bus stands, stadiums/sports grounds, National Games village housing complex, residential layouts etc., as indicated in **Appendix 3.4**.

Exhibit 3.3: Conversion of lakes for other purposes



Kamakshipalya lake converted as sports ground

Evidently, these conversions were authorised by the State Government, BBMP, BDA *etc.*, over a period of time. Reference is invited to Paragraph 4.4 of the Report on Lakes which refers to instances of irregular grant of lake land to various Government bodies, private parties and others in violation of the codal provisions which resulted in reduction of lake area. These conversions also resulted in the change in land use from being pervious to impervious thereby leading to higher runoff of rain water and consequent flooding.

Further, a Legislative Committee constituted to study the status of encroachments on lakes/water bodies in Bengaluru and its surroundings covering the period 2014-2017, in its report revealed unabated encroachments of most of the water bodies, both by government departments/agencies and private parties, leading to diminishing boundaries of wetlands besides pollution of the ecosystem. The major encroachers among government agencies were Education department, Forest department, BDA, BWSSB, BBMP, Bengaluru Electric Supply Company, Bengaluru Metropolitan Transport Corporation, Karnataka Housing Board, Railways, Defence authorities, National Highways, *etc.*

The Committee expressed dissatisfaction particularly with the BDA, forming residential layouts over as many as 23 water bodies without obtaining the permission from the Revenue Department. The Committee attributed degeneration of water bodies to the negligence and irresponsibility of officials and recommended action against the offenders through a judicial investigation.

Appendix 3.2**(Reference: Paragraph 3.3.1/Page 16)****List of disused lakes under BBMP**

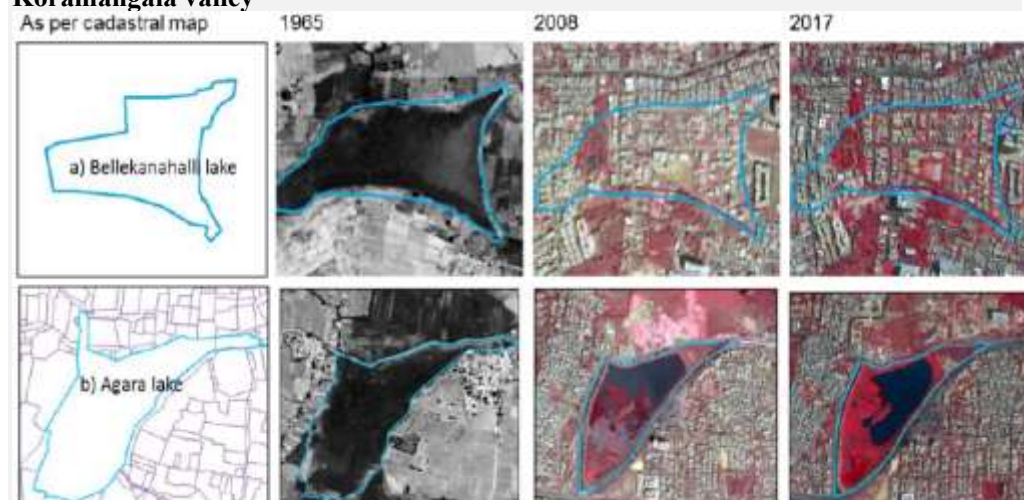
Sl. No.	Zone	Name of the lake	Village and survey number	Area	
				Acres	Guntas
1	Bangalore	Tavarekere	Tavarekere -74	10.00	16.00
2	South	Karisandra Lake	Karisandra - 7	13.00	0.00
3		Nandi Shettappa Lake	Jaraganahalli-53	0.00	0.00
4		Chikkalsandra Lake	Chikkalsandra - 76	12.00	26.00
5		Ittmadu	Ittamadu-17	5 4.00	0.00
6	Bommanahalli	Belakahalli (Lingannana Kere)	Belakahalli-172/2A	7.00	0.00
7		Doresani Palya	Belakahalli-167	56.00	37.00
8	East	Konena Agrahara lake	Konena Agrahara -60	20.00	10.00
9		Byatagunte Palya Lake	Byatagunte palya - 14	5.00	25.00
10		Geddalahalli Lake	Geddalahalli-03	21.00	18.00
11		Lingarajapura Lake	Lingarajapura-49	16.00	14.00
12	Mahadevapura	Vijanapura Lake	Vijanapura-42	29.00	15.00
13	RR Nagar	Bovimaranahalli	Halagevaderahalli - 124	22.00	34.00
14		Gundopanth Lake	Pantharapalya-59	2.00	1.00
15	West	Anche ramana kere (Gangondanahalli)	Gangondana halli - 8	0.00	17.00
16		Sanigoruvahalli	Sanigoruvahalli - 120	15.00	24.00
17		Shivanahalli (Agrahara Dasarahalli)	Agrahara dasarahalli - 72	9.00	25.00
18		Kamakshi palya	Sanegoruvanahalli - 60	6.00	35.00
	Total			247.00	297.00

Appendix 3.3

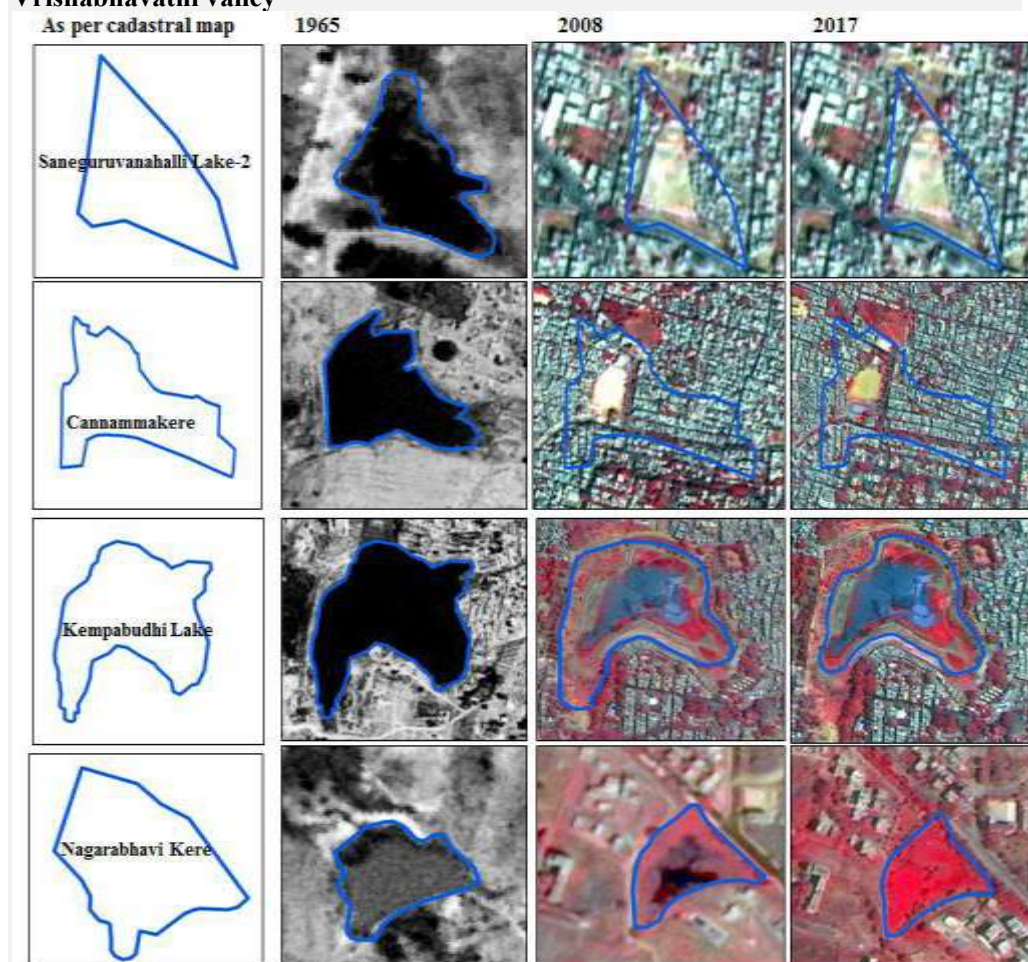
(Reference: Paragraph 3.3.2, Page 20)

Typical changes (change in size and shape) noticed in lakes

Koramangala valley



Vrishabhavathi valley



Appendix 3.4

(Reference: Paragraph 3.3.2, Page 20)

Table showing a few lakes converted for other purposes

<i>Name of Lake</i>	<i>Status now</i>
<i>Akkithimmanhalli Lake</i>	Hockey Stadium
<i>B.Channasandra Lake</i>	BDA Layout - East of NGEF Layout
<i>Baalayyana Lake</i>	Sports Ground
<i>Bagalagunte Hosa Lake</i>	Residential Layout
<i>Challaghatta Lake</i>	Golf ground - Karnataka Golf Association
<i>Challakere Lake</i>	BDA Layout - HBR 2nd Block
<i>Channasandra-2 Lake</i>	BDA Layout - HRBR 1st Block
<i>Chennamma Lake</i>	BDA Layout - Banashankari 2nd Stage
<i>Dasarahalli Lake</i>	Dr. B.R Ambedkar Stadium
<i>Dharmambudhi Lake</i>	Majestic Bus Stand
<i>Domlur Lake</i>	BDA Layout - Domlur 2nd Stage
<i>Government Lake</i>	BDA Layout - Nagarabhavi 2nd Stage
<i>Government Lake</i>	BDA Layout - RMV 2nd Stage
<i>Government Lake</i>	BDA Layout - HBR 1st Block
<i>Government Lake</i>	BDA Layout - Banashankari 6th Phase
<i>Hennur-Nagavara Lakes</i>	BDA Layout - HBR Layout 1st Stage, 5th Bk
<i>Jakkarayana Lake</i>	Sports Ground
<i>Kadugondanahalli Lake</i>	Ambedkar Medical College
<i>Kamakshipalya Lake</i>	Sports Ground
<i>Karanji Lake</i>	Gandhi Bazar Residential area
<i>Kempambudhi Lake</i>	Sewerage collection tank
<i>Kethmaranahalli Lake</i>	BDA Layout - Rajajinagar 1st Block
<i>Kodihalli Lake</i>	Residential Layout
<i>Koramangala Lake</i>	National Games Village housing complex
<i>Krishnarajapuram Lake</i>	BDA Layout - East of NGEF Layout
<i>Kurubarahalli lake</i>	Residential Layout
<i>Lingannana Lake</i>	BDA Layout - Sarakki Dollars Colony
<i>Manganahalli Lake</i>	BDA Layout - Sir MV Layout 6th Block
<i>Marenahalli Lake</i>	Residential Layout
<i>Miller's Tank</i>	Guru Nanak Bhavan, Badminton Stadium
<i>Nagashettihalli</i>	Lake Space Department
<i>Sampangi</i>	Lake Sports Stadium
<i>Shivanahalli</i>	Lake Playground, Bus stand
<i>Shoolay Lake</i>	Football Stadium
<i>Siddikatte Lake</i>	KR Market
<i>Srinivagalu Lake</i>	BDA Layout - ST Bed Layout Koramangala
<i>Subhashnagar Lake</i>	Residential Layout
<i>Thippasanda Lake</i>	BDA Layout - HAL 2nd and 3rd Stage
<i>Venkatarayana Lake</i>	BDA Layout - Banashankari 6th Phase
<i>Yellugunte Lake</i>	BDA Layout - HSR 3rd Sector

Source : Sri Koliwad Committee Report,
<http://parisaramahiti.kar.nic.in/lostlakes.html>, and
https://en.wikipedia.org/wiki/Lakes_in_Bangalore

Chapter 3

Storm Water Management - CAG Report

The Audit Report of CAG-2021 gives a detailed account of the storm water management at the city. Its few sections reproduced below present a fair picture of its management.

1.7 Organisational structure for management of storm water drains

The Urban Development Department (UDD) headed by Additional Chief Secretary (ACS) to Government of Karnataka is the controlling department for all Urban Local Bodies in the State, including BBMP. The Commissioner, BBMP is responsible for enforcing and overseeing the implementation of storm water management under the BBMP area. The Commissioner is assisted by Special Commissioner (Projects), Engineer-in-Chief and Chief Engineer (SWD). For the management of SWDs, the BBMP is divided into nine zones each headed by an Executive Engineer (EE).

4 Planning, designing and construction of storm water drains

4.1 Planning

Section 9 of the Karnataka Town and Country Planning Act, 1961, stipulates that every planning authority should carry out a survey of the area within its jurisdiction and, prepare and publish a comprehensive development plan (CDP)/revised master plan (RMP) consisting of a series of maps and documents indicating the manner in which the development and improvement of the entire planning area within the jurisdiction of the Planning Authority is to be carried out and regulated.

Further, Section 13-D of the Act also provides for revision of the master plan at least once in 10 years from the date on which the master plan had come into force. The plans should indicate areas reserved for parks, play grounds and other recreational uses, public open spaces, public buildings and institutions etc. The Act does not explicitly describe the area preserved as tanks or lakes in the CDP/RMP.

4.1.1 Deficiencies in the Master Plans prepared by BDA

The Bengaluru Development Authority (BDA) prepares master plans. The details of plans brought out by BDA are as shown in the **Table 4.1**.

Table 4.1: Details of plans prepared by BDA

Plan	Identified as	Date of approval	Valid till
First	Outline Development Plan	22-05-1972	11-10-1984
Second	Comprehensive Development Plan	12-10-1984	04-01-1995
Third	Revised Comprehensive Development Plan	05-01-1995	24-06-2007
Fourth	Revised Master Plan	25-06-2007	Till the approval of RMP-2031

Source: Information furnished by BDA

BDA did not have on record the first two development plans prepared for the periods May 1972 to October 1984 and October 1984 to January 1995. In the revised CDP approved in January 1995 and valid till June 2007, there was no consistency in the representation of water bodies which were shown as tanks and also as parks and valleys; and drains (water ways) were not exhibited explicitly. The CDP was, therefore, incomplete and deficient. In the absence of clear data on the width/type of the drains at any stage, encroachment /disruption of flow could not be analysed.

The Revised Master Plan - 2015 for Bengaluru approved in June 2007 and valid till the approval of RMP 2031 was also deficient for the following reasons.

- Though RMP-2015 recognised the importance of having buffer zones for different types of SWDs, it did not classify the drains as required and hence, did not notify 'no development area' along the drains. As a result, the required buffer zone around/along the water bodies/water ways were neither marked nor maintained.
- Though drains were mapped in the RMP-2015, many existing drains and water bodies (as per the Master Plan of SWDs prepared by BBMP) were not shown in the maps published (**Exhibit 4.1**).

Exhibit 4.1: Illustrative photographs showing drains/water bodies not mapped in RMP-2015 (shown in red rectangles)

Master Plan of SWDs



Shown as a water body

RMP-2015



Shown as a park



SWD shown between Bellandur and Ibbalur tanks



No connectivity between two tanks

Buffer zones are areas of land adjacent to a drain or water body which are meant for providing utilities such as power, pipelines for water/oil/gas *etc.*, and also to facilitate easy maintenance of drains. The RMP stipulated buffer of 50, 25 and 15 mtrs (measured from the centre of the drain) on either side of primary, secondary and tertiary drains respectively. (page 34)

The lapses indicated above facilitated unabated construction along the drains without allowing the required buffer area. Photograph captured during joint inspection showing construction without allowing buffer area along SWDs are shown in **Exhibit 4.2**.

Exhibit 4.2: Illustrative photographs showing construction of buildings allowed without buffer area along storm water drains



Bommanahalli Zone



Koramangala Zone



Rajarajeshwarinagar Zone



South Zone (Covered Drain)



West Zone



West Zone

4.1.3 Preparation of master plan of drains by BBMP

The master plan prepared by BDA was to be followed by all the authorities for taking up any development work. However, we observed that BBMP got a separate master plan of drains (including the expanded area comprising of City/Town Municipal Councils and 110 villages that was integrated during 2007), water bodies, bridges/culverts, low lying areas, *etc.*, prepared by M/s. STUP Consultants at a cost of 3.62 crore during 2010-11.

The master plan of drains of BBMP was incomplete as

- it was restricted to identifying only the primary and secondary drains in contradiction of the NDM guidelines which stipulated preparation of comprehensive database of all drains.
- many of the drains shown in RMP-2015 were not mapped in the master plan of drains (**Exhibit 4.3**). Besides, a large number of drains which were in existence but not found in RMP-2015 were also not mapped. This raises questions on the validity and reliability of the database.
- the master plan did not provide for buffer zones along the different kind of drains, despite being clearly spelt out in RMP-2015, which was in force.

Further, the records relating to preparation of the master plan of drains containing tender conditions, tendering process, award of contract and payments made to the agency were recorded 'to have been lost' and thus, not furnished to audit. The CE, SWD also did not possess on record the detailed volumes of master plan pertaining to individual zones, except for Yelahanka and Rajarajeshwari Nagar (RR Nagar). Hence, the correctness of the preparation of the master plan by the agency as well as compliance to tender conditions could not be verified.

BBMP did not initiate action to reconcile its master plan with that prepared by BDA. It also did not conduct any physical inspection of the drains to update its master plan and to ensure inclusion of all the drains under its jurisdiction in the RMPs. Significant discrepancies between the two sources of data deprived planners of a single source of truth for planning/development of the city.

The State Government replied (January 2019) that RMP-2015 which was in place at the time of preparing SWD master plan had not captured the drainage networks. The SWD master plan has been shared with BDA and now finds its place in the draft RMP-2031 and the anomalies are getting ironed out.

However, verification of draft RMP-2031 showed that drains shown as primary and secondary in BBMP's map were exhibited as secondary and tertiary in the draft RMP-2031. These discrepancies assume higher significance in light of the judgments of the National Green Tribunal (NGT) enhancing the buffer area along the drains. Moreover, the basis on which BDA identified, classified and exhibited the tertiary drains in the RMP when BBMP, the authority for construction and maintenance of SWD, does not have the data on tertiary drains was not explained.

The State Government accepted (August 2020) that the nomenclature of primary, secondary and tertiary drains in draft RMP-2031 are different from that mentioned in SWD master plan and action would be taken to discuss the issues with BDA for proper reconciliation and corrective measures.

4.1.4 Storm water drain inventory

Paragraph 4.5 of NDM guidelines stipulate that all ULBs/States/UTs shall prepare an inventory of the existing storm water drainage system on a Geographic Information System (GIS) platform. The inventory was to be both watershed based and ward based with clear mapping of the major as well as minor systems. Further each road was supposed to have drains on both sides for collecting storm water which would ultimately lead into primary/secondary SWDs to allow runoff. Hence, the tertiary/road side drains form the major contributor to urban drain runoff. For quantification of runoff in different kinds of drains and their upkeep without allowing for clogging/flooding, the SWD authorities should have on record comprehensive data of different types of roads (length, width, type of surface, perviousness, gradient, *etc.*) collected at regular intervals.

However, the CE, SWD, the authority for construction and designing SWDs thereon, did not possess comprehensive data of different roads and tertiary/surface road side drains within the jurisdiction of BBMP. The lack of comprehensive data on runoff is bound to have an adverse impact on the design, construction and management of drains.

The absence of comprehensive inventory of drains with BBMP and its failure to classify them properly contributed to lack of clarity on critical issues including the extent of buffer zone to be maintained. Buffer of 50, 35 and 25 mtrs (measured from the edge of the drain) on either side primary, secondary and tertiary drains respectively. This, in turn, would

- hamper regular maintenance of the drains.
- impact one of the purposes of creating a buffer zone *i.e.*, to provide space for laying of utilities. Audit observed that many utility lines like water pipes, sanitation pipes, electrical, telephone, optical cable, *etc.*, were laid across the drains in many locations obstructing the flow in drains and overflows (**Exhibit 4.4**). The absence of buffer zone also results in encroachments as indicated in Paragraph 5.1.3.

Exhibit 4.4: Photos showing the presence of utility lines in SWDs



West Zone



Bommanahalli Zone



Koramangala Zone



Bommanahalli Zone



NGV campus, Koramangala Zone



South Zone

4.2.1.6 Multiple authorities within BBMP for construction/management of drains leading to lack of coordination

For an efficient SWD system with due discharge of all the runoff into definitive locations, the interconnectivity of all types of drains is essential. The IRC guidelines specify that urbanisation of any locality and population needs a well-engineered surface and subsurface drainage system. In the present day context of depletion of water table, the storm water drainage should be effectively utilised for ground water recharging. It should be ensured that water from the road flows to the roadside drains through inlets and gratings. As per paragraph 4.13.4.1 of NDM guidelines, inlets should be provided on the roads to drain water to the roadside drains. For effective drainage, this should join the peripheral drains, which in turn should join the main or trunk drain for ultimate discharge to the natural drain or detention facility or retention facility.

BBMP has in place different authorities for construction and maintenance of different types of drains/roads under its jurisdiction. Audit observed from the joint physical inspection of a few drains that BBMP had constructed roads/drains without ensuring that the inlets to drain water from the roads were properly aligned with the roadside drains/underground drains leading to water logging on roads. Evidently, there was lack of coordination among these authorities to ensure proper cambering/ gradient during formation of roads, regular cleaning of bell mouths/kerb vents provided to surface roads as well as proper linkage with SWDs resulting in choking and clogging of water on roads (**Exhibit 4.8 and 4.9**). The tweets of the various stations of traffic department (**Exhibit 4.10**) attribute water logging to choking and blockage of drains.

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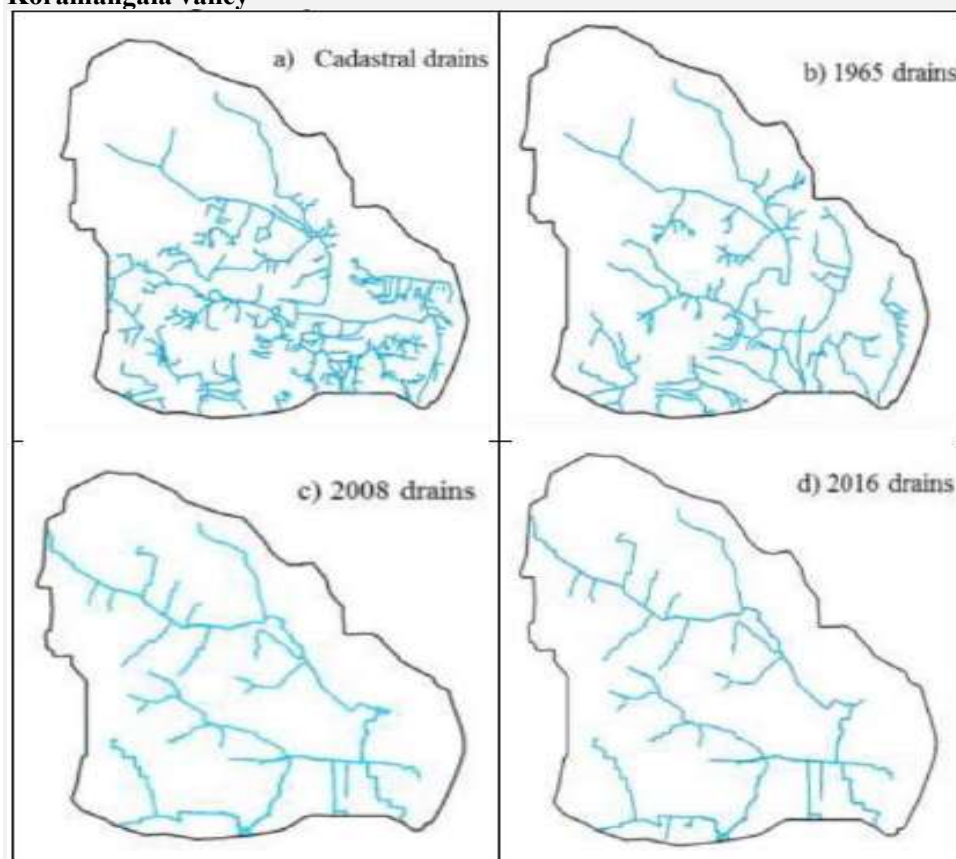
3.3.2 Decrease in water bodies and length of drains

Reduction in length of drains

Natural drainage/stream systems were concretised into storm water drains with increase in urbanisation. As per the study, total length of drains (primary and secondary) as per cadastral maps (early 1900s) was 113.24 km and 226.29 km in Koramangala and Vrishabhavathi valleys respectively, which was reduced to 62.84 km and 111.72 km by 2016/2017. The time series drainage maps showing the changes is indicated in **Chart 3.2**.

Chart 3.2: Time series drainage maps

Koramangala valley

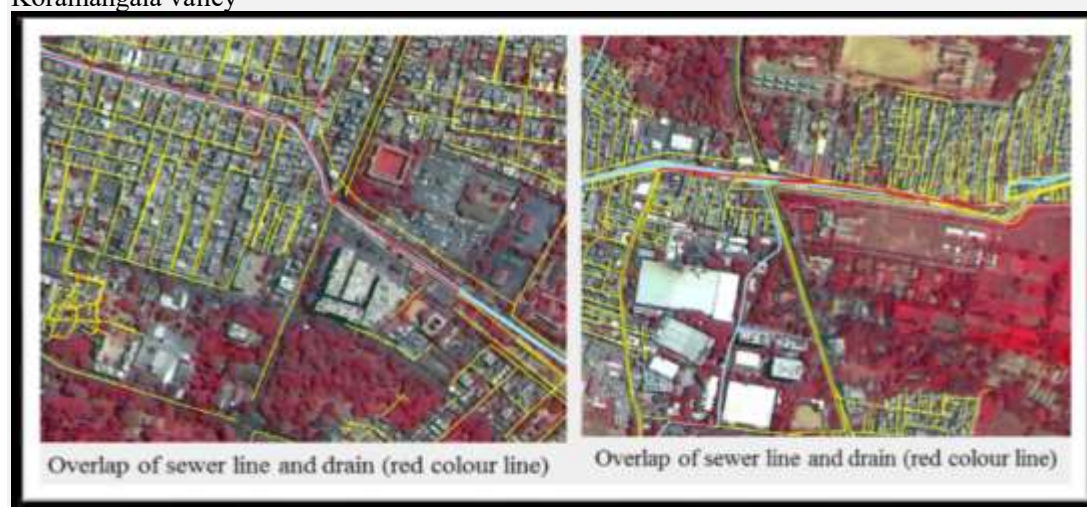


5.1.1 Flow of sewage in storm water drains

The SWDs are meant to carry only the runoff from the rain water, and thus were to be generally dry during off monsoon period. Section 230 of the Karnataka Municipal Corporation Act, 1976 (KMC Act, 1976) and Section 72 of the Bengaluru Water Supply and Sewerage Board Act, 1964 specifically prohibit laying sewerage lines inside SWDs. The IRC guidelines (Paragraph 8.1) prohibit sewerage drains and its content entering the SWDs. The study conducted by Audit with RRSC to analyse the time series land use changes and status of drains (as mentioned in Paragraph 2.4) revealed intersection or overlapping areas between drains and sewer lines that were the likely areas/ zones of possible intermixing of rain water and sewage, a few of which were validated by field visits. The sewer line layer obtained from BWSSB was overlaid on drainage maps and the intersection of the two layers was extracted to identify the overlapping areas. Audit observed that there were continuous stretches where the drains and sewer lines intersect/overlap each other. The total length of such stretches was about 16.0 km with about 728 points of intersection in Koramangala valley and about 28.09 km with about 342 points of intersection in Vrishabhavathi valley. Typical example of overlapping of drains and sewer lines is depicted in **Exhibit 5.1**.

Exhibit 5.1: Intersection of SWD and sewer lines

Koramangala valley



Audit also observed that sewerage lines were laid inside the SWDs and large quantity of sewage was invariably let into SWDs in complete disregard of the codal provisions. Out of 1,440 MLD of sewage generated in BBMP areas, about 780 MLD (54 *per cent*) was discharged into SWDs/water bodies without treatment. BWSSB, which is responsible for sewage disposal within the jurisdiction of BBMP area currently operates 27 Sewage Treatment Plants (STPs) with an operational capacity of 1,073 MLD. The utilisation efficiency was only 60 *per cent* with 644 MLD of total sewage reaching these STPs. Works in respect of 11 STPs with an operational capacity of 520 MLD were in progress as at the end of March 2021. Further, though 110 villages were made part of the BBMP area in 2007, BWSSB was yet to provide sanitation facilities to these villages.

The joint inspection of drains confirmed the existence of sewerage lines within SWDs. Further, audit noticed sewage being discharged into the SWDs directly or through fractured manholes at many places (**Exhibit 5.2**). This can be attributed to the absence of regular inspections by BBMP as indicated in Paragraph 5.2.1. Consequently, none of the SWDs were dry irrespective of the rainfall.

Exhibit 5.2: Photos showing the flow of sewage in SWDs



Direct discharge of sewage into SWD at Pattanagere, RR Nagar Zone



Discharge of sewage into SWD through fractured manhole at Seshadripuram, West Zone



Sewage flowing in SWD at Ejipura, Koramangala Zone



Discharge of sewage into SWD through fractured manhole at Yeshwanthpur, RR Nagar Zone



Discharge of sewage into SWD through BWSSB pipeline at Silk Board Junction, Bommanahalli Zone



Sewage flowing in the SWDs due to chain of fractured manholes in the drain connecting to Herohalli Lake, Dasarahalli Zone

5.1.3 Survey of encroachments and their removal thereon

Section 234 of the Karnataka Municipal Corporation Act, 1976 (KMC Act, 1976) clearly prohibits construction/erection of any building, wall, fence or other structure on SWDs and empowers the Commissioner, BBMP to remove such structures and recover the cost from the offenders. Paragraph 4.23.1 of NDM guidelines clearly states that the drains should be delineated and boundaries fixed. Further, the Karnataka High Court directed (2011) BBMP to conduct a detailed survey of encroachments on SWDs and to clear all encroachments.

BBMP initially provided a list of 1,988 encroachments identified under its jurisdiction but did not furnish the source/period of information. Hence, audit could not ascertain whether any survey was actually conducted to identify the encroachments.

During the joint inspection of about 70 drains²⁵, Audit observed 23 cases of significant encroachments (Exhibit 5.4) as listed in Appendix 5.1. Out of these, 16 cases were not in the list provided by BBMP. Thus, the completeness and reliability of the data on encroachments available with CE, SWD was doubtful.

Exhibit 5.4: Photographs showing the encroachments on SWDs



5.2.2 Absence of regular maintenance of drains

The BBMP did not take up maintenance of drains regularly and continuously leading to blockage and growth of vegetation in the drains, consequently resulting in drains overflowing during rains, particularly where the utility lines were laid. Absence of regular inspection and maintenance also facilitated damage to fencing/ walls of drains allowing inordinate dumping of debris. The lack of maintenance can be linked to the absence of periodical physical inspection of SWDs and documentation of the findings of the inspections by field engineers. This was compounded by the fact that BBMP failed to prepare action plans for regular maintenance activity and also the maintenance manual indicating the roles and responsibilities of the concerned.

Audit observed during joint inspections of drains that SWDs were filled with debris, vegetation and were heavily silted indicating absence of regular maintenance of drains (**Exhibit 5.9**).

Exhibit 5.9: Photos showing the status of drains due to inadequate maintenance



Vegetation inside SWD at Magadi Road Railway Bridge, West Zone



Dumped debris/wastes and garbage inside SWD at Ejipura, Koramangala Zone



Vegetation inside SWD Akshaya Nagar, Bommanahalli Zone



Dumped wastes and garbage inside SWD near Football Stadium, Koramangala Zone



Vegetation inside SWD at Vyalikaval, West Zone



Vegetation inside SWD at Peenya, Rajarajeshwarinagar Zone

Executive Summary

The city of Bengaluru is a victim of a paradoxical situation - urban flooding on one hand and depletion of ground water table levels, on the other. There is an urgent need for urban managers to address this issue from the water security/environment and urban planning perspectives.

Rapid increase in frequency of flooding in the city over the last few years, leading to destruction of roads, traffic congestion lasting several hours and extensive damage to public property and health has highlighted the need for a thorough examination into the design, agility/adaptability of the infrastructure for the management of storm water and other relevant issues; hence this **performance audit (PA)**.

The scope of this PA is not limited to the SWD infrastructure. We have attempted to address the larger conceptual question of redefining storm water as a critical natural resource worthy of conservation. The PA which covers the period 2013-14 to 2017-18 involved test-check of records in the Office of the Commissioner, Bruhat Bengaluru Mahanagara Palike (BBMP) and Chief Engineer, SWD along with joint physical inspections of drains. Audit conducted an independent study of long term changes in land use patterns utilising geospatial inputs with technical support from Regional Remote Sensing Centre-South, Indian Space Research Organization, Bengaluru to obtain details of changes across specific time periods which affect ground water recharge, simultaneous variations in SWD infrastructure in order to identify weaknesses in their management/monitoring.

The PA revealed that Bengaluru witnessed large scale encroachment of lakes/drains and depletion of natural drainage systems. The changes in land use such as decrease in vegetation cover and open spaces and increase in built up area resulted in loss of inter-connectivity between water bodies impacting effective recharge of ground water and increase in runoff of storm water. A study by the Indian Institute of Science states that the city (covering an area of 741 sq km) had 1,452 water bodies with a total storage capacity of 35 TMC during early 1800s. By 2016, the number of water bodies in the same area reduced to 194 with a storage capacity of 5 TMC. The current storage capacity which has further declined due to siltation is merely 1.2 TMC (2016). Out of 210 lakes under the jurisdiction of BBMP as at the end of December 2020, 18 lakes with a total area of 254 Acres and 17 guntas were identified as disused lakes; making them vulnerable to encroachments and future conversions.

A robust policy governing storm water management does not exist. The State Government and BBMP failed to consider urban surface runoff (average annual rainfall being 969 mm during 2013-19) as a water resource despite the growing scarcity of water in the State/city. More than 40 *per cent* of properties under the purview of Bengaluru Water Supply and Sewerage Board (BWSSB) failed to adopt mandatory rain water harvesting structures.

The storm water drains of Bengaluru are documented in two different documents prepared by two different agencies – Revised Master Plan 2015 by Bengaluru Development Authority (BDA) and Master plan of drains by BBMP. There were significant discrepancies between these two plans with regard to identification and classification of drains. This deprived the planners of a single source of truth for planning/development of the city. BBMP did not possess fool-proof data on the total number/length and nature of different types of drains under its jurisdiction. The absence of a comprehensive inventory of drains and their proper classification contributed to lack of clarity on critical issues such as the extent of buffer zone to be

maintained, *etc.* This in turn hampered maintenance of drains as many utility lines like electrical, telephone, optical cable, *etc.*, were laid across the drains in many locations obstructing flow in the drains.

Comprehensive Detailed Project Reports (DPRs) prepared for improvement of SWDs were reportedly not available with BBMP and the only available revised DPR was deficient. Works executed were either incomplete or were abandoned due to non-availability of required land and poor performance of the contractors. This resulted in BBMP not submitting the Utilisation Certificates and consequent loss of financial assistance of `83.59 crore under the Jawaharlal Nehru National Urban Renewal Mission.

Though a large number of works were abandoned and rescinded due to poor performance of contractors, risk and cost clause were not invoked. This led to extension of undue financial benefit of `35.31 crore to the contractors. BBMP also resorted to payment of `94.93 lakh to an agency without the original records and proper reconciliation under questionable circumstances.

BBMP failed to prepare a SWD manual specifying the design, construction and maintenance of the SWD infrastructure of the city. It failed to factor in reasons for high intensity rainfall due to rapid urbanisation and did not adhere to the provisions of Indian Road Congress and the guidelines of National Disaster Management Authority while designing and constructing roads/drains. Ground water recharge structures were not taken up due to flow of sewage in SWDs. Water bodies and drains were not interconnected and linkage between different drains was absent. This affected free flow of storm water leading to frequent flooding in various parts of the city.

BBMP executed construction/remodeling of 332 km and maintenance of 308 km of SWDs (primary and secondary) respectively till 2017-18 from out of its own funds and funds received from Central/State Government. The performance audit showed instances of improper and delayed execution of works and execution of SWD works by multiple authorities within BBMP which was fraught with the risk of duplication of claims. It incurred an expenditure of `8.51 crore on items of works, which were objected to by audit earlier and though the Committee on Local Bodies and Panchayat Raj Institutions had endorsed Audit's observations. Records such as 'completion plans' and 'as built drawings' and works history registers, which are vital for subsequent planning were absent and basic financial records such as grants register, deposit register, register of securities, schedule of work expenditure, register of advance *etc.*, were not maintained. This was indicative of a weak financial control mechanism within BBMP.

Rampant mixing of sewage (780 MLD) with storm water is a serious problem. Sewerage lines were drawn inside the SWDs and large quantity of sewage was illegally let into SWDs. Joint inspection of drains confirmed the existence of sewerage lines within SWDs and sewage being discharged into the SWDs directly or through fractured manholes at many places. Since the water in SWDs is not treated in the same manner as sewage, the untreated sewage is flowing into water bodies and affecting the quality of ground water adversely. This carries substantial risk of spurt in vector/water borne diseases such as dengue, typhoid, cholera, hepatitis, *etc.*, and adverse environmental outcomes including disappearance of biodiversity and aquatic ecosystems. The Chief Health Officer (Public Health), BBMP confirmed an outburst of cholera in the city during March 2020 and stated that seven out of the 25 suspected cases had been confirmed as cholera. He, *inter alia*, attributed sewage flowing in open SWDs to the spreading of the epidemic.

The Revised Master Plan 2015 and NGT directions stipulated a buffer zone on either side of primary, secondary and tertiary drains and the statutory provisions stipulated putting in place boundary marks for such descriptions. None of the test-checked drains, though had such boundary markings. This led to encroachment of drains as well as construction in buffer zone. BBMP was yet to take action on 714 out of the 2,626 identified encroachments. The completeness and reliability of the data on encroachments available with BBMP was low as audit noticed significant instances of encroachments, in addition to those recognised by BBMP. Removal of encroachments was incomplete.

Severe blockages of surface drains/SWDs indicated absence of periodical inspections as well as its regular maintenance of drains. BBMP failed to adopt quality monitoring measures and install Sewage Treatment Plants, despite Court directives leading to continuous contamination of water bodies. BBMP did not take up any information, education and communication activities/awareness camps for educating people regarding importance of SWDs and their proper upkeep and did not enforce penal provisions for violation/dumping of debris in SWDs. Consequently, BBMP failed to protect and maintain the drain infrastructure resulting in continuous misuse of the drains.

Chapter 4

September, 2022 floods

The city faced unprecedented floods on September 4 and 5, 2022. Two days of torrential rain left a large part of the city under water. The citizens were stranded for over 20 to 30 hours without any access to medical help and cut off from the outside world. Boats were employed to evacuate the stranded citizens. The offices were forced to issue work-from-home orders to employees.

The following reports describe the scenario that prevailed at different parts of the city during the flood.

www.reuters.com

India's Bengaluru hit by flooding, traffic snarls after heavy rain

By Nivedita Bhattacharjee and Nandan Mandayam
September 6, 2022



Members of a rescue team row their boat past submerged vehicles following torrential rains in Bengaluru, India, September 5, 2022. REUTERS/Samuel Rajkumar

BENGALURU, Sept 5 (Reuters) - Large parts of India's tech capital Bengaluru were under water on Monday after torrential rains uprooted trees, caused crippling traffic and forced offices to issue work-from-home orders to employees, raising fears of further disruptions through the week.

The Bangalore Urban district on Sunday received 28.1 mm rainfall, 368% more than the average, according to data compiled by the state-run India Meteorological Department (IMD). The city has received 141% more rainfall than average since the start of monsoon season on June 1.

Bengaluru, home to various global companies as well as home-grown startups, has established itself as an important business centre in the country's southern region. "Bengaluru and south interior Karnataka are likely to get heavy rainfall until Friday. From Saturday, rainfall activity will go down," said a senior weather official with IMD based in Pune.



People pull a car through a water-logged road following torrential rains in Bengaluru, India, September 5, 2022. REUTERS/Stringer



People commute in tractors due to water-logged roads following torrential rains in Bengaluru, India September 6, 2022. ANI/REUTERS TV via REUTERS.

On Monday, sources told Reuters that Morgan Stanley (MS.N) and Goldman Sachs Group Inc (GS.N) had asked their employees to work from home. IT major Wipro Ltd (WIPR.NS) and Walmart's Flipkart said they had asked employees to work from home.

"After two years of work from home, companies are coming back and infrastructure has completely collapsed," said Krishna Kumar, General Manager at the Outer Ring Road Companies Association (ORRCA). "Authorities need to focus on scaling up the infrastructure," Kumar said.

Monday's flooding forced citizens to empty basements and parking lots, spend more time on the roads in traffic and face power outages in places. "Right now, we are all living in fear," said Deepa Babu, a psychologist who counsels adolescent kids and lives in Rainbow Drive, an enclave near the city's major tech hubs.

India's Bengaluru Flooded After Days of Torrential Rains

Life for many in the southern Indian city of Bengaluru is disrupted after two days of torrential rains set off long traffic snarls, widespread power cuts and heavy floods that swept into homes and submerged roads.

By Associated Press

Sept. 6, 2022, at 2:24 a.m.



Firefighters rescue an elderly woman from a flooded apartment after heavy rainfall in Bangalore, India, Monday, Sept. 5, 2022. Life for many in the southern Indian city of Bengaluru was disrupted on Tuesday after two days of torrential rains set off long traffic snarls, widespread power cuts and heavy floods that swept into homes and submerged roads. (AP Photo/Kashif Masood)

BENGALURU, India (AP) — Life for many in the southern Indian city of Bengaluru was disrupted on Tuesday after two days of torrential rains set off long traffic snarls, widespread power cuts and heavy floods that swept into homes and submerged roads.

With several parts of the city still heavily waterlogged, videos on social media showed people hopping onto tractors to get to work. Many schools were shut on Monday over the heavy downpours and authorities have warned of interruptions to the water supply. Boats were deployed to rescue people submerged in floodwaters.

The city, dubbed India's tech capital, is home to several technology companies and many of their offices have been lashed by the rains, prompting employees to work from home.

Even though September is usually the wettest month in Bengaluru, this year has seen more rains than normal.

The two zones that make up the city, Bengaluru Urban and Bengaluru Rural, have seen 141% and 114% excess rainfall respectively. On Monday night, 131.6 millimeters (5.2 inches) of rain was recorded, making it the wettest September day in the last eight years.

According to the India Meteorological Department, Karnataka, the southern state that's home to Bengaluru, is among the regions that have received maximum rainfall this year. It has seen 34% more rainfall in the past three months than what it usually receives this time of the year.

While there is no direct connection between the excessive rains in Bengaluru and climate change, there is growing evidence that the monsoons, the most important weather system for the Indian subcontinent, are being altered due to climate change. Scientists say this is making extreme events such as excess rainfall the new normal.

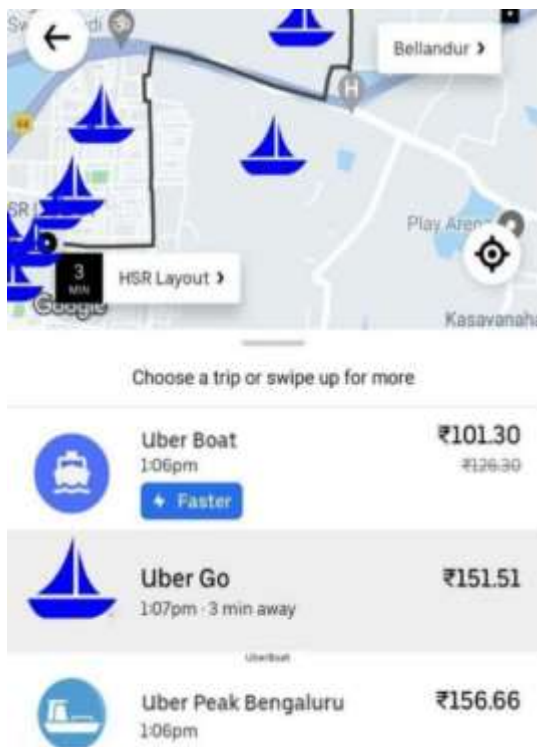
<https://www.bbc.com/news/world-asia-india-62806237>

Bengaluru floods: Boats and tractors replace cars in 'India's Silicon Valley'

By Imran Qureshi

BBC Hindi, Bangalore

6 September



A poster that has gone viral in the past 24 hours is a graphic parody on the state of affairs in a city famously described as India's IT capital - Bangalore (also known as Bengaluru).

A social media user has altered the bookings page of Uber to make it look like it's offering boats instead of cars. The meme has gone viral.

Many parts of the city have been flooded after heavy rainfall since Sunday. Experts say rapid construction, fast erosion of famous lakes and gardens, and poor urban planning are to blame for the crisis. The Times of India website tweeted the photo of a swanky villa that resembled a pool.

Bangalore Times

Pool villa? A villament in Yemalur inundated following heavy rains in #Bengaluru, on Sunday



And one woman shared a video claiming that adventure vacations were now available "at your front door". "I went to the office to pick up something important for a meeting. It was only after I reached that I realised that no-one else was there. So, I decided to go back home and my heart was in my mouth and my legs were shaking [as the streets had flooded by then]," said an IT professional who didn't want to be named.

yesodharan

@hype_atia

Adventure vacations now at your front door, right here in Whitefield



The Indian Meteorological Department (IMD)'s local office said the last time it rained so heavily in the month of September was in 1998. And people who drove through flooded streets described the horrors they felt.

Many IT companies have asked their employees to work from home for the rest of the week and many schools have gone back to online classes. India's army has deployed boats to rescue stranded people.

Why this havoc?

IMD officials told the BBC that unusual weather patterns had caused the heavier-than-usual rainfall in the city and many other parts of the state.

The city's water supply has also been impacted after heavy rains in Mandya district flooded the pumping stations that send water from the Cauvery River to Bangalore.

Tushar Girinath, chief commissioner of the city's main civic organisation, blamed the floods on the unusually heavy rainfall which, he said, could make life difficult anywhere. "For the first time, all the lakes in the city are full. They cannot take any more water. We simply do not have the capacity to take the water out because the storm water drains are already overflowing. We need to see the flooding in this context."

But, Mr. Girinath, said there were signs of relief as water levels had started receding in many areas. "We are also removing encroachments over the storm water drains. This work has already begun and we are going full steam ahead," he added.

Horror on repeat

Officials are facing heavy criticism for the recurring problem of floods in the city.

"The flooding this time is really bad, but it has also happened in an area where many people have access to social media, so it has been escalated a lot more this time than in the past. Before it settles down, there is bound to be one more heavy rainfall that can compound the situation," said Srinivas Alavalli, head of civic participation in Janagraha charity.

Experts say frequent flooding in many Indian cities in recent years is because many water bodies have been built over.

"The development model has not respected nature's contours. If you think that you can block how nature operates [water finding its own levels], you are deluding yourself. We chose to convert water sheds into IT parks. Nature is not going to respect that," urban planner V Ravichander said. And the problem, he added, would only get worse in the coming years.

"It is not going to get any better. The city has overgrown. The storm water drains are shrinking because we have not desilted them. This is where the problem of incompetence in governance comes in." The way forward, he said, was for communities to come together in the neighbourhood to find solutions. "People have no option but to co-operate and find technical solutions and take it to the government."

<https://www.business-standard.com/>

Is brand Bangalore being washed away by flood?

Bangalore's authorities had no plan B ready when the city was flooded recently. Visuals emerging from India's Silicon Valley were shocking! We ask if brand Bangalore is taking a hit

Bhaswar Kumar Sourabh Lele & Viveat Susan Pinto | New Delhi

Last Updated at September 12, 2022 07:00 IST

Just a few days of rain changed everything, exposing the crumbling infrastructure of crammed Bangalore and washing away the tall claims of authorities.

Roads disappeared under muddy flood water and motorboats came roaring rescuing people from their own homes. Fish were spotted on highways and people were seen swimming in their living rooms in posh colonies of Bangalore -- the capital city of Karnataka and IT hub of India. Their luxury cars, from BMWs and Range Rovers to Bentleys, were left behind, half-submerged. The stark visuals of flooding inundated social media too.

Life came to a standstill in Bangalore's key Outer Ring Road area, which houses more than 700 companies, including offices of big ones like Microsoft, Intel, Goldman Sachs and Morgan Stanley. Many of these companies have had to allow work from home or pass on important work to locations outside of the city. While companies usually have business continuity plans involving such steps, they have still paid a heavy price.

The Outer Ring Road Companies Associations has reportedly claimed that the flood caused a loss of Rs 225 crore to the IT and banking companies it represents on the 30th of August alone. In a September 1st letter to Chief Minister Basavaraj Bommai, the industry body asked the government to come up with short-term and medium-term plans to address the issue. Stating that concerns about Bengaluru's future growth were being raised across the world, the letter warned that companies may leave if the infrastructure issues were not addressed.

IT and IT-enabled services remain one of the top services exports from India. Close to 50 per cent of the total services exports comprise IT exports. India's services exports set a new record of \$254.4 billion in FY22, beating the previous high of \$213.2 billion in FY20.

According to the Karnataka economic survey, the state contributed an estimated 38 per cent to India's FY22 software exports, which were estimated to be worth \$170 billion.

According to a recent report by Inc42, Bangalore has clinched the title of India's 'top start-up hub'. It added that within India, Karnataka topped the list of states in terms of start-up ecosystem development with \$62 billion in funding. Meanwhile, Delhi-NCR attracted \$39 billion and Maharashtra 20.

The stakes are high when it comes to brand Bengaluru's future. India's start-up success story has been touted across international forums by the government, which is also counting on the ecosystem to generate much-needed high-quality jobs. India also has an ambitious services export target of \$1 trillion by 2030. The continued growth of the IT

industry, and by extension that of Bengaluru, will be a necessary condition for achieving this goal.

But now, there are fears that the city's technology capital image could be in deep water. And, the brand image has indeed been dented.

Mohandas Pai, Chairman of Manipal Global Education Services and Advisor at Manipal Education & Medical Group, says brand Bengaluru is hurt by flooding and bad governance. Authorities taking the city's IT ecosystem for granted. Illegal constructions blocking waterways needed to stop flooding. CM should set up a high-level panel to fix issues.

But, Bangalore can still salvage the situation. A senior IT services veteran based in Bengaluru told *Business Standard* that when he spoke to five large Fortune 500 firms, none said that the recent flood would make them move out of the state or the city. The National Association of Software and Service Companies also appeared to be committed to staying put.

Last week, a Nasscom-led consortium of software companies met Karnataka IT and BT Minister C N Aswathanarayana to discuss the way forward. State officials have agreed to have IT sector participation in governance mechanisms planned for Bengaluru's tech corridors. The officials will also hold a meeting with the industry to discuss plans once every fortnight.

Advertising guru Prahlad Kakkar said that Bengaluru's image as a cosmopolitan city had taken a bit of a beating. But, he added that this was a problem that also plagued other cities in India, including Mumbai, where flooding is routine. Kakkar concluded that the underlying causes would have to be addressed seriously if Brand Bengaluru was to emerge again as a vibrant city. What is clear is that if the authorities have the will, the city can definitely brave the deluge and emerge stronger than before.

Chapter 5

Reasons for the floods - Experts' views

The following articles present Experts' views on the reasons for frequent and severe floods at the city.

The New Indian Express

Scientist says sudden, heavy rain in Bengaluru due to global warming

By Bala Chauhan

Published: 09th September 2022 08:07 AM

BENGALURU: Large swathes of land, mostly in the IT hub in Bengaluru, have been inundated following unprecedented torrential rains since September 4.

Unlike in the past, it is being observed that the pattern of rainfall has changed — from an occasional drizzle and shower to a sudden and heavy downpour now. There are increasing instances of heavy and incessant rainfall leading to overflowing of lakes, flooding and water logging, which have stumped weathermen and town planners.

While there is a lack of detailed research corroborating link between climate change and monsoon, meteorologist and Professor at the Centre for Atmospheric and Oceanic Sciences and the Chair of Divecha Centre for Climate Change, IISc, S K Satheesh said that extreme weather events like heavy rainfall over a short period could be because of a “gradual increase in the number of cloud condensation nuclei (aerosols) in the atmosphere and global warming.”

Atmospheric aerosols are known to play a crucial role in cloud formation processes by acting as cloud condensation nuclei (CCN). They define the droplet size distribution forming rain pattern.

For a given amount of water vapour content in the atmosphere, a lesser number of large cloud droplets will form if the number of aerosols is less. For the same amount of water vapour content, a large number of smaller cloud droplets will form when the number of aerosols is high, which in polluted conditions can lead to heavy rainfall within a short period of time.

Speaking to TNIE, the scientist said that the “aerosol loading in India is increasing in the last 10 years because of anthropogenic activities. There is a two per cent per annum increase in the loading of the aerosols in the atmosphere in the Indian subcontinent. This could be because of anthropocentric activities coupled with unsustainable landscape planning and urbanisation etc,” said the meteorologist.

Satheesh cautioned, “Increase in the number of aerosols in the atmosphere will not only alter the rainfall pattern, it will also adversely impact public health, leading to an increased number of people falling sick with respiratory disorders, block solar radiation hitting the earth thereby impacting climate and agriculture, by default,” added Satheesh.

“Presence of a large number of aerosols in the atmosphere will also impact satellite remote sensing of the earth’s surface,” he added.

Satheesh said that though there are enough policies for pollution control and regulation of construction activities, enforcement is a problem. “Research should provide a scientific basis for governments at all levels to develop climate-related policies. Linking disaster management with artificial intelligence (AI) is another aspect, which needs attention with a changing climate,” said the noted scientist.

The Hindu

Bengaluru’s Rain Nightmare: ‘Zenrainman’ Shares 6 Ways to Stop Floods from Repeating

By Rinchen Norbu Wangchuk
September 6, 2022

S Vishwanath, a civil engineer and urban-regional planner with over three decades of experience in the water and sanitation sector explains what Bengaluru must do to prevent flooding and water-logging.

Overnight rains in Bengaluru have left “one person dead and 75 localities inundated, displacing residents of nearly 2,000 flooded houses, marooning 10,000 other residences and damaging 20,000 vehicles”, a Times of India report notes. These are astonishing figures for a city like Bengaluru following heavy rainfall.

The worst affected parts were the eastern, southeastern and northeastern parts of the city. According to the Indian Meteorological Department (IMD), the city recorded 13.16 cm (131.6 mm) of rainfall, most of which fell in less than 12 hours on Sunday night. This was the third heaviest rainfall the city has witnessed in September in the past 75 years.

What we saw on Sunday night is a continuation of heavy rainfall from last month. In August, the city recorded 370 mm, according to IMD. The figures for August 2022 fell a little short of the city’s record of 387.1 of rainfall in August 1998.

Despite unprecedented rains, the reason why parts of the city are horribly inundated, flooded and water-logged is primarily because of man-made factors. Speaking to **The Better India**, S Vishwanath, a civil engineer and urban-regional planner with over three decades of experience in the water and sanitation sector, discusses why Bengaluru has witnessed such unprecedented floods and the potential solutions to address them.

‘Tyranny of small decisions’

In 1966, the American economist Alfred E Kahn wrote an essay explaining a phenomenon he called ‘the tyranny of small decisions’. The essay presents a scenario in which several decisions, individually small and insignificant in

size and time perspective, cumulatively result in significant outcomes that are both sub-optimal and undesirable.

Bengaluru, according to S Vishwanath, has reached this juncture of excessive flooding and water-logging because of what he calls 'the tyranny of small decisions'.

Having worked on policy teams for writing the State (Karnataka) Water Policy, rainwater harvesting policy and bye-laws for Bengaluru and wastewater reuse policy for Karnataka, he says, "Every individual act by a landowner to level a site, encroach on a stormwater drain or fill up debris into a lake doesn't have an immediate impact. An individual doesn't think that it's a problem if they encroach on a drain. But like that one person, 50 others are doing the same thing because of individual decisions taken by the authorities. When you're suddenly confronted with a 120mm a day rainfall event, you have a massive problem."

More specifically, however, Vishwanath highlights three key reasons behind recent events:

- 1) **Climate crisis is clearly upon us:** We have a distinct increase in rainfall compounded by the urban heat island effect – caused by exponential growth and concretisation – which creates excess heat and thus causes more intense rainfall across shorter durations. Typically, the city would endure a rainfall intensity of 60 mm per hour, but now we are talking about 180-200 mm per hour for short durations. Bengaluru is just not prepared for this sort of rainfall intensity.
- 2) **Parts of the city worst-affected have relatively flat terrain:** These areas used to be old rice paddy fields. When we overlaid infrastructure here, we built roads without any cross-drainage works adequate for this kind of rainfall. These roads have essentially become dams.
- 3) **Underinvestment in critical infrastructure:** There are feeder channels that take the water from the surface to a nearby lake or a channel. During agricultural times, these feeder channels used to move about depending on which land was cultivated, which lake was being used and so on.

They have not been identified for modern requirements of runoff. Therefore, governments have underinvested in stormwater drainage infrastructure. To compound that problem, we have inadequate sewage infrastructure, leaving sewage to float about in these drainage channels and lakes.

Sustainable solutions

However, Vishwanath does believe that there are both short and long-term solutions available.

- 1) **Road audit:** One of the things we need to do immediately is a road audit for drainage. The authorities need to ask themselves whether all roads have stormwater drains on adjacent sides. Are these drains networked? Most drains today simply stop at a dead end. These drains should go in a hierarchy and empty into a lake or a river. Are the cross drainage works adequate like the box culverts and Hume pipes, and can the water move from one side of the road to the other? This kind of audit has to be done for all roads in the city.
- 2) **Protect stormwater channels:** Stormwater channels should be lined with concrete so that they are demarcated and authorities must ensure they are not allowed to be encroached upon. That network should make sure that it disposes of the water finally into a river like Vrishabhavathi.

If a government wants to build a road or metro station, they acquire the land, pay compensation to the previous owners and clear up the space for those purposes.

Similarly, the authorities need to now acquire land for stormwater drains. The city will have to fairly compensate those who lose their land, but then they will have to design the stormwater drains adequately for these climate change rainfall events. This process has to be systematic as a hydrological design which is watershed-based. We've not been engaging in the process systematically and instead have been increasing the number of drains on an ad-hoc basis.

- 3) **Invest rapidly in sewage networks:** We need to saturate the city with sewage networks so that all sewage is picked up and sent to sewage treatment plants (STPs).
- 4) **Improve our prediction skills:** The Karnataka State Natural Disaster Monitoring Centre has 99 automatic weather stations in Bengaluru, which is the largest density of weather stations for any city in India. Now, they are very good at identifying heavy rainfall incidents and sending a signal out to a given area in the city that it can expect floods.

We need to improve their predictive powers to a minimum of three days in advance and make sure that this information is acted upon as soon as possible. This will require more investment in the modelling process and improving the automatic weather stations.

- 5) **What can citizens, communities and apartment complexes do?** People must invest heavily in rainwater harvesting. We need to make sure that we collect as much of the rainwater as possible, use that to recharge our aquifers and try to design our plots and apartments and gated communities as zero rainfall discharge areas. The idea is to hold onto the rain and ensure its positive use rather than letting it run about and flood the place.

The rainwater harvesting by-law in Bengaluru states that for every square metre of roof area, you have to create a storage or recharge of 60 litres of water. For every square metre of the paved area around the building, you must create a storage or recharge of 30 litres of water.

Broadly, a one-day rainfall event of 60 mm can be completely managed by even a small house let alone an apartment or large gated communities.

If the law is followed, we'll be reducing the impact of flood by a factor of three. About 65% of the land use of any particular area is housing or residential. If these residential areas manage rainwater appropriately through rainwater harvesting, there will be no external water to create floods.

- 6) **Design on a watershed basis:** Every lake is a micro or mini watershed. Authorities must understand that a watershed is a hydrological unit, understand how water flows and ensure that it's drained into channels and flows into the lake. When the lake overflows, it connects to the next downstream lake, for which they will need sluice gates that they can operate.

If you anticipate heavy rainfall, you open up sluice gates, allow the water to flow out, and the lake is ready to receive the flood waters and retain it without a problem. Essentially, you prevent flooding by taking precautionary action. If there is a rainfall event which exceeds 60mm, like the ongoing one, that excess water should be drained to reach lakes which are connected to different parts of the city.

Why is fixing Bengaluru's flood problem a national issue?

For a city like Bengaluru, it takes a special kind of skill to create floods.

"The city has an average elevation of 920 metres and cascades down into valleys and rivers on all sides. Unlike Chennai, Mumbai or Kolkata, this city is on a hill or a ridge line. It calls for great incompetence if the city can create floods," argues Vishwanath.

"However, the western or southern parts of the city aren't flooded except for small pockets where stormwater drains or lakes have been encroached on. The flood has particularly affected the eastern part (its southeast side) of the city. And that is a peculiar hydrological construct. It's a flat terrain with very limited percolation because the clay layer at the bottom doesn't allow the water to percolate. We must learn that the city isn't one unit. It comprises many units of rainfall and hydrology and we should design adequate infrastructure for context," he adds.

One reason why the recent floods in Bengaluru have garnered such media attention is that it struck the Outer Ring Road, which connects the city to all the major software tech and e-commerce offices. "That's why there is such a

massive public hue and cry. Since the floods have directly hit the city's economic engine, there is all this hullabaloo," he notes.

Besides all the damage, the ongoing floods have sparked an unusual public discourse online where 'locals' have started blaming the influx of migrants for what's happening in the city.

Vishwanath says, "But this city can provide good employment opportunities to many Indians. This is something everyone from Bengaluru should be proud of. Commensurate with the value the city generates in terms of GDP, both the Centre and State must invest special funds to build up the requisite infrastructure and prevent these floods. It's not a local-city problem, but a national one and the authorities need to invest a lot more than they're doing right now."

Bengaluru floods because it is 78% concrete: IISc study

Unchecked concretisation and acute loss of wetland and vegetation over the years are the two key reasons causing frequent flooding during a spell of heavy rain or monsoon in Bengaluru. High-density urban development in catchment areas leads to an increase in impervious areas in the city.

The IISc study -Frequent Floods in Bangalore: Causes and Remedial Measures -says paved surfaces in the city have increased up to 78% due to years of unplanned urbanisation. Researchers have established the extent of concretisation by comparing the spatial maps of the city over the years and contrasting them with the ground situation.

The study, conducted with assistance by researchers Vinay S and Bharath H Aithal of IISc, says there was no need to concretize the channel of storm water drains as vegetation in these drains bears the load in peak monsoon. "Drains with vegetation and without any bottlenecks would be the best option to mitigate floods. Narrowing channels and concretizing will increase the quantum of water and velocity,

T V Ramachandra, Vinay S., Bharath H. Aithal

Frequent Floods in Bangalore: Causes and Remedial Measures

Environmental Information System, CES, Indian Institute of Science, Bangalore
August 2017

Floods in an urbanised landscape refer to the partial or complete inundation from the rapid accumulation or run-off resulting in the damage to property and loss of biotic elements (including humans). Urban flooding is a consequence of increased impermeable catchments resulting in higher catchment yield in a shorter duration and flood peaks sometimes reach up to three times. Thus, flooding occurs quickly due to faster flow times (in a matter of minutes). Causal factors include combinations of loss of pervious area in urbanising landscapes, inadequate drainage systems, blockade due to indiscriminate disposal of solid waste and building debris, encroachment of storm water drains, loss of inter connectivity among lakes, housing in floodplains and natural drainage and loss of natural flood-

storages sites. Flood mitigation in urban landscape entails integrated ecological approaches combining the watershed land-use planning with the regional development planning. This includes engineering measures and flood preparedness with the understanding of ecological and hydrological functions of the landscape.

Bangalore is experiencing unprecedented urbanisation and sprawl in recent times due to concentrated developmental activities with impetus on industrialisation for the economic development of the region. This concentrated growth has resulted in the increase in population and consequent pressure on infrastructure, natural resources and ultimately giving rise to a plethora of serious challenges such as climate change, enhanced green-house gases emissions, lack of appropriate infrastructure, traffic congestion, and lack of basic amenities (electricity, water, and sanitation) in many localities, etc. This study shows that there has been a growth of 1028% in urban areas of Greater Bangalore across 45 years (1973 to 2017). Urban heat island phenomenon is evident from large number of localities with higher local temperatures. The study unravels the pattern of growth in Greater Bangalore and its implication on local climate (an increase of ~2 to 2.5 °C during the last decade) and also on the natural resources (88% decline in vegetation cover and 79% decline in water bodies), necessitating appropriate strategies for the sustainable management.

Frequent flooding (since 2000, even during normal rainfall) in Bangalore is a **consequence of the increase in impervious area with the high-density urban development in the catchment and loss of wetlands and vegetation**. This is coupled with narrowing and concretising storm water drains, lack of appropriate drainage maintenance works with the changes in enhanced run-offs, the encroachment and filling in the floodplain on the waterways, obstruction by the sewer pipes and manholes and relevant structures, deposits of building materials and solid wastes with subsequent blockage of the system and also flow restrictions from under capacity road crossings (bridge and culverts). The lack of planning and enforcement has resulted in significant narrowing of the waterways and filling in of the floodplain by illegal developments. Causal factors and remedial measures to mitigate impacts of flooding are:

Reasons

1. Loss of interconnectivity among lakes due to encroachment of drains or dumping of solid wastes, Construction and Demolition (C & D) wastes
2. Encroachment of flood plains and wetlands (construction in valley zones, flood plains and lake bed) and de-notifying lakes (under the guise of 'dead lakes' – no lake can be dead as it does the job of ground water recharge)
3. Narrowing and concretising storm water drains impairing hydrological functions of the natural drains
4. Loss of pervious areas - reduction of open spaces, wetlands and vegetation cover
5. Increased paved surfaces in the city (78% paved surface and likely to be 94% by 2020) due to unplanned irresponsible urbanisation by senseless decision makers.

Solutions: Ecological Management of Storm Water Drains and Wetlands to Mitigate Frequent Flooding in Bangalore

1. Reestablish interconnectivity among lakes by removing all blockades (encroachments, solid waste dumping)
2. Protect Valley zones and Buffer regions of wetlands: protect valley zones considering ecological function and these regions are 'NO DEVELOPMENT ZONES' as per CDP 2005, 2015
3. Stop narrowing and concretising natural drains
 - Vegetation in the drain takes the load during peak monsoon, there is no need to concretise the channel.
 - Vegetation allows groundwater recharge while treating the water (bioremediation);
 - Drains with vegetation without any bottlenecks (hindrances) would be the best option to mitigate floods.
 - Narrowing channel and concretizing would only increase the quantum of water and velocity, which would be disastrous.
 - Objective should be towards mitigation of floods and not to generate high overland flows (with increased quantum and flow velocity)
 - Experts should think sensibly with holistic knowledge (considering all subject knowledge) than fragmented narrow sectorial knowledge. Advice by pseudo experts would be detrimental as the society would be deprived of ground water, frequent floods and unnecessary livelihood threats.
4. Decongest Bangalore
 - Shift major installations to other cities in Karnataka,
 - Stop further industrialisation and commercial establishments in Bangalore.
 - Protect open spaces – lakes, parks, etc.
 - Stop further growth of dying city – with water and oxygen scarcity
 - BWSSB should stop issuing senselessly NOC (no objection certificate) to major building projects as there is not sufficient water in the city.
 - Environment clearance as per the norms of Environment Protection Act (2016), Wetlands (Conservation and Management) Rules, 2016, SWM 2016, C & D Wastes, 2016, Air act 1981, Water (prevention of Pollution) Act, 1974.

Chapter 6

Decoding of waterlogging at RBD Layout

6.1 Waterlogging at RBD Layout

The Rainbow Drive (RBD) layout is one of the popular residences in the Sarjapur road neighbourhood of the city. When Wipro Limited, the tech giant shifted its headquarters to the outskirts of Bengaluru in the 1990s, the RBD layout that was just developed became the choice of residence for many techies. It has been in the news again in the recent years, but for different reasons. It is inundated every year during rainy season. It was inundated completely on May 5, 2022 too, earlier this year just by a stray summer shower and reportedly left its residents stranded for more than 20 hours.

The following Newspaper report gives an account of its vulnerability for the flood.

Indian Express

Bengaluru: Rainbow Drive Layout residents approach MLA, seeks solution to flooding during rains

Bengaluru | May 11, 2022

Residents of Rainbow Drive Layout at Sarjapur in Bengaluru, which has more than 35 houses, have written to Mahadevapura MLA Arvind Limbavali apprising him of the constant flooding in their area during rains. In an open letter, the MLA was alerted of the flooding on May 5 which reportedly left the residents stranded for more than 20 hours.

“This is to bring to your urgent notice about the plight of 1000+ people living in fear of rain, flooding, power outages and cut-off from emergency help and at Rainbow Drive Layout, Sarjapur Main Road. We had major flooding during pre-monsoon rain on 5th May 2022 and citizens were stranded for over 20 hours without any access to medical help or cut off from the outside world, with elderly people suffering the most,” the letter dated May 10 read.

Listing out the issues, the residents wrote that the fire brigade had to pump out water for more than 12 hours after the rain stopped on May 5.

“This gives an indication that water is coming into our residences from outside. Rainbow Drive entrance is at a low-lying area and hence the Sarjapur main road stormwater drainage must be lowered and/or an additional outlet should be provided on an urgent basis. Urgent pathway construction (is) needed for the entry and exit for residents in order to access medical help.

There are many straightforward alternative outlets which can be provided along the way to nearby stormwater drainage along roads,” the letter said and added that the mixing of rainwater with sewage and untreated water can lead to diseases like dengue and malaria.

The residents told the MLA that they had already written to Bruhat Bengaluru Mahanagara Palike (BBMP), Bangalore Water Supply and Sewerage Board (BWSSB), panchayat and Karnataka State Pollution Control Board but to no avail.

A resident Darshana Minnoor said: “The rainwater from surrounding villages of Junnasandra and Halanayakanahalli are being diverted into the RBD layout from which it exits to the stormwater drain (SWD) on the Sarjapur Road. Due to level differences between our drain and the SWD on Sarjapur Road, the egress is extremely slow and water stagnates for hours and most of the time overnight, causing flooding and water entering into houses at the entrance. Ideally, the drains along Junnasandra Road should be carrying the rainwater directly to Sarjapur Road SWD.”

6.2 The BBMP view

The residents’ argument is that the rainwater from the two surrounding villages is diverted into the layout. But, the corporation has a different view. The layout is built right on the Rajakaluve; about 40 houses are built on it. The following report in Times of India gives its details.

Times of India

Karnataka: Revenue department papers show 40 houses standing on drains in flooded layout

Shreyas HS, Aug 6, 2022

Residents of the 25-year-old layout have been contending there are no SWDs in their enclave. They’ve been blaming neighbouring villages for having breached a wall, resulting in diversion of rainwater into their layout. Flooding of the layout — the fifth in years — prompted residents to get a tractor to ferry kids to school.

Encroachments of stormwater drains (SWDs) in Rainbow Drive Layout, off Sarjapur Road, are leading to flooding, Bruhat Bengaluru Mahanagara Palike (BBMP) officials said. According to documents from the revenue department, about 40 houses have come up on the drains in the 35-acre layout.

Residents of the 25-year-old layout, several parts of which were under water for the third consecutive day on Friday, have been contending that there are no SWDs in their enclave. They have been blaming neighbouring villages for having breached a wall, resulting in diversion of swirling rainwater into their layout. Flooding of the layout - the fifth in the past six years - prompted residents to get a tractor to ferry their children to school.

However, the revenue map shows that two SWDs scythe through the layout, one leading to Halanayakanahalli Lake and another to Junnasandra Lake. In 2014, the KR Puram tahsildar had ordered clearing of properties erected on drains in the layout.

Following a complaint from a farmer, Ambareesh Reddy, about the encroachments in the layout, the revenue department carried out a survey. It found the layout was developed by encroaching on Kaluve Kharab (kharab land with stormwater drain).

Reddy told TOI that he filed a complaint as his farmland was regularly getting flooded because of this encroachment by the layout developers. After the order was issued, residents challenged this in court, saying that the government did not serve any notice on them. The high court disposed of the case in 2014, directing the government to issue notice to the owners and gave the residents a week to submit objections.

The court also stated that the government can only take action after considering the objections. However, no action has been taken by the authorities since then for reasons unknown. Six years later, the revenue department yet again marked the properties for demolition, said N Rajendra, one of the first persons who bought plots in the layout. Besides admitting that builders have encroached on stormwater drains and residents have built houses, he said, some stretches of the drain fall under BBMP.

The properties marked for encroachment fall under kharif Halanayakanahalli gram panchayat. A part of the layout comes under the gram panchayat and the rest under BBMP.

'No rajakaluves here'

Meanwhile, Anand Natarajan, president, Rainbow Drive Plot Owners Association, argued that there are no rajakaluves in the layout and the allegations are untrue. Even BBMP officials have confirmed the same. The water is coming from neighbouring villages, he claimed.

Mahadevapura special commissioner Trilok Chandra said properties have come up on drains and they will soon carry out a survey to mark encroachments in their limits. Chandra confirmed that encroachment of SWDs was responsible for the flooding in Rainbow Drive Layout.

6.3 The actual reasons for the waterlogging at the layout

The residents and BBMP have their own arguments on the reasons for the waterlogging at the layout. The residents argue that the rainwater from the two surrounding villages is diverted into their layout. According to BBMP, the layout is built right on the Rajakaluve. They could be the reasons for a certain extent. But they are not the main reasons, as discussed below.

- i. Figure 6.1 shows the location map of the RBD layout. It is located on Sarjapur road between Halanayakanahalli and Junnasandra lakes and Saul Kere. The total catchment area of Sarjapur road at the reach is 10.1 sqkm. There are four lakes, viz., Hado Siddapura, Choodasandra, Halanayakanahalli and Junnasandra in the above catchment.
- ii. The layout was developed **during 1980s**, the first to be developed in the area. Prior to its development, it was an agriculture land in the two village limits, viz., Halanayakanahalli and Junnasandra. A large part of the rainfall was percolated into the ground. The lakes too had significant storage capacities. The discharge from them was not significant.

- iii. Figure 6.2 shows the Google image at the reach in 1982. There were no defined stream paths at the reach. The spillover from the lakes was thinly discharged over the agriculture land and Sarjapur road.
- iv. The villages were administered by the local bodies. Neither panchayats nor BDA then seriously thought of preparing the master plan for the stormwater drainage at the reach.
- v. The layout was developed right **on the stream path** without any idea of its existence. The authorities too did not notice it seriously.
- vi. Over the years, the entire catchment was developed into a residential area as shown in Figure 6.3. Near entire rainfall on it turned into run-off and was directed towards the layout.
- vii. Initially, there was a small road at the reach. It was widened and reconstructed. Its level was increased. The adjacent lands were filled up. As a result, the RBD layout turned out to be a low-land locked between Sarjapur road and its adjacent layouts and apartments, as shown in Figure 6.5.
- viii. The Bhoomi website of GoK presents a KMZ file as shown in Figure 6.4, which attempts to trace the nalla courses/stream paths at the reach on Google earth, that existed 40 years ago. Two streams take off at a little distance from Halanayakanahalli lake as traced in the above file. Both are broken at lengths in between and do not connect to Saul Kere. The left stream abruptly ends at the layout. It then continues from Sarjapur road opposite to an open land between Saket Calipolis and Keerthi Regalia Apartments as shown in the figure.
- ix. There was a small culvert on the road opposite to the open space between Saket Calipolis and Keerthi Regalia Apartments as shown in Figure 6.7 which was planned earlier to drain off the discharge from the lakes. But, the land on its upstream is completely filled up now and apartments are built up.
- x. In the absence of a planned drainage system at the location, two attempts are made to drain off the discharge from the lakes at the reach, as below.

Attempt-1: The sewage drain of the layout is connected to the drain on Sarjapur road and then to the culvert opposite to the open space between Saket Calipolis and Keerthi Regalia. But, there is no drain continued from the culvert at its other end. Therefore, it is completely filled up with soil and buried under the road as shown in Figure 6.6.

Attempt-2: It is then attempted to take the discharge along the road and connect it to another culvert opposite to Salarpuria SANCITY and then to a sewage drain near Bank of Baroda as shown in Figure 6.8. Figure 6.10 shows the sewage drain near Bank of Baroda. It runs in a narrow space between the buildings. It is completely filled with jungle.

- xi. In addition, the drain on Sarjapur road is built at a significantly higher level compared to sewage drain at the RBD layout as shown in Figure 6.11. As a result, the sewage drain can not push its discharge into the drain on the road. The spill-over from the lakes is struck at the layout itself as shown in Figure 6.13.
- xii. Since both attempts to drain off the discharge at the reach fail, it is held back at the layout. What we see at the layout during the rain is not just the stormwater from its two neighbouring villages, viz., Halanayakanahalli and Junnasandra; it is actually obstructed run-off from a **catchment area of 10.1 sqkm** at Sarjapur road as shown in Figure 6.1, finding no escape route.

A tour through Figures 6.1 to 6.15 will give a clear picture of the phenomenon of the waterlogging at the layout.

Thus, the RBD layout is inundated mainly due to,

- i. Land development at the then outskirts of the city **without stormwater masterplan**.
- ii. Development of the layout without an idea of the stream path at the location, and
- iii. Later attempts to discharge it through sewage and road drains.

The layout was developed in the village limits, which were later absorbed in BBMP. Therefore, the BBMP can not be blamed. Similarly, it was developed with due approvals from concerned authorities. The developers too can not be blamed. But, an attempt now to trace the stream path from revenue records and proposing to dismantle certain structures at the layout may serve a little purpose. It may not facilitate free discharge of the run-off at the reach. The road level is raised. The adjacent lands are filled up. Huge structures are built up on them. We cannot propose to dismantle them. We cannot recreate the topography that existed 40 years ago. There would be absolutely no change in the waterlogging at the layout and its surrounding locality, unless a new drain of adequate capacity is proposed between Halanayakanahalli lake and Saul Kere considering its existing topography.



Figure 6.1: The location map of the Rainbow Drive layout. It is located on Sarjapur road between Halanayakanahalli and Junnasandra lakes and Saul Kere. The stormwater from a catchment area of 10.1 sqkm drains at Sarjapur road at the reach. There are four lakes viz., Hado Siddapura, Choodasandra, Halanayakanahalli and Junnasandra in the above catchment.



Figure 6.2: The Google image of the location in 1982. Prior to the development of the layout, it was an agriculture land in the two village limits, viz., Halanayakanahalli and Junnasandra. There were no defined water courses at the reach. The spill-over from the lakes was thinly discharged over the agriculture land and Sarjapur road.

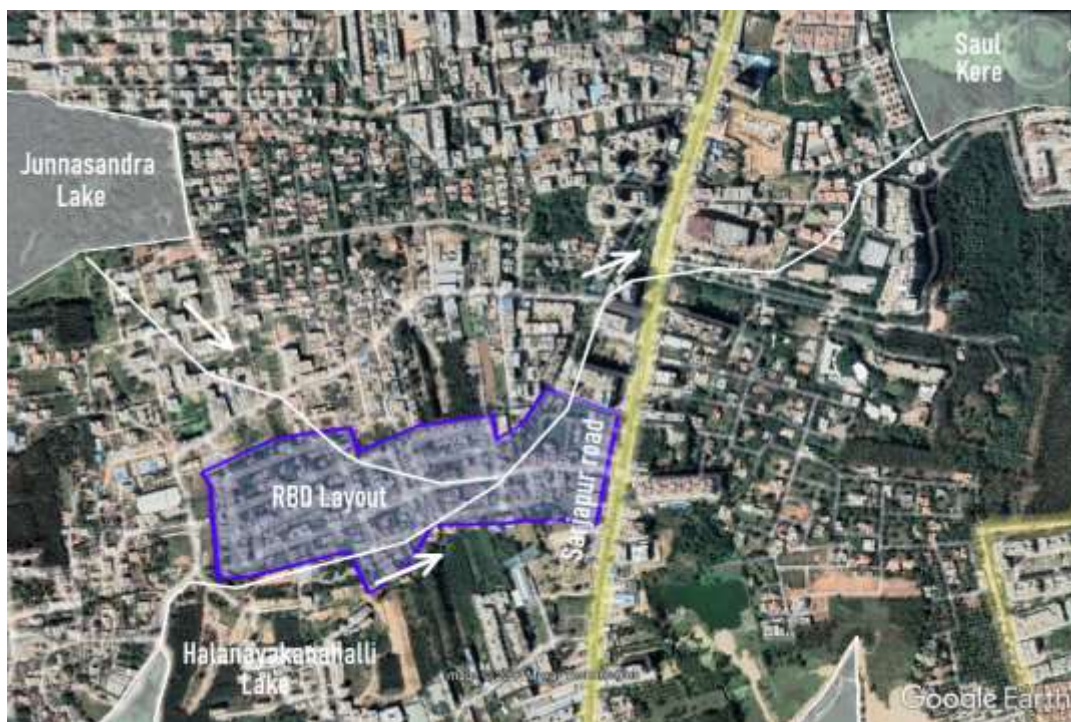


Figure 6.3: The Google image of the reach in 2022. The entire land was developed. The road was widened and its level was significantly raised over the years.

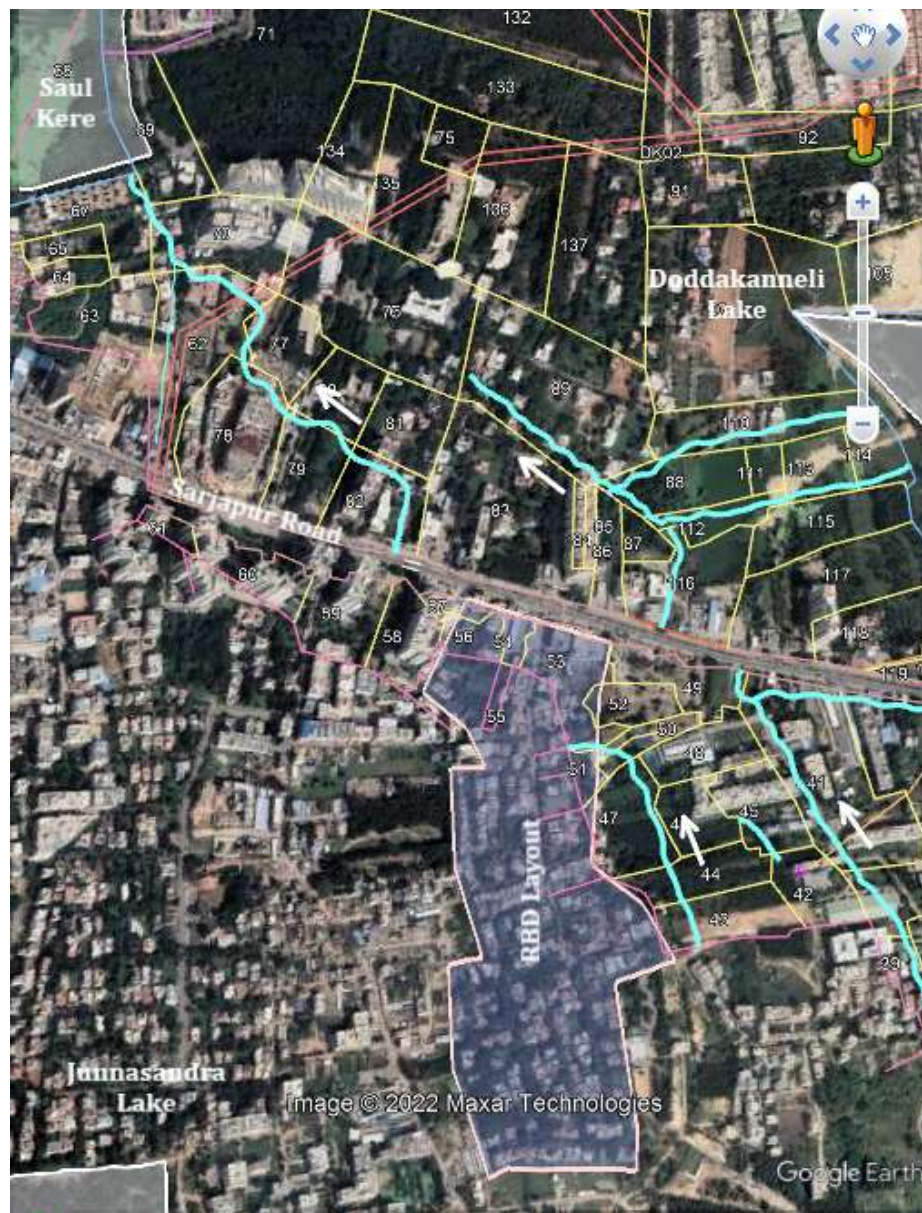


Figure 6.4: A close view of the KMZ file of the revenue department. It attempts to trace the stream paths on Google earth that existed before the land was developed in the vicinity. Two streams start at a certain distance from Halanayakanahalli lake as traced in the file, but abruptly stop at RBD layout and Sarjapur road. The right stream runs for a certain distance after Sarjapur road. The left stream continues from Sarjapur road opposite to Keerthi Regalia Apartment and joins Saul Kere.



Figure 6.5: A view of Sarjapur road from the entrance gate of RBD layout. Its level is significantly raised.



Figure 6.6: The attempted stormwater drainage-1: In the absence of a planned stormwater drainage system, an attempt is made to regulate the discharge from Halanayakanahalli and Junnasandra lakes through sewage drain of the RBD layout, the drain on Sarjapur road and then connecting it to the culvert on it opposite to the open space between Saket Callipolis and Keerthi Regalia.



Figure 6.7: A view of the culvert on Sarjapur road in front of the open land between Keerthi Regalia and Saket Callipolis Apartments. There existed a pipe culvert earlier. A new one was built adjacent to it in 2020. There is no drain continued from it at the other end. It is completely filled up with soil and buried under the road.



Figure 6.8: The attempted stormwater drainage-2: Since the Attempt-1 could not realize, an attempt is then made to continue the discharge through the side drain on the road, a culvert on it in front of Salarpuria Sanctity and then trough the sewage drain adjacent to Bank of Baroda.



Figure 6.9: A view of the side drain on Sarjapur road in front of Keerthi Regalia. It is too small to carry the discharge from a catchment area of 10.1 sqkm.

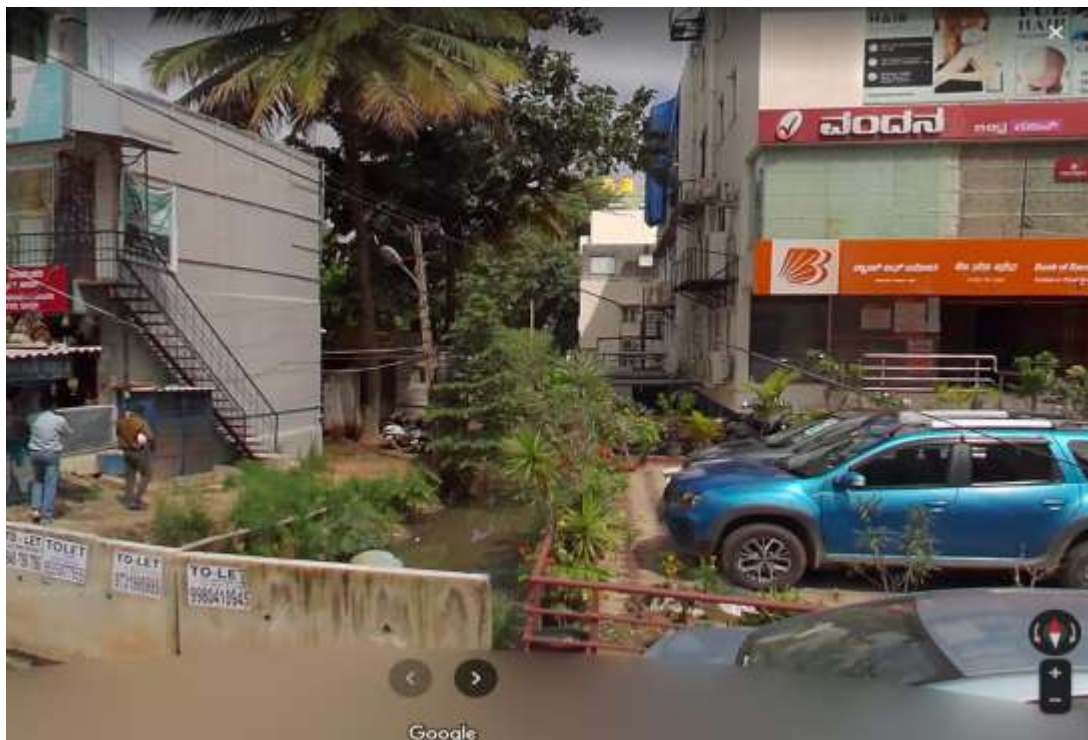


Figure 6.10: The view of the sewage drain adjacent to Bank of Baroda. It passes through a narrow space between the buildings and is completely occupied by jungle. It has a capacity just adequate to discharge sewage from its surrounding locality.



Figure 6.11: A view of the side drain on Sarjapur road in front of RBD layout through which attempt is made to regulate the discharge from the layout. It is at a significantly higher level than the sewage drain at the layout.



Figure 6.12: The pocket of low-land near RBD layout. Since it was the first layout developed at the reach and the surrounding lands were developed by filling up the area, a large area near the entrance of the layout turned into a lowland. The stormwater from the surrounding area is drained at it.

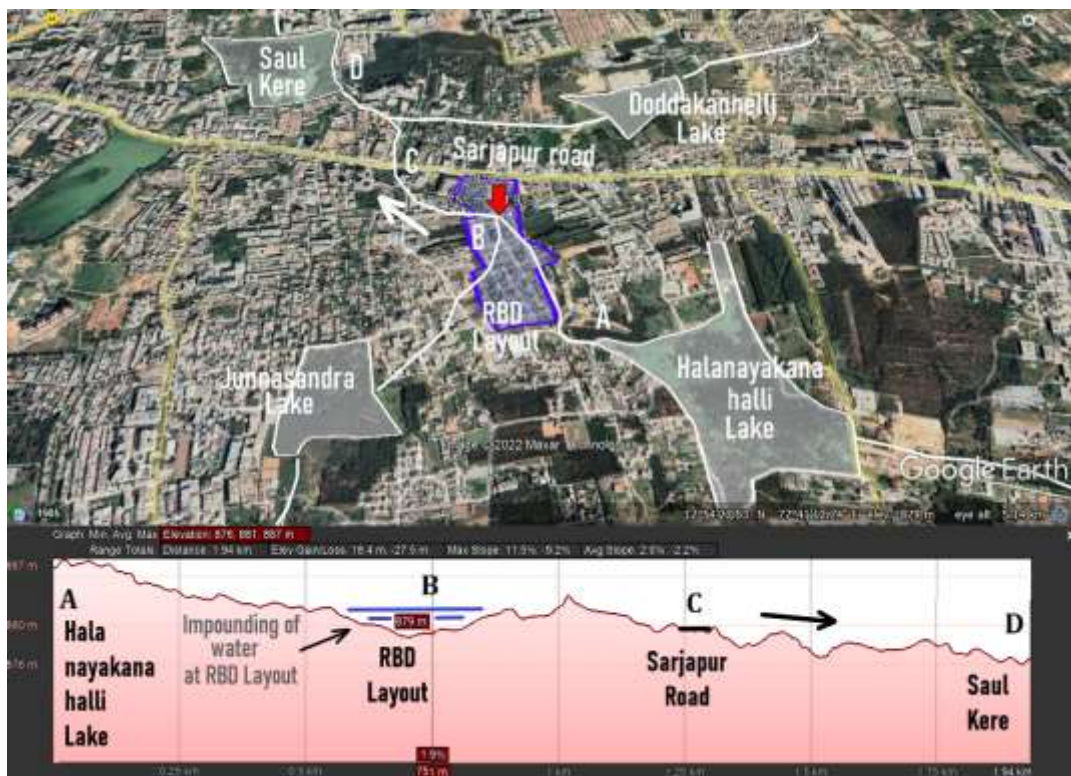


Figure 6.13: The ground profile from Halanayakanahalli lake to Saul Kere, showing the pocket of low-land at the reach near RBD layout.



Figure 6.14: A view of the waterlogging at RBD layout on 5h September, 2022. (Photo courtesy: The Hindu)



Figure 6.15: Another view of the waterlogging at RBD layout on 5th September, 2022. It is not just the stormwater from its two neighbouring villages, viz., Halanayakanahalli and Junnasandra, but the obstructed and held back run-off from a catchment area of 10.1 sqkm at Sarjapur road, unable to find any escape route.

Chapter 7

Decoding of Waterlogging at Wipro Campus on Sarjapur Road

The Wipro Campus on Sarjapur road was completely inundated during the flood of September, 2022. The BBMP observed that it was inundated mainly due to laying of the slab on the drain at the campus and adjacent Greenwood residency. However, a detailed study reveals that it is not the main reason. It was inundated mainly due to recent construction of divider on Sarjapur road and construction of Sattva Tech Park on the open land adjacent to its main gate, as discussed below,

- i. The Wipro Corporate office is located on Sarjapur road between Junnasandra, Halanayakanahalli and Doddakannelli lakes and Saul Kere as shown in Figure 7.1.
- ii. Figure 7.2 shows the Google image of the area between RBD layout and Saul Kere in 2002. It was entirely an agriculture land. The Wipro campus was built right **on the stream paths** between Halanayakanahalli and Doddakannelli lakes and Saul Kere.
- iii. Figure 7.3 shows the Google image of the area at the reach in 2019. The entire area was developed. A large number of structures were built on the stream paths.
- iv. The road has a falling gradient at the reach between RBD layout and Wipro campus. Figure 7.4 shows its Google image in January 2020. There were **only trees planted at its centre**. The spill over from Halanayakanahalli and Doddakannelli lakes was discharged through the open lands adjacent to Sarjapur road, open space at the main gate of the Wipro campus, open land on its right and then through Greenwood residency, as shown in Figure 7.5.
- v. Figure 7.6 shows the Google image of the area at the reach in March 2022. The **Sattva Tech Park** was built in the open land on the right side of the Main Gate of Wipro campus.
- vi. Around the same time, Sarjapur road was re-asphalted. A large camber was given for the road sloping from its centre to its edges as shown in Figure 7.7.
- vii. Figure 7.8 shows an ariel view of the road at the reach. The divider was laid all along the stretch of the road with an opening opposite to Main Gate of Wipro campus. The road surface at the opening was flattened and given a gentle slope towards the gate as shown in Figure 7.9..
- viii. Thus, there were two developments just before the rainy season of 2022, viz.
 - (a) The Sattva Tech Park was built in the open land on the right side of Wipro main gate; it totally obstructed the discharge from upstream lakes to Saul Kere at the reach. the campus.
 - (b) The **road divider** was constructed at the centre of Sarjapur road.

- ix. The spill over from Doddakannelli lake was directed into Wipro campus from right track of the road. The spill over from Halanayakanahalli and Junnasandra lakes was directed along its left track **by the divider** and then into Wipro gate. The combined discharge joining at the centre of the campus was stuck and accumulated at the campus in front of its tall structures on its rare end and villas of the Greenwood residency, as shown in Figure 7.10.
- x. From the video footages of the flood at the reach on September 5, 2022, we could clearly see the spill-over from RBD layout directed on the left track of Sarjapur road by the divider and then into Wipro Gate and accumulating at the centre of the campus. Figure 7.11 shows a still from a video footage in which the discharge on the road could be seen directed into Wipro Gate.
- xi. What we were seeing at RBD layout during peak flood was the discharge from Doddakannalli, Halanayakanahalli and Junnasandra lakes struck at Wipro campus and stretched upto it.

Here, we may observe that,

- i. The stormwater from a catchment area of about **11.5 sqkm** is directed at Wipro Campus.
- ii. The drain at Wipro complex and Greenwood residency is just their stormwater or sewage drains connected together.
- iii. Whether it is covered or open, it could just discharge the sewage from its surrounding locality.
- iv. The discharge from the lakes invariably spills over it and runs on the ground.
- v. It is held back, if there is any obstruction in its direction and does not find any escape route.
- vi. Therefore, clearing the slab on the drain would not significantly alter the flow phenomena at the reach. It would not direct the entire discharge from a catchment of 11.5 sqkm to run through it. It invariably demands a free overbank area for a certain width on it.



Figure 7.1: The location map of Wipro Corporate Office. It is located on Sarjapur road between Junnasandra, Halanayakanahalli and Doddakannelli lakes and Saul Kere.



Figure 7.2: The Google image of the area between RBD layout and Saul Kere in September 2002. It was entirely an agriculture land. The Wipro campus was built right on the stream paths between Halanayakanahalli and Doddakannelli lakes and Saul Kere.



Figure 7.3: The Google image of the area in February 2019. There was development in the entire area and all along the stream paths.

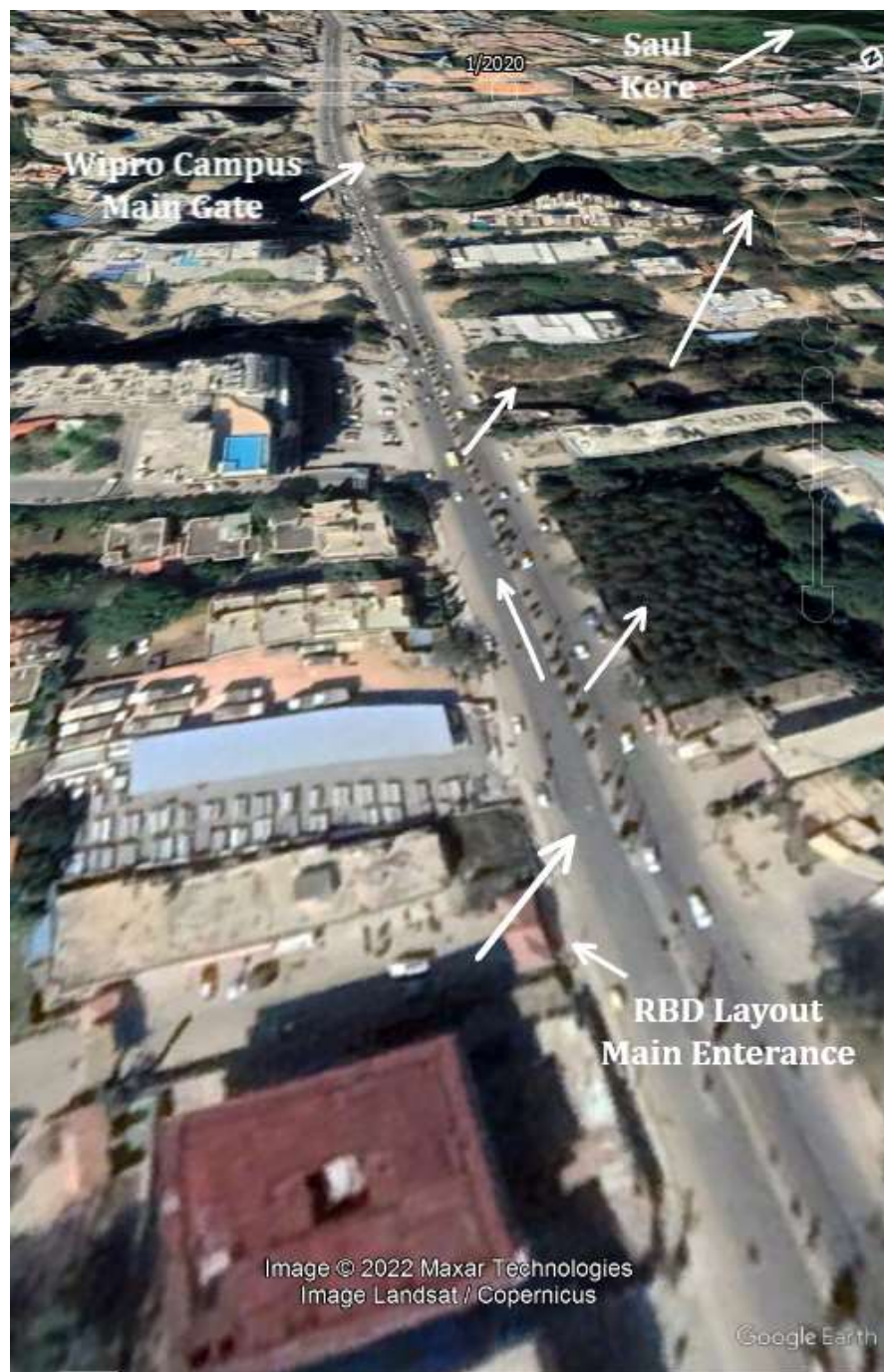


Figure 7.4: The Google image of Sarjapur road near Wipro Campus. It has a falling gradient towards Wipro campus from main entrance of RBD layout. Until December 2021, there were just trees planted at the middle of the road. The discharge from Halanayakanahalli and Junnasandra lakes was spilling over the road and finding its route to Saul Kere through the open lands in between, as shown in the figure.



Figure 7.5: The discharge path from Halanayakanahalli and Doddakannelli lakes until December 2021. The discharge from Doddakannelli lake was directed to Sarjapur road and joined the discharge from Halanayakanahalli and Junnasandra lakes. The combined discharge was then finding its route through the open lands adjacent to Sarjapur road, open space in Wipro campus near its Main Gate, open land on its right and then through open space between the villas of Greenwood residency.



Figure 7.6: The Google image of the area in March 2022. Sattva Tech Park was built on the open land adjacent to Main Gate of Wipro campus. It totally obstructed the discharge from upstream lakes to Saul Kere.



Figure 7.7: The divider on Sarjapur road in front of RBD layout. It was constructed during December 2021 replacing trees. A large camber was given for the road sloping from its centre to its edges.



Figure 7.8: An ariel view of Sarjapur road near Wipro campus in March 2022. The road divider is constructed all along its stretch with an opening at the Main Gate of the campus.



Figure 7.9: A close view of Sarjapur road and divider opening infront of Wipro Gate. The road surface at the divider opening is flattened and given a gentle slope towards the gate.



Figure 7.10: The phenomenon of the waterlogging at Wipro campus. There were two developments just before the rainy season of 2022, viz. (i) The Sattva Tech Park was built; it totally obstructed the discharge of the lakes at the reach (ii) The road divider was constructed on Sarjapur road. It directed all the spill over from RBD layout on the left) The track of Sarjapur road and then into the gate of Wipro campus. The spill over from Doddakannelli was directed into the campus from its right track. Both discharges were stuck at the campus, in front of its tall structures at its rare end.



Figure 7.11: A view of flood water at the Main Gate of the Wipro complex on 5th September, 2022. It clearly shows the spillover from Halanayakanahalli lake directed into it from the left track of Sarjapur road. It is then accumulated inside the campus and held back upto the road.



Figure 7.12: The view of the flood water at RBD layout on 6th September, 2022. During peak flood, the discharge from three lakes viz., Halanayakanahalli, Junnasandra and Doddakannelli lakes struck at Wipro campus is held back upto the layout. Therefore, what we see in the above picture is actually the obstructed flow at Wipro campus stretched upto the layout and beyond it.

Chapter 8

Decoding of Waterlogging at Outer Ring Road

8.1 The Ring Road is flooding since 2017

The Outer Ring Road near RMZ Ecospace was one among the locations which were severely affected during September, 2022 flood. It is inundated quite frequently since 2017 as the following article of October 2017 reveals.

The News Minute

Bengaluru's Outer Ring Road completely inundated with only 2mm rainfall, here's why

BENGALURU RAIN MONDAY, OCTOBER 16, 2017



Employees of RMZ Ecospace were advised to work from home due to severe waterlogging. The area surrounding Savalkere or Saul Kere in Bengaluru received just 2mm of rainfall on Sunday night. Yet, the Outer Ring Road was completely inundated. Savalkere was filled to the brim and the overflowing water turned the Outer Ring Road into a river, resulting in waterlogging at RMZ Ecospace, the tech park located on the arterial road.

Vehicular movement along the stretch was also grindingly slow.

“The water from Savalkere overflowed and the water went into the nearby sewage drain. This sewage water is now flowing into the Outer Ring Road. The HSR Layout Police have had to literally push two wheelers to safety. They have been working since last night to ensure the safety of commuters. The water has now flooded the service road and vehicular movement is happening only on one side of the road. Pedestrians have been advised to use the skywalk,” said DCP (East) Traffic, Abhishek Goyal.



Employees of Ecospace were advised to work from home and the pick-up facility was cancelled due to severe waterlogging.

“Many employees have been coming in for work. The waterlogging has occurred only near the Ecospace gate. There has been no flooding inside the building. Ecospace security personnel and the Bengaluru Traffic Police have been helping vehicles enter and exit the area without any difficulty. Buses are also able to ply,” a spokesperson from Ecospace told TNM.

What caused a major arterial road to flood with just 2mm rain?

Spread across 62 acres, Savalkere or Saul Kere is one of the largest lakes in the area. Activists say that more than three acres of the lake were encroached upon by illegal occupants, with houses built on the lake bed.

Despite issuing notices to vacate the land, the encroachers paid no heed. In 2013, they were cleared and the restoration process of the lake was started. “This is also done shoddily. If you see the height of the inlet and the outlet, it is uneven. Water from a lake must not overflow, the height of the inlet and outlet (which themselves are storm water drains) must be equal. In Savalkere, the height of the outlet is lower than that of the inlet. Obviously, the water will overflow.

Besides, so many restaurants and houses have encroached on the interconnected system of storm water drains. The BBMP has built several concrete drains but this is not enough,” said Ramprasad, convenor of citizens’ group called Friends of Lakes. Nagesh, a lake activist from Bellandur area, who is working for the rejuvenation of Savalkere, said that currently, there is only one tertiary sewage drain which is connected to the lake’s outlet.

“The outlet is connected to a drain which in no way is capable of withstanding the outflow from the lake. The main issue is that the storm water drain system has been encroached upon. BBMP has constructed several secondary drains but these are made of concrete and there is no scope for the water to seep into the ground. Besides, none of these storm water drains have been desilted. That’s why just 2mm rainfall has resulted in waterlogging,” Nagesh added.

8.2 It was flooded during last week of August 2022 too

The Ring Road was also affected severely during the last week of August 2022, as the following article reveals.

The Hindu

Bengaluru rains: Outer Ring Road completely inundated after overnight heavy downpour

Darshan Devaiah B.P.

BENGALURU: AUGUST 30, 2022 17:13 IST



Many cab drivers parked their cars by the side of the road and helped people to get across the flooded stretch. | Photo Credit: Murali Kumar. K

Traffic movement was crippled on ORR due to water from Savalakere lake overflowing into a nearby drain following incessant rains, resulting in flooding on the roads

Hundreds of employees working in and around RMZ Ecospace on the Outer Ring Road (ORR) in Bengaluru had to experience the horrors of peak-hour traffic, exacerbated by inundation due to heavy overnight rain on Tuesday.

Speaking to The Hindu, Ajith Kumar, a techie working at RMZ Ecospace said, "There was water from last two days, but today the level is more than two to three feet and my bike couldn't move at all. So, I parked and walked to my office to reach on time."

Pavan Kumar, a cab driver working for a private firm, was seen helping senior citizens cross the road as well as bike riders who were struggling to push their vehicles due to gushing water on the road.

"I left my work now to help people here who are not able to cross the roads. The main reason for this situation is the overflowing of the lake. The drains are small and the capacity of rainwater could not be handled by the drains built by the civic agencies," Mr. Pavan told The Hindu.

8.3 Reasons for waterlogging at the road – Experts' views

Experts mainly attribute the excess construction at the valley and encroachment of lakes for the frequent flooding of the road. They view that the infiltration of soil is stopped due to too much tar and concrete. The capacities of the lakes have been reduced. The run off from the catchment is increased. The Rajakaluves prove to be too inadequate to discharge the resultant run-off. The following article presents their views.

news9live.com/

Bengaluru Outer Ring Road floods: Experts blame excess construction on valleys, lake encroachment

Prajwal D'Souza

1 Sept 2022 6:20 PM

'Venice of Bengaluru', 'Lakes on road', 'New branch of Wonderla in Bellandur' — these were some of the terms used by frustrated netizens of Bengaluru as they encountered a heavily waterlogged stretch of Outer Ring Road (ORR) near Ecospace service road near Bellandur in the city on Tuesday and Wednesday following a bout of heavy rains on Monday.

There was heavy traffic congestion as office-goers spent hours crossing an 8-kilometre stretch connecting Bellandur, Sarjapur and ORR as rainwater overflowed from the 'rajakaluve' (storm water drain) onto the street. On Wednesday, despite it not raining for nearly 48 hours, the condition remained more or less the same in the ORR area. Locals say there was nearly 4-5 feet of water on the roads.

As the traffic police diverted the traffic and rescuers came to the aid, the state government and the Bruhat Bengaluru Mahanagara Palike (BBMP) faced the heat from the angry residents, who questioned the infrastructure works and monsoon preparedness of both the state and the city corporation. According to IMD's data on rainfall recorded in Bengaluru Urban for August, the district saw a departure of 192 per cent – the normal was 116.8 mm, while the actual rainfall was 341.5 mm.

What went wrong?

RMZ Ecospace, the city's tech park, which is near the Bellandur ORR is surrounded by Savalkere, one of the largest lakes in the region. While many pointed out that the overflowing of Savalkere caused the SWD to flood resulting in waterlogging of the ORR, Raj Bhagat Palanichamy, Geo Analytics expert at World Resources Institute (WRI) India said that the problem was much bigger. According to Palanichamy, the root of the problem lies in the topography of the region.

"When rainwater flows, it erodes certain places creating valleys. Across these valleys, where the water flows, lakes were built as dams. They were built for irrigation purposes and existing drains in the valleys were modified for the same purpose. However, the old villages were close to the ridges but not in the valley itself. Once Bengaluru started growing, rapid construction happened in the valleys as well. Either farmers sold the land or they themselves started the constructions," Palanichamy told News9.

The expert also pointed out that too much tar and concrete was also to blame for the watery mess. "When the houses were built, roads were also constructed and widened- for example, this is why the ORR was constructed. Because of these constructions in the valleys, the water flow has been severely blocked. Only the government-owned remnants of the irrigation canals – Rajakaluves – remained. However, it is not sufficient for carrying floodwaters. There is too much tar and concrete, which also stops the infiltration of soil. Because of the blockage, the areas – Manyata Tech Park, RMZ Ecospace, Rainbow Drive and others – are flooding," the expert added.

The problem is that there is no regulation to control the construction, Raj Bhagat said, adding that construction is taking place in every valley in Bengaluru, where there is the action of the market forces. Outer Ring Road Companies Association (ORRCA) General Manager Krishna Kumar said that since the width of Rajakaluves was reduced due to construction, the water flow was affected.

"Rainwater fills the lakes and then it flows to the Rajakaluves – it is the natural system. Let's say the width of the Rajakaluve is reduced from 10 metres to metres, when the force of the water is normal, it flows through it, but when the force is higher, the water will seep into the roads and surroundings. This in turn will hugely affect the traffic – especially in a high-density corridor like ORR," Kumar told News9. Kumar added that officials from the BBMP informed them that due to the encroachment of Rajakaluve, excess water is flowing into the roads.

Is there a solution?

Palanichamy, the Geo Analytics expert, said that the solution is difficult but not impossible. "We have to think of creating new drains for these places. More culverts need to be created and more space needs to be found for the existing Rajakaluves. The government has to procure a little more land and for building more drains based on scientific calculations on how much discharge will be there during extreme bouts of rainfall," the expert said.

For the new areas, he added, there need to be proper restrictions on the usage of land. "Since the city is expanding, there is a likelihood that more places will be impacted because of this. When new roads or buildings come up, it will create more issues. Those places need to be identified and the authorities have to figure out how growth can be regulated without affecting the aspirations of the public. But for existing ones, new drains are needed and one has to think about reducing concrete space and increasing more percolation into the soil," he added.

The art of passing the buck

If passing the buck is an art, then civic authorities are the Picassos of it – Bangaloreans have seen this happening with BBMP, Bangalore Water Supply and Sewerage Board (BWSSB) and Bangalore Development Authority (BDA). When BBMP started to face the heat for the problems in ORR, the blame was immediately moved towards the BDA.

"Due to the heavy rains, the waste weirs of a few lakes may have flowed over their capacity. But there was no lake breach specifically. We may have to install more gates and ensure better management of water flowing through the waste weir. We

will take up such works within the next monsoon," KV Shashikumar, BBMP Chief Engineer (Lakes), was quoted as saying by Deccan Herald while another engineer pointed the finger at the BDA for not "providing a proper drainage system" when ORR was constructed in the area.

But the BDA is of the argument that the existing problems "is because of poor maintenance and the (BBMP's) failure to clean the stormwater drains". "The ORR was handed over to the BBMP at least six years ago. There have been heavy rains but such problems had never come up. This is only because of poor maintenance and the (BBMP's) failure to clean the stormwater drains. There is no design flaw and the road has been constructed as per the standards," HR Shantharajanna, Engineer Member, BDA told the daily.

Earlier in May, it was reported that the BBMP had decided to desilt nearly 800 kilometres of drains, where close to Rs 83 crore was set aside by the civic authority to remove drain blockages. However, flooding in the city casts doubts on the BBMP and its so-called monsoon preparedness and efforts to desilt drains.

8.4 The actual reasons for the inundation of the road

According to the experts, Saul kere is one of the largest lakes in the region. Its overflowing is causing the stormwater drain to flood resulting in the waterlogging of the Outer Ring Road. However, a detailed study reveals that there is indeed no defined stormwater drain at the reach. The actual reasons for its inundation could be analyzed as below,

- i. Figure 8.1 shows the location map of the Outer Ring Road near RMZ Ecospace. It is located across the valley between Saul Kere and the main stream from Bellandur lake. A total catchment area of **27.9 sqkm** drains at the road at the reach. There are 14 lakes in the above catchment as shown in the figure.
- ii. Figure 8.2 shows the Google image of the valley between Saul Kere and the Outer Ring Road in March 2002. The spill over from Saul Kere was thinly discharged at the valley. There was **no defined waterway**.
- iii. Figure 8.3 shows its Google image of the valley in March 2003. There was a small road at the reach before 2003. The Outer Ring Road was newly constructed with a culvert at the deepest point of the valley opposite to the location, where EPZ Ecospace is built now.
- iv. Figure 8.5 shows its Google image of the valley in May 2004. The land at the upstream of the culvert was filled up upto Saul Kere and levelled for the construction of Ecospace complex. The construction of the Ecospace complex was commenced.
- v. A close view of the image of the valley in May 2004 reveals that there was **no specific stormwater drain** planned between Saul Kere and the main stream from Bellandur lake. The stormwater drain at the border of Ecospace is joined to the culvert on Outer Ring Road to regulate the discharge from Saul Kere as shown in Figure 8.7. The drain is completely covered under the slab along the road.

- vi. The construction of Metro rail was commenced in January 2022. Until its commencement, the discharge from Saul Kere was collected at the open land adjacent to Ecospace, spilling over the Outer Ring Road and finding its route to the main stream from Bellandur lake as shown in Figure 8.8.
- vii. Figure 8.12 shows the Google image of the catchment at the upstream of the Outer Ring Road in 2003. The entire catchment was **agriculture land**. A significant portion of rainfall was percolated into the ground. The lakes were virgin. The discharge from them was not significant.
- viii. But over the years, the entire catchment was concretized as the Google image in 2022 in Figure 8.10 reveals. Near entire rainfall is turned into run-off and directed to Outer Ring Road.
- ix. The Outer Ring Road has a total width of about 50 m at the reach. The entire discharge from a catchment of 27.9 sqkm is attempted to be regulated through a small culvert of above length. Figures 8.8 and 8.9 show the close views of the culvert near Ecospace.
- x. Before the commencement of Metro rail work, the road had just dividers at its centre and between it and service road as shown in Figure 8.14. The discharge from Saul Kere spilling over the road could easily negotiate them.
- xi. Figure 8.15 shows the **median wall** constructed at the centre of the road as a part of Metro rail work. The discharge from Saul Kere spilling over the road is totally obstructed and is held back by it, as shown in Figure 8.16.

Figure 8.18 shows the view of the waterlogging at the road during peak flood. We could see that the lane towards Ecospace is completely waterlogged, while it is relatively less waterlogged on the opposite lane. The traffic could negotiate it. It clearly indicates the reason for the increased severity of waterlogging at the reach during 2022. The free flow of flood over the road is totally obstructed by median wall constructed at the beginning of the year.

It is interesting to observe that the road is inundated at the reach since quite few years. But there are no efforts to investigate into the reasons for its inundation and initiate any measures for it. On the other hand, the Metro rail work is commenced. It demands construction of a large dedicated stormwater drain between Saul Kere and the main stream of Bellandur lake and a large culvert across it on Outer Ring Road depending on the present topography. Once the Metro work is completed, it would be too difficult to take up their construction.

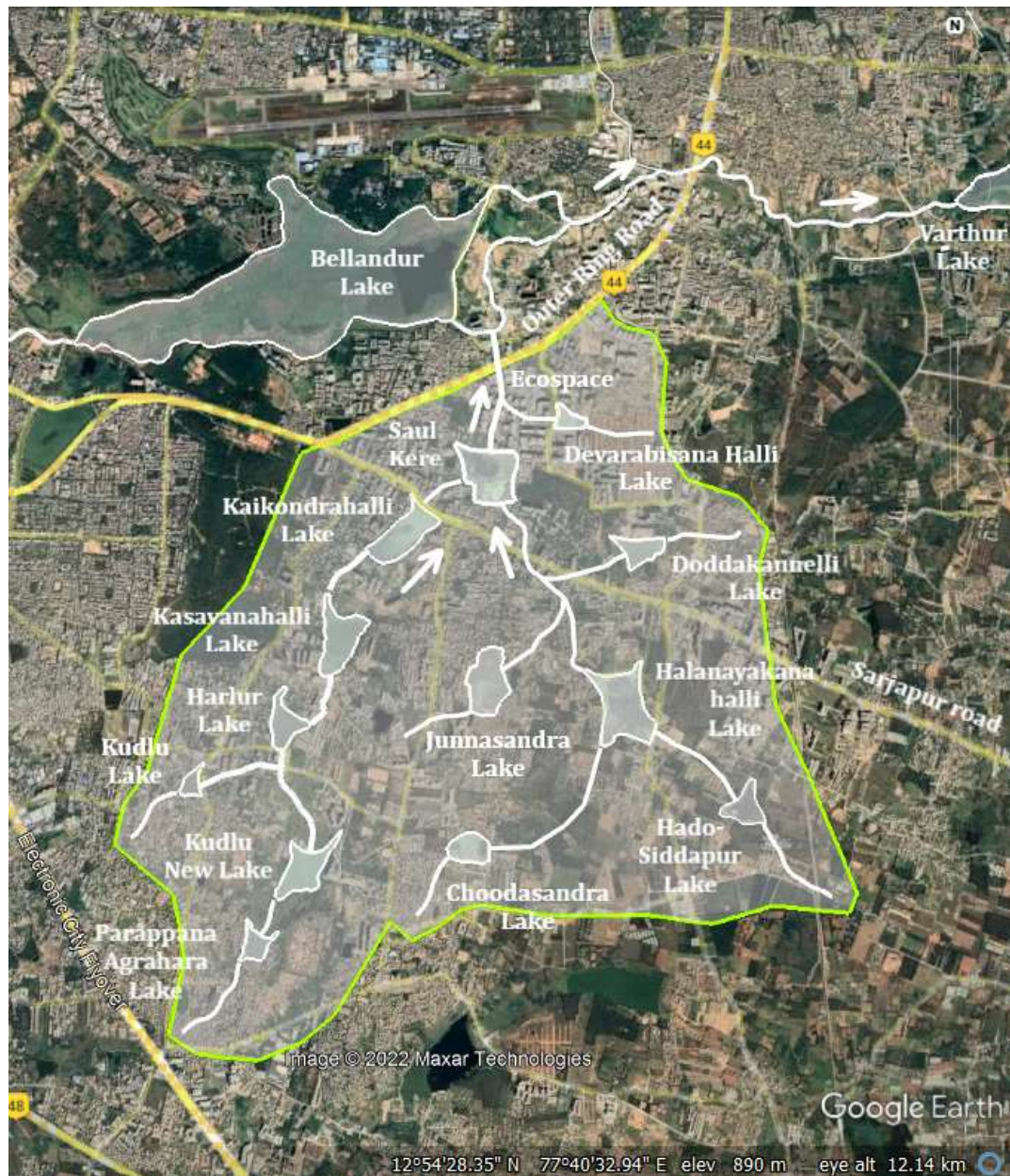


Figure 8.1: The location map of Outer Ring Road near RMZ Ecospace. It is situated across the valley between Saul Kere and the main stream of Bellandur lake. A total catchment area of 27.9 sqkm drains at the road at the reach. There are as many as 14 lakes in the above catchment area as shown in the figure.



Figure 8.2: The Google image of the valley between Saul Kere and Outer Ring Road in March 2002. There was no defined stream path. The spillover from Saul Kere was discharged on the agriculture land over a wide area.



Figure 8.3: The Google image of the Outer Ring Road in March 2003. Earlier, there existed only a small road at the reach. The Ring Road was newly constructed with a culvert at the deepest point at the valley, opposite to the location, where EPZ Ecospace is built now.



Figure 8.4: The Google image of Outer Ring Road and the valley in February 2004. The land at the upstream of the culvert was filled up and levelled for the construction of Ecospace complex.



Figure 8.5: The Google image of the road and the valley in May 2004. The entire stretch of the land upto Saul Kere was filled up and levelled. The construction of the Ecospace complex was commenced.



Figure 8.6: The Google image of the valley in January 2022. The entire land is developed.



Figure 8.7: A close view of the Outer Ring Road near EPZ Ecospace. There is no specific stormwater drain planned at the reach. The drain at the border of Ecospace is joined to the culvert on the Outer Ring Road and it is attempted to regulate the spill over from Saul Kere through it. It is completely covered under slab adjacent to the road.



Figure 8.8: A close view of the culvert near Ecospace. It has the capacity just to discharge sewage from its surrounding locality.



Figure 8.9: Another close view of the Culvert.



Figure 8.10: The culvert runs for a length of 50 m. The entire runoff from a catchment area of 27.9 sqkm is attempted to be regulated through it.



Figure 8.11: The Google image of the Ring Road and the valley in January 2022. The Metro rail construction had just commenced. Until its commencement, the discharge from Saul Kere was collected at the open land adjacent to Ecospace, was spilling over the road and finding its route to the main stream from Bellandur lake as shown in the figure.



Figure 8.12: The Google image of the catchment area at Outer Ring Road near Ecospace in 2003. It was completely an agriculture land. A significant portion of rainfall was percolated into ground. The lakes were virgin. Another its large portion was absorbed by them.



Figure 8.13: The Google image of the catchment area in 2022. The entire catchment area is concretized except few pockets of open land. Near entire rainfall is converted into run-off and directed at the Outer Ring Road.



Figure 8.14: The dividers at the centre of the road and between the road and service road near Ecospace before commencement of Metro rail construction. The discharge from Saul Kere spilling over the road could easily negotiate them.



Figure 8.15: A view of the Metro rail construction at the road. The discharge from Saul Kere spilling over the road is totally obstructed by the wall constructed all along its length and is held back. We could clearly observe that there is a large waterlogging at the lane opposite to Ecospace and only at the above side, while the lane on its other side is completely free from any waterlogging. There is free traffic movement on it. (Photo courtesy: India Today)

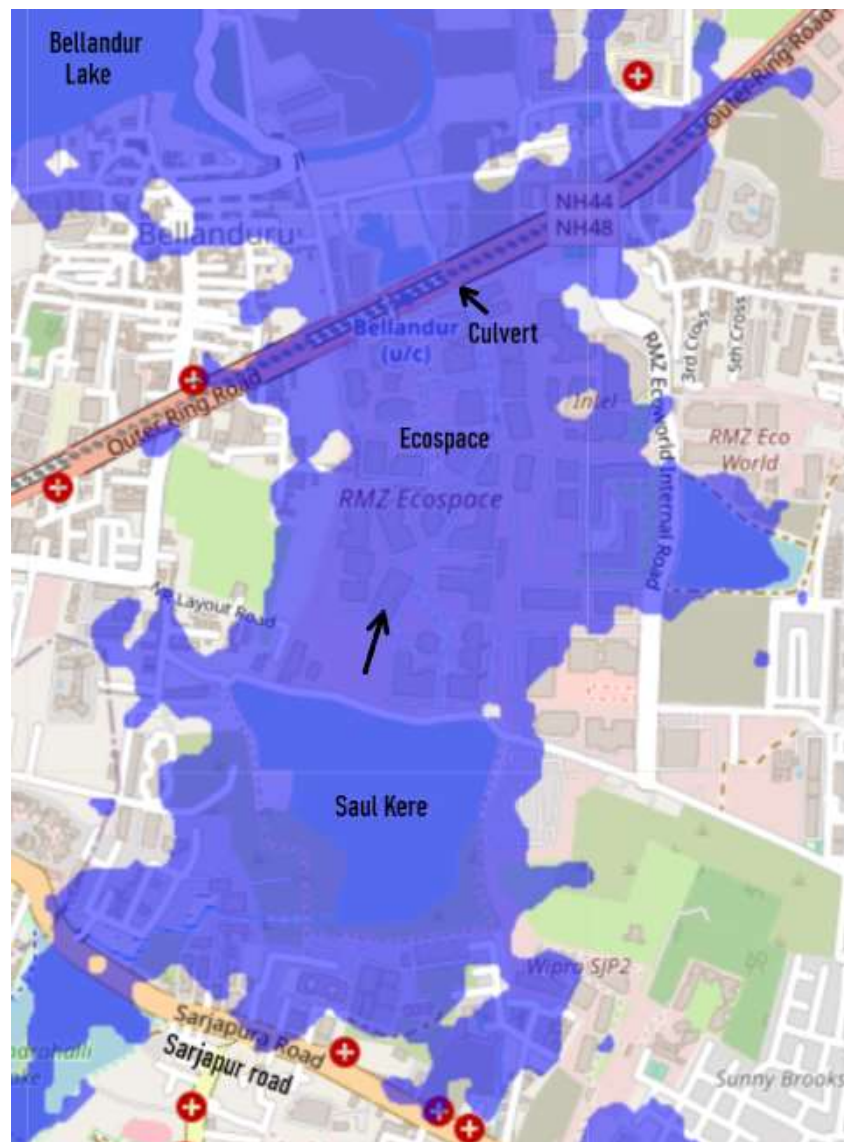


Figure 8.16: The inundation map at the Outer Ring Road near Ecospace. The obstructed discharge at the road is impounded at its upstream on a large area in the valley and is stretched upto Sarjapura road. The Ecospace is inundated in the above water.



Figure 8.17: An image of the flood water striking the wall of Metro rail in front of Ecospace on 6th September, 2022. (Courtesy: twitter@_maqtab)



Figure 8.18: Another view of the waterlogging at the Ring Road during the flood. We could clearly see that the lane towards Ecospace is completely waterlogged while the lane on its opposite side is relatively free. The traffic could negotiate it. It clearly indicates the reason for the increased severity of waterlogging at the reach during 2022. The free flow of flood over the road is completely held up by the construction of the wall. (Photo courtesy: The Print)

Chapter 9

Decoding of Waterlogging at Epsilon Layout and Yemalur Road

9.1 Waterlogging at Epsilon Layout and Yemalur Road

The upscale apartment complexes and gated communities like DivyaSree 77° Town Centre, DivyaSree 77 Place, Sobha Palladian, Epsilon Residential Villas, Rohan Jharoka and Sai Gardens in Yemalur were under water for over three days following the September 4-5, 2022 rain. Besides homes, hundreds of luxury cars including luxury brands Bentley, BMW, etc. were submerged and had to be towed by tractors, as the following reports reveal.

Bengaluru: Homes flooded, CXOs take tractors to safety
Times of India
TNN | Sep 6, 2022, 04.18 AM IST



BENGALURU: The flooding caused by rain on Yemalur-Bellandur corridor in east Bengaluru gave rise to an unusual sight on Monday morning. The overnight rain had rendered CEOs, COOs and CFOs living in villas and upscale apartments dotting the stretch completely vulnerable to the nature's wrath.

With their swanky cars under water and residences flooded, they found saviours in tractors which were rushed to their help by people from neighbourhood Kempapura.

Fix all drainage systems, not just on main roads: Resident

Many corporate honchos were spotted abandoning their plush homes and climbing onto tractors to leave for star hotels and houses of their friends. Their upscale apartment complexes and gated communities like DivyaSree 77° Town Centre, DivyaSree 77 Place, Sobha Palladian, Epsilon Residential Villas, Rohan Jharoka and Sai Gardens took the brunt of rain fury.

Vinod Kaushik, founder and CEO of a startup catering to the construction and remodeling industry, climbed a truck with his family members and two dogs for a trip to the main road.

A resident of DivyaSree 77 Place, Kaushik said: "Pretty much, all storm water drains have been messed up. If you close all down streams, this is what happens. They should fix all drainage systems, not just on main roads."

Sanjeev Kumar Ghai, a member of 77 Place task force, said: "This time, the intensity of rain was very high and there was a lot of water in the basements too. Pumps could not enter the area because of the water. BBMP should ensure water flows to the drainage." He said they arranged tractors for residents. "But it is difficult, especially for those who need emergency items like medicines," he added.

<https://www.moneycontrol.com/>

Bengaluru floods: Residents may have to cough up crores to fix drowned homes, cars

With the ground floor submerged, it will take about three months for the homes to be fixed, while many of the cars may have to be written off for good.

NIKHIL PATWARDHAN & BHAVYA DILIPKUMAR

SEPTEMBER 08, 2022 / 06:04 PM IST



Many luxury villas were submerged in the flash floods that accompanied the downpour.

If you thought a million-dollar house would at least guarantee a safe roof over your head, you'd be wrong, as many affluent residents in India's technology hub Bengaluru found out this week.

Billionaires and millionaires of the city, dubbed India's Silicon Valley, commuted on tractors and took shelter in hotels as their luxury villas and cars were drowned in the deluge that accompanied the downpour.

Epsilon Villas, a housing complex in Yemalur in Bengaluru, home to entrepreneurs like Wipro chairman Rishad Premji, Unacademy co-founder Roman Saini, Britannia CEO Varun Berry, BigBasket co-founder Abhinay Choudhari, and Page Industries MD Ashok Genomal, among others, has been under water for over three days now.

The complex has about 100 villas, each costing somewhere around Rs 11-12 crore. Around 80 of the 100 villas have been completely flooded since Sunday evening, according to residents. "We have had floods here earlier, where 5-6 houses were affected, so we were prepared for that. But this time the ground floors of about 80 percent of the houses have been submerged. The loss will be around Rs 150-200 crore," said Anees

This was corroborated by at least four other residents, who said they would have to spend somewhere around Rs 70 lakh to 1 crore to fix their houses. Moreover, residents were told that it will take at least two-three months to get their homes back in shape.

Besides homes, hundreds of cars, including luxury brands like Bentley, Mercedes, Audi, BMW, etc., were submerged and had to be towed by tractors on the morning of September 7. Residents said most were beyond repair. "We are staying at a friend's place. Looking at the damages —the flooring, wiring, and more — it seems like it will take at least two-three months," said Diwakar Gupta, a resident at Epsilon Villas for more than 18 years.

Residents said they had expected the water to recede on Monday morning (September 5), when the rains stopped. But since all the sewage lines were cluttered, the water could not flow out. Moreover, it rained heavily again that evening, making the situation worse.

Ultimately, residents had to rescue themselves and take shelter elsewhere. Residents claimed that on September 6 they rescued about 150 people, including children as young as five and senior citizens, some of who had mobility problems. They said that the army and fire brigade helped in the rescue operations with trucks, tractors, buses, and boats.

"About 70 were rescued on boats, while the rest were rescued in buses," Gupta said. "The situation got slightly better only on September 7, after Karnataka chief minister (CM) Basavaraj Bommai visited the area the prior evening. Residents are apprehensive of what will happen if it rains heavily again.

Bengaluru is home to millions of IT employees, thousands of start-ups, and about 40 of the country's 105 unicorns. "This impacts brand Bengaluru. If the situation doesn't improve, it will impact investments. Bengaluru has a lot of youth. Young people from all over come here. They are the ones who will be paying taxes for at least another 25 years. But if this is how it is going to be, they will move out soon, and that will hurt the city and the state's revenues," said Pawan Mehta, another resident of the Epsilon Residential Villas in Yemalur.

9.2 Decoding of the waterlogging at Epsilon Layout and Yemalur road

The actual reasons for the waterlogging at Epsilon Layout and Yemalur road may be analyzed as below,

- i. Figure 9.1 shows the location map of Epsilon, DivyaSree and other housing complexes in Yemalur which were severely affected during the flood of September, 2022. They are situated on the either side of a stream in Yemalur and at just upstream of its confluence with the main stream of Bellandur lake near Outer Ring Road in Kadubeesanahalli. There is a bridge on the main stream of Bellandur lake on Outer Ring Road near in Kadubeesanahalli Marathahalli Police Station.
- ii. Figure 9.2 shows the catchment area at the bridge on the Outer Ring Road. It is about 191 sqkm. i.e., the sewage and stormwater from about **1/4th area of Bengaluru city** is drained at it. Besides direct discharge from Bellandur lake, two large streams join the main stream at its downstream and drain at the road.
- iii. Figure 9.4 shows the Google image of the area upstream of the road in March 2002. The spill over from Bellandur lake was discharged in two streams at the reach. **Two separate bridges** were built on each of them.
- iv. The sewage treatment plant of BWSSB was built in 2005, by filling up the wet land adjacent to the left bridge of the road. The entire land between the two streams was also developed all along the stretch except for a small strip near the road. However, the right bridge was functioning.
- v. Figure 9.7 shows the Google image of the reach in January 2022. Between March 2021 and January 2022, the open land between the streams and adjacent to the road was also filled up for the construction of ancillary treatment plant. The discharge through the right bridge was **disconnected**.
- vi. Figure 9.10 shows a view of the left bridge. It has the capacity just to discharge sewage from Bellandur lake and its downstream catchment. The entire stormwater from nearly 1/4th of the city is attempted to be regulated through the above bridge.
- vii. The bridge gets completely choked up by the weed during flood. As a result, the flow is obstructed and impounded at its upstream as shown in Figure 9.12. It nearly reached the level of the Outer Ring Road during the September, 2022 flood. A JCB could be seen at the bridge clearing the weed for four days after the rain stopped to facilitate its discharge.
- viii. The obstructed flow is impounded at the valley on the upstream of the road and it is stretched upto Bellandur lake and into the valley of the stream in Yemalur as shown in the above figure. The Epsilon and other layouts in its vicinity were inundated in it. What we were seeing at the layouts was just the impounded water at the bridge on Outer Ring Road stretched upto them.

As a follow up measure to ease the discharge at the reach, a small bridge that existed at the entrance of the bridge was dismantled immediately after the flood. The drain was cleaned at the reach and widened. But it may facilitate free discharge of sewage. The bridge would invariably be choked up by weed during flood due to its small vent openings. The discharge through it would invariably be obstructed. Similarly, the inundation at Yemalur road and housing complexes and apartments in its surrounding is thought to be due to encroachments of drains in their vicinity and they are attempted to be cleared. It would in no way help in preventing the waterlogging at the layouts and road, until free discharge of flood is facilitated at the bridge on Outer Ring Road.

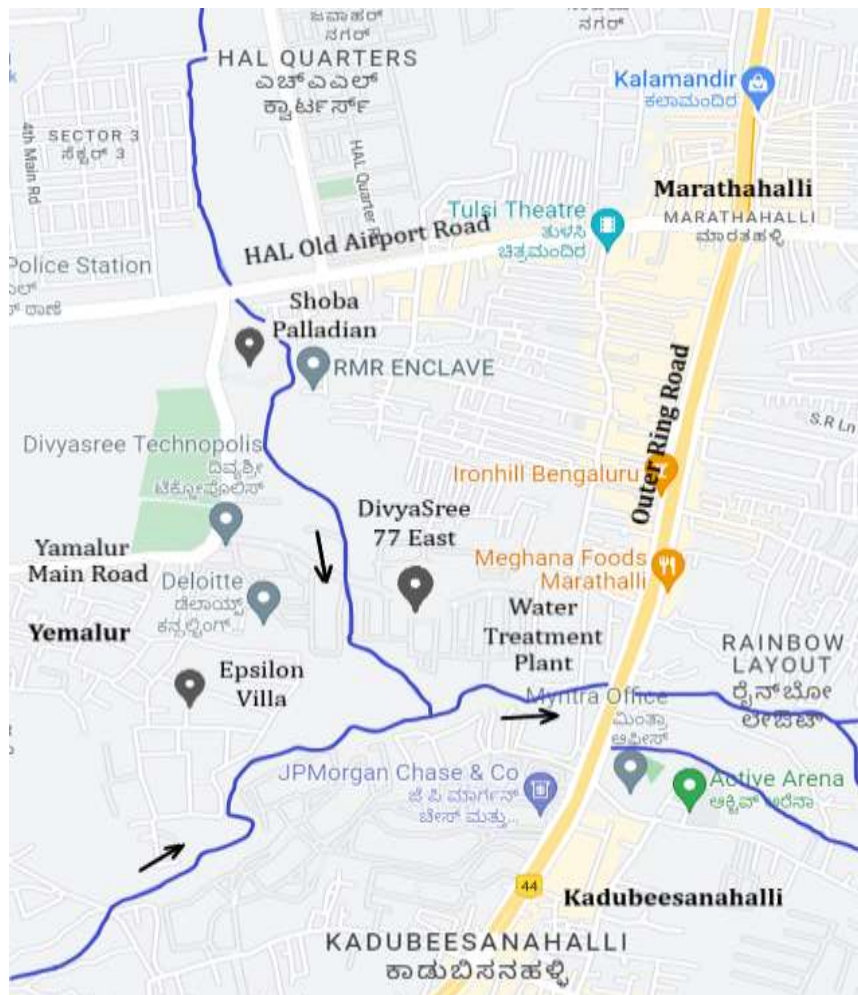


Figure 9.1: The location map of Epsilon layout and other upscale housing complexes in Yemalur which were severely affected during September, 2022 flood. They are situated on the either side of a stream in Yemalur and at just upstream of its confluence with the main stream of Bellandur lake near Outer Ring Road in Kadubeesanahalli. There is a bridge on the main stream of Bellandur lake on Outer Ring Road near Marathahalli Police Station.

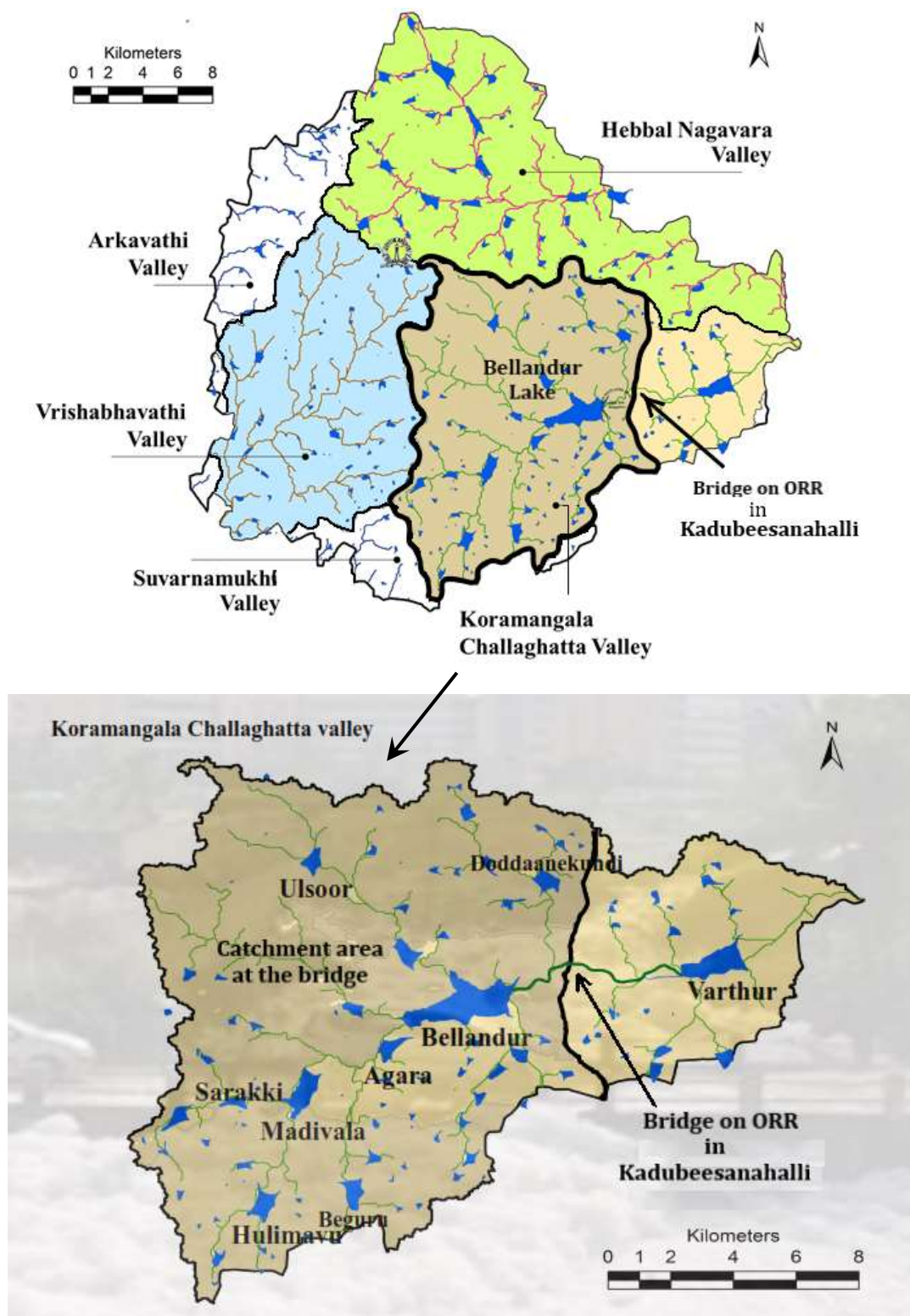


Figure 9.2: The catchment area of the bridge on Outer Ring Road in Kadubeesanahalli near Marathahalli Police Station. It is about 191 sqkm or about 25% of the total area of the city.

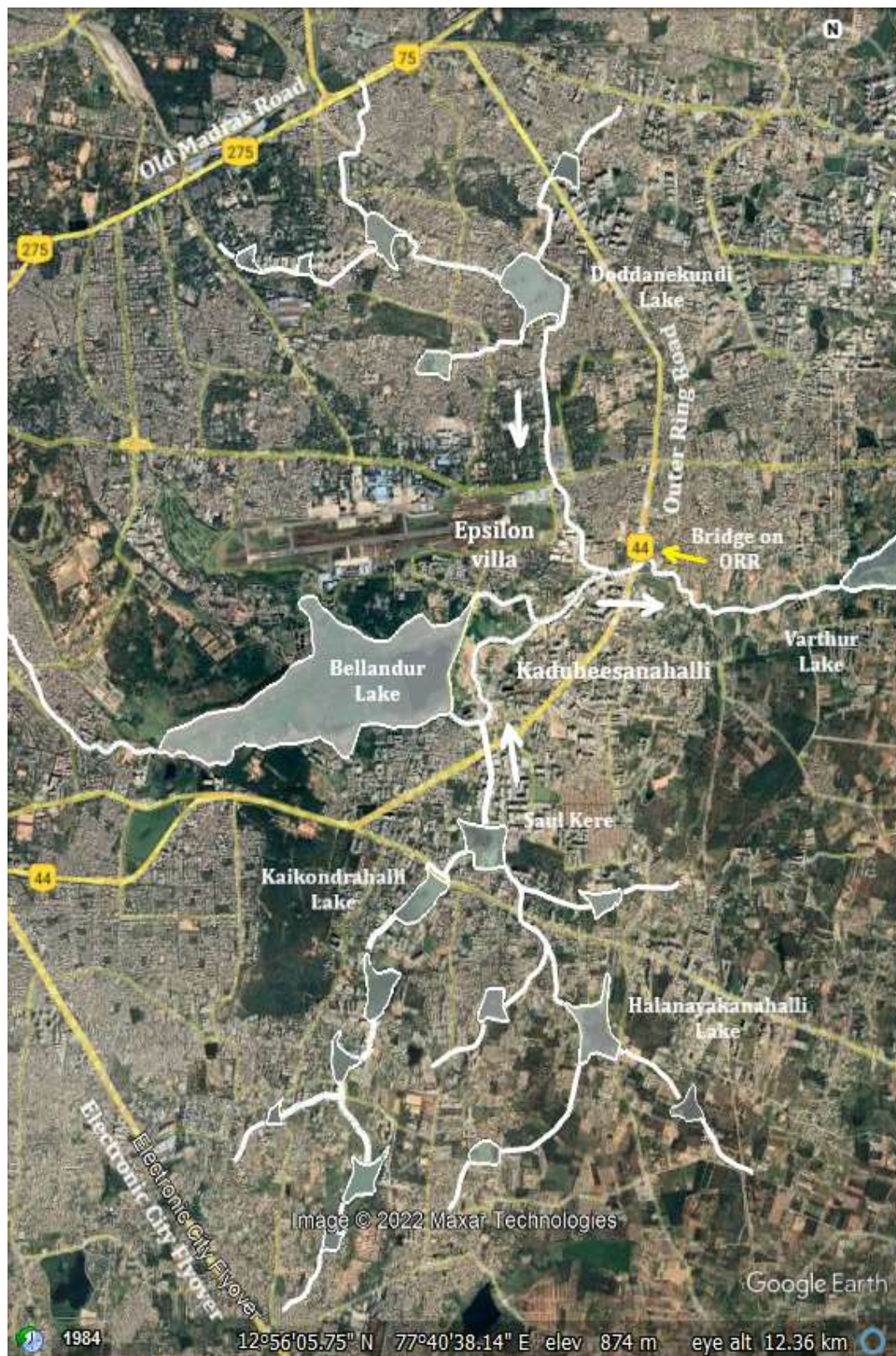


Figure 9.3: The lake system and the streams at immediate upstream of the bridge on Outer Ring Road near Marathahalli Police Station.



Figure 9.4: The Google image of the valley at the upstream of the bridge near Marathahalli Police Station in March 2002. The spillover from Bellandur lake was discharged through two streams. There were bridges on each of them on Outer Ring Road. It was completely a wet land at the both upstream and downstream of the bridges.



Figure 9.5: The Google image of the valley in February 2009. The BWSSB sewage treatment plant was constructed between left stream and Outer Ring Road.



Figure 9.6: The Google image of the valley in March 2021. The entire land between the two streams was filled up and developed except for a small strip near the road. However, the right bridge was still functioning.



Figure 9.7: The Google image of the valley in January 2022. Between March 2021 and January 2022, the open land between the streams was filled up for construction of the ancillary treatment plant. The discharge through the right bridge was disconnected.



Figure 9.8: The close view of the bridges on Outer Ring Road near Marathahalli Police Station in March 2021. Both were functioning. The spill over from Bellandur lake and its downstream catchment was discharged through both of them.



Figure 9.9: The close view of the bridges on Outer Ring Road near Marathahalli Police Station in January 2022. The open land between the two bridges was filled up. The flow through the right bridge was disconnected.



Figure 9.10: A close view of the left bridge. It has the capacity just to discharge the sewage from Bellandur lake and its downstream catchment area. It gets completely choked up by the weed during flood. The sewage and stormwater from a catchment area of about 191 sqkm or about 1/4th of the total area of the city is attempted to be regulated through it. It is totally held back.



Figure 9.11: The view of the right bridge, which has become defunct after the land between the two bridges was filled up for the construction of the ancillary treatment plant.

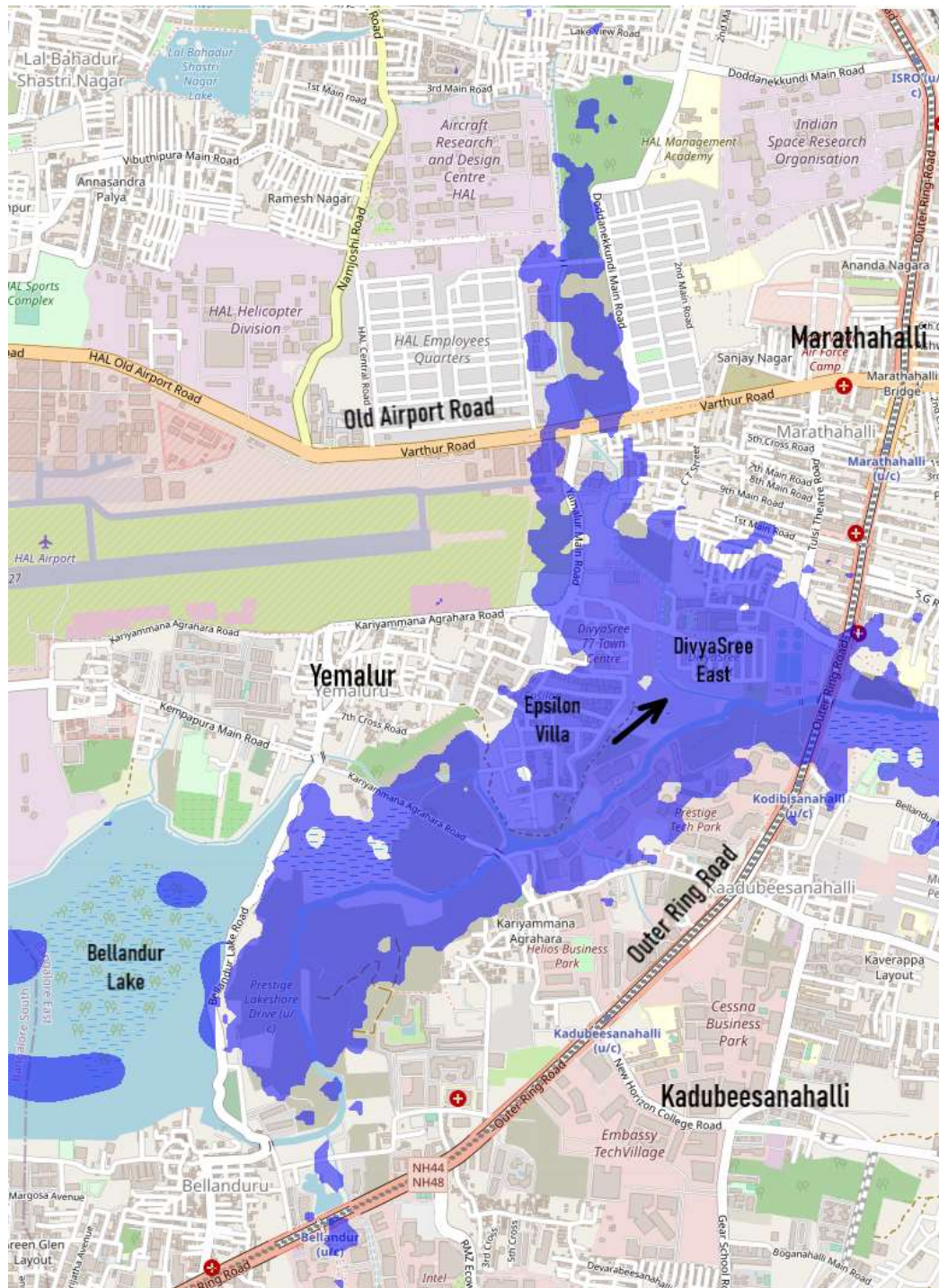


Figure 9.12: The inundation map at the upstream of the bridge on Outer Ring Road near Marathahalli Police Station. The obstructed discharge is impounded at the valley on the upstream of the road and is stretched upto Bellandur lake and into the valley in the stream in Yemalur. The Epsilon and other layouts in Yemalur were inundated in it.



Figure 9.13: A view of the waterlogging at Yemalur road on September 5 and 6, 2022. It is just the held-up water at the bridge on Outer Ring Road stretched upto the road. (Photo curtesy: Times of India)



Figure 9.14: The view of the waterlogging in front of LCA Tejas gate on 6th September, 2022. It is the same held up water at the bridge on Outer Ring Road in Kadubeesalanahalli stretched upto the gate. (Photo curtesy: Times of India)



Figure 9.15: The view of the waterlogging at the DivyaSree 77 East on 6th of September 2002. It is not the stormwater spilling over from the drains in its vicinity. It is the impounded water at the bridge on Outer Ring Road in Kadubeesalanahalli stretched up to it. (Photo curtesy: India.com)

Chapter 10

Decoding of Waterlogging at Borewell Road

10.1 Floods at Borewell Road

The Borewell road in Whitefield was another place at the city, where many apartment complexes and homes were submerged in the chest-deep water, during the flood of September of 2022. The following article gives the picture of the waterlogging at the road and its aftermath.

Times of India

Ladders, boats & rafts: How residents fled flooded homes in Bengaluru

SANIKA ATHAVALA,
Sep 7, 2022

About 100 families, with children and luggage in tow, fled the pools of sewage and rainwater that had stagnated around their homes.

The scene on Borewell Road in Whitefield on Tuesday morning was akin to a shot from a catastrophic movie. Hundreds of residents were leaving flooded homes on boats, rafts, ladders, and by forming human chains. Following two consecutive nights of heavy rain, many apartment complexes and homes on Borewell Road were submerged in chest-deep water.



It had receded to about knee-high late on Tuesday night. About 100 families, with children and luggage in tow, fled the pools of sewage and rainwater that had stagnated around their homes. Four apartment complexes — DNA Apartments, BM Glorietta, SS Felicity and GSR Srinivasam — appeared to bear the brunt.

Residents spent the morning trying to drain rainwater from their basements and retrieve submerged vehicles. Others distributed essentials and helped neighbours evacuate. Waseem Akram, a cab driver and resident of the area for the past 28 years, has helped at least six people since Monday, to move out to safer areas.

On Tuesday too, he was loading luggage onto dry parts of the footpath to pack into his car. An old resident of Nallurahalli area, which later became Whitefield, Akram said waterlogging has choked Varthur Lake. Residents' associations of apartments have told members that the situation would take 2-3 days to resolve. Since most apartments neither have power nor water supply, residents decided to relocate.

"Residents have been jumping over the space between apartment blocks and using a ladder to climb down to the top of the incline that leads to the basement," said Manish Jain, a resident of SS Felicity. Some even rowed out on makeshift boats and rafts.

Durgaprasad, a father of two and an IT employee, has been living in BM Glorietta for the past six years. He said he has never seen such flooding before. "The wall behind us fell yesterday when Nallurahalli lake overflowed," he said. "It has been raining heavily and regularly in the area for the past week and residents have been pumping out water, most of which collected in our basement. People just act and don't think about the consequences."

In the night, to help some of those who were stuck, locals fashioned a bridge out of rope and used it to help them cross over to a safer area. Less-affected residents have been volunteering for relief work and were seen running around bringing food and water to those enclosed in ponds of sewage.

Over the past week, residents across these affected complexes have shelled out anywhere between Rs 3-each.

The Hindu

Rain may have stopped in Bengaluru, but woes continue

Jahnavi T. R.

SEPTEMBER 07, 2022

Although the intensity of rainfall has not been much in the last two days, the aftermath of the two heavy showers on Sunday and Monday continued to trouble the residents of the IT corridor even on Wednesday. Inundated roads and apartment basements, JCBs clearing encroachments and cleaning up lakes, and tractors carrying people were a common sight around the zone. However, the situation was getting slightly better at many gated communities due to the efforts of the fire and emergency personnel.

As the compound wall of one of the apartments had collapsed, the residential building in which Mutthurayappa and his wife Sujatha reside had been flooded and the access to their home almost cut off. "All the IT professionals in apartments have moved to hotels to stay safe, but what should people like us who cannot afford it do? We have called the Bescom and BBMP too many times in the last three days, and yet they have not paid any heed to our problems. We have been here for far too long than apartment residents and yet our problems get neglected," the couple bemoaned.



Residents moving out of apartments on Borewell road in Whitefield, after it was flooded due to Nallurahalli lake breach, in Bengaluru on Wednesday. | Photo Credit: K. Murali Kumar

The grim aftermath of Bengaluru rains

The residents of this area alleged that along with the overflow from Nallurahalli lake and Sheelvantharakere lake, the demolition of a stop wall built next to the Rajakaluve, by the civic authorities to clear encroachments, led to the flooding of their apartments.

“The apartment has been here for nearly 10 years and it has been flooded like this only after that wall was taken down. This has been the situation for over three days now. We are paying from our pockets for pumping machines. We have barely received any help from BBMP,” said Chandra Kumar, resident of SS Felicity.

10.2 Decoding of the waterlogging at the road

The phenomenon of the waterlogging at the Borewell road could be analyzed as below,

- i. The Borewell road is located on the stream path between Nallurahalli lake and Sheelvanthakere at about 800 m from Nallurahalli lake in Whitefield. The Sheelvanthakere joins the main stream of Varthur lake at its downstream near Varthur circle.
- ii. The Nallurahalli lake has a small catchment area of about **3.3 sqkm**. Before the development of the area, Its spill- over was thinly discharged over the agriculture land. There was no defined stream path, as it was case with the most of the lakes in the area. However, no drain was planned either to discharge its the spillover at the later years also, as the locality developed into a residential area.

- iii. The some of the stormwater and sewage drains at the borders of the layouts were connected together to regulate the spillover from the above lake.
- iv. Figure 10.4 shows the drain between SS Facility Apartment and Ananya Farm. It is connected to a culvert on the Borewell road. The culvert crosses the road at an angle and is connected a drain between an open land and GSR Srinivasam Apartment.
- v. There is **no stormwater or sewage system** planned for the locality. The sewage from its layouts and apartments is conveniently discharged into the above drain.
- vi. The drain always carries sewage and runs at its maximum capacity. Therefore, the discharge from Nallurahalli lake invariably spills over it and Borewell road and runs in the open land between GSR Srinivasam and Malibu Paloma Apartments as shown in Figure 10.7. It is then discharged through a narrow space between the compound walls of DNA Apartments and Malibu Paloma.
- vii. Figure 10.9 shows the Google image of the open land between GSR Srinivasam and Malibu Paloma Apartments in December 2021. Certain **labour sheds** were built on the above land and right at the mouth of the narrow space between the compound walls.
- viii. They totally obstructed the discharge at the reach. Since there are structures built on either side of it close to one another, it could not find any escape route too. Therefore, it was held back and impounded at the upstream of the sheds. The Borewell road and the apartments in its vicinity were inundated in it.

Thus, what we were seeing at the Borewell road and DNA Eden View, BM Glorietta and other apartments during the flood was the above held up water at the open land between GSR Srinivasam and Malibu Paloma Apartments stretched upto them.

We could observe that the drain is not encroached upon at its any reach. It starts from an open land at about 200 m downstream of Nallurahalli lake and runs for a length of just about 600 m between it and Borewell road. The discharge from Nallurahalli lake is not significantly large. The locality has a significant falling gradient. But it did still experience severe waterlogging. It is mainly due to an attempt to route the discharge of lake through sewage drain. It invariably spills over. It is freely discharged as long as there is no obstruction on its overbank. If there is any new construction on it, it is held back and impounded at its upstream. We could also see that if the labour sheds were not built at the open land in the last December, there would not have been any waterlogging at the Borewell road and layouts and apartments in its vicinity during the year.



Figure 10.1: The location map of the Borewell road. It is situated on the stream path between Nallurahalli lake and Sheelvanthakere, at about 800 m from Nallurahalli lake. The Sheelvanthakere joins the main stream of Varthur lake at its downstream near Varthur circle.

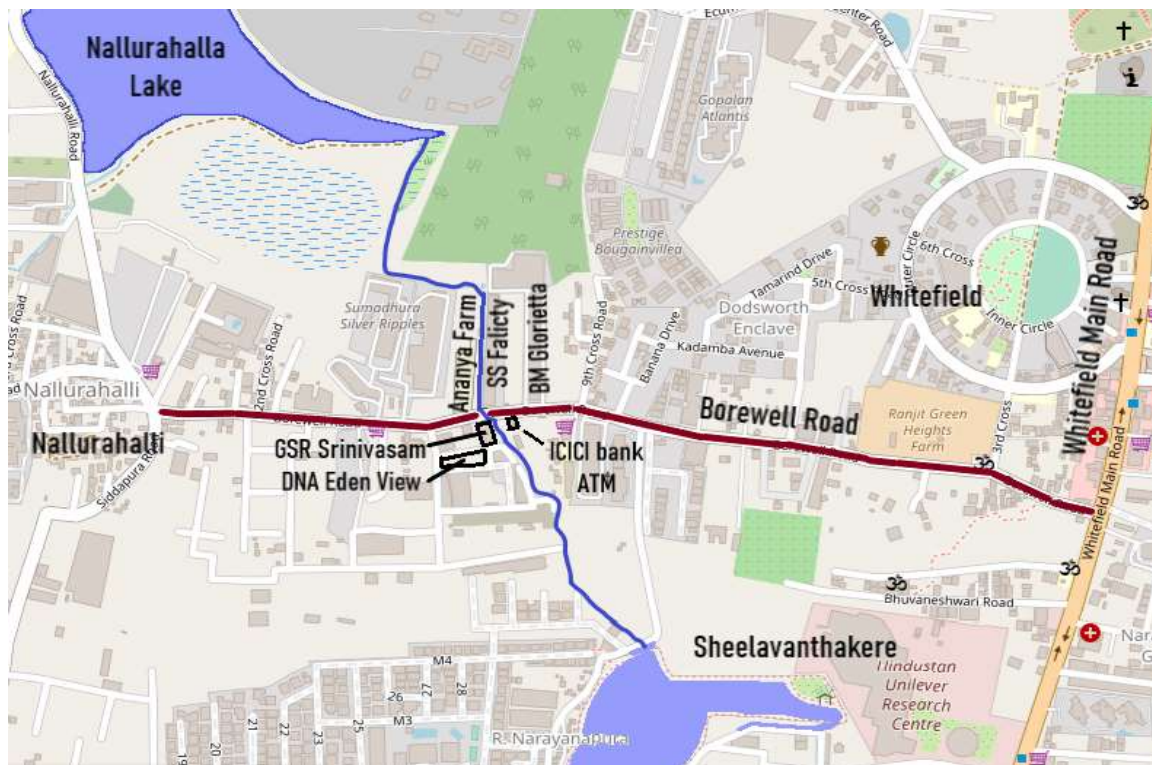


Figure 10.2: The map showing the apartment buildings on Borewell road which were severely affected during the flood of September, 2022.



Figure 10.3: The Ananya farm and SS Facility Apartment on the upstream of the Borewell road.



Figure 10.4: The close view of the drain between Ananya farm and SS Facility Apartment. It is connected to a culvert on Borewell road. The culvert runs at an angle and joins the drain at an open land between GSR Srinivasam and Malibu Paloma Apartments.



Figure 10.5: The view of the open land between GSR Srinivasam and Malibu Paloma Apartments.



Figure 10.6: A close view of the drain between GSR Srinivasam and Malibu Paloma Apartments. It is covered with stone slabs. It is infact a sewage drain. The drain at the upstream of Borewell road and particularly the culvert has the capacity just adequate to discharge it. Therefore, the discharge from Nallurahalli lake spills over the road. It then runs on the open land between GSR Srinivasam and Malibu Paloma Apartments.



Figure 10.7: The Google image of the drain at the reach near Borewell road in March 2021. The flow from Nallurahalli lake was freely discharged in the Ananya farm and the open land between GSR Srinivasam and Malibu Paloma Apartments and then through a narrow space between the compound walls at the end of the open land.



Figure 10.8: A close view of the drain between GSR Srinivasam and Malibu Paloma Apartments in March 2021. The discharge from the lake and runoff from the surrounding locality was collecting at the open land and then negotiating the narrow space between the compound walls at its end. It was freely discharged at the reach.



Figure 10.9: The Google image of the drain at the reach near Borewell road in December 2021. Certain labour sheds were built on the open land DNA and Malibu Paloma Apartments.



Figure 10.10: A close view of the labour sheds on the open land between DNA and Malibu Paloma Apartments. The labour sheds are constructed right at the mouth of the narrow space between the compound walls. It totally obstructed the discharge of Nallurahalli lake and runoff from the surrounding locality at the reach.



Figure 10.11: The spill over from the Nallurahalli lake which was freely discharging during 2021 is now obstructed by the construction of the labour sheds at the very mouth of the narrow space between the compound walls of DNA and Malibu Paloma Apartments. It is held back and impounded at the upstream of the sheds.

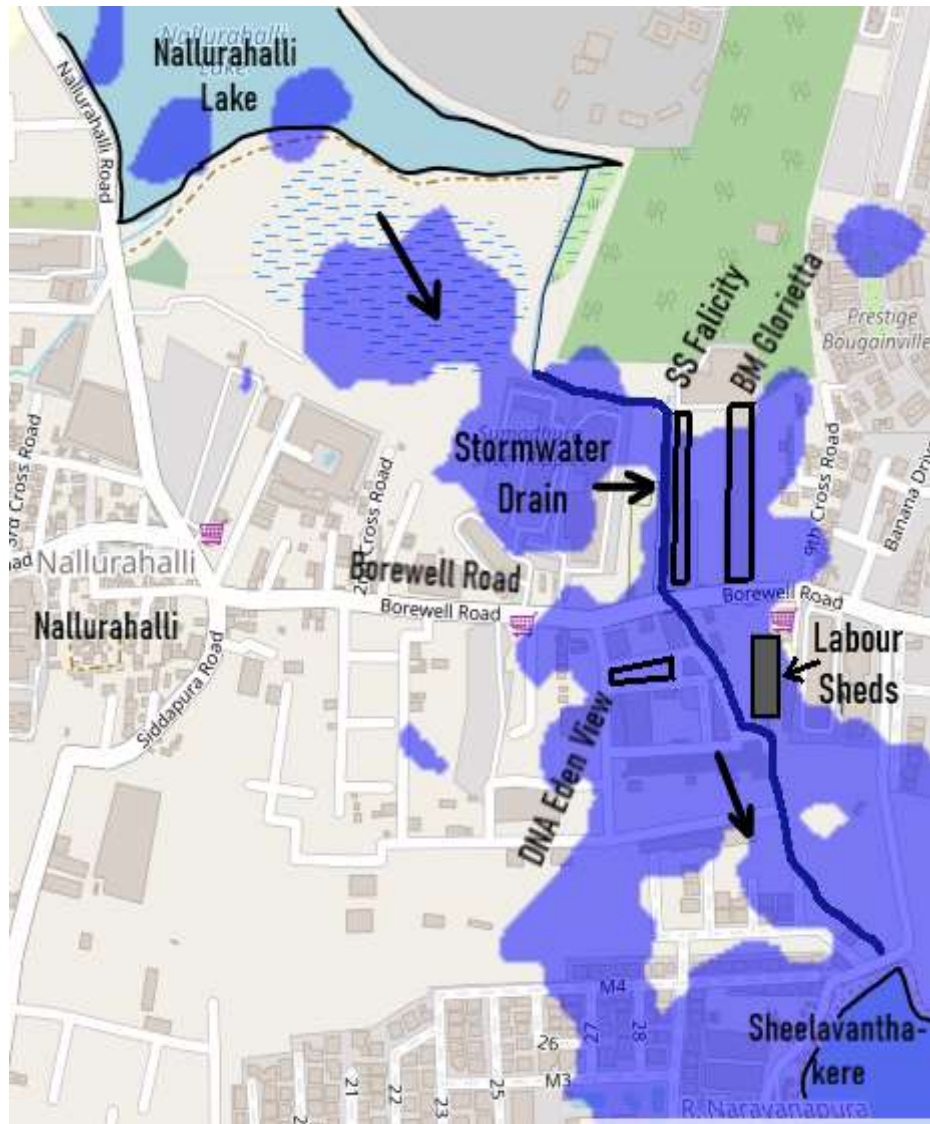


Figure 10.12: The inundation map at the Borewell road



Figure 10.13: The view of the waterlogging at the basement of BM Glorietta Apartment on 7th September, 2022. It is just the held-up water by the construction of labour sheds on the open land between GSR Srinivasam and Malibu Paloma Apartments stretched upto the apartment.



Figure 10.14: The view of DNA Eden View Apartments during peak flood. It is also the same obstructed water at the open land between GSR Srinivasam and Malibu Paloma Apartments stretched upto it. (Photo courtesy: The Hindu)

Chapter 11

Decoding of Waterlogging at Panathur-Balagere Road

11.1 Waterlogging at Panathur-Balagere Road

The Panathur-Balagere road was another location at the city, where boats were deployed to evacuate the citizens trapped during the flood. The following article gives an account of the scenario at the location during the flood.

www.hindustantimes.com

Bengaluru rain: Boats & tractors rescue locals as cars float on flooded roads

Published on Sep 05, 2022

Reported by Pathi Venkata Thadhagath | Edited by Sohini Goswami

Chief minister Basavaraj Bommai has instructed officials to speed up the relief works in flood affected areas.



Some inflatable boats were deployed in the Varthur area, while many such boats were spotted near Sekhar Bellevue apartment in Balagere-Panthur area. (Screengrab from Twitter video)

Several parts of Bengaluru are inundated with boats being taken out to rescue locals stuck in their houses since the overnight downpour. Tractors are also being sent to bring people out of their houses.

Some inflatable boats were deployed in the Varthur area, while many such boats were spotted near Sekhar Bellevue apartment in Balagere-Panthur area. According to reports, some locals are buying small boats for the temporary commute.

Heavy rain & poor infrastructure turns Bengaluru roads into lakes. In Sarjapur Road, with vehicles of many residents submerged, tractors were hired as the mode of transport to reach workplaces. Many gated communities in the area are under water.

Chief minister Basavaraj Bommai has instructed officials to speed up the relief works in flood affected areas. "There has been a lot of rain in Bengaluru. I have spoken to the BBMP commissioner. The flooding has happened in the same places as before and these include Bommanahalli and Mahadevapura. Nearly 30 places have been impacted. I have asked that two SDRF (State Disaster Relief Force) teams be sent to Mahadevapura and Bommanahalli," Bommai said.

The Sampangiramanagar area in Bengaluru reported a rainfall of 148mm on Sunday night, according to IMD data. Areas like Mahadevapura, Marathahalli, Doddenakundi, Varthur and HAL road in the city reported more than 100mm rainfall during the same period.

11.2 Decoding of waterlogging at Panathur-Balagere Road

The waterlogging at the Panathur-Balagere road could be analyzed as below,

- i. Figure 11.1 shows the location map of the Panathur-Balagere road and the reach near Disha Central Park on it, which was severely affected during September, 2022 flood. The road runs parallel to the main stream of Bellandur lake and the severely affected reach is located at the upstream of the newly constructed road near BWSSB Sewage Treatment Plant near Varthur lake.
- ii. Figure 11.2 shows the catchment area of the main stream of Bellandur lake at the newly constructed road near the BWSSB Sewage Treatment Plant. It is about 209 sqkm or about **28% of the total area of the city**. The discharge from Bellandur and a large number of lakes at its downstream drain at the road as shown in Figure 11.3.
- iii. Figure 11.4 shows the Google image of the stream at the upstream of Varthur lake near BWSSB Sewage Treatment Plant in April 2015. It was entirely a wet land at the reach. The sewage treatment plant was constructed in the above wet land in the year 2014-15.
- iv. A new road was proposed on the upstream of Varthur lake near the Sewage Treatment Plant during the year 2017-18, connecting Thubarahalli Ext. road to Balagere.
- v. Figure 11.6 shows the Google image of the road in March 2021. The construction of the bridge was just commenced. The stream was still **discharged freely** at the reach.
- vi. Figure 11.7 shows the Google image of the road in March 2022. The construction of the bridge was completed during the year.

- vii. Figure 11.8 shows the view of the bridge. The runoff from about 28% of the total area of the city is directed at the bridge, besides sewage from Bellandur lake. But, it appears that it is designed just for discharge of sewage. Its vent openings are too small and they get choked up by the weed during flood. The flow is invariably obstructed.
- viii. A **GI sheet barricade** is put up at the upstream edge of the road all along its length as shown in Figure 11.9. The free discharge above the road is too prevented by it. As a result, the discharge from Bellandur and a large number of lakes at its downstream is held back and impounded at the upstream of the road.
- ix. Figure 11.10 shows the inundation map at the upstream of the road. The Balagere-Panathur road and the apartments in its vicinity are inundated in the above obstructed water.

Thus, what we were seeing at the Balagere-Panthur road during the flood was the held-up water at the newly constructed road near BWSSB Sewage Treatment Plant. We could observe that there would not have been any waterlogging at the reach, if a large bridge was constructed on the new road and an adequate waterway was provided for it for discharge of stormwater.

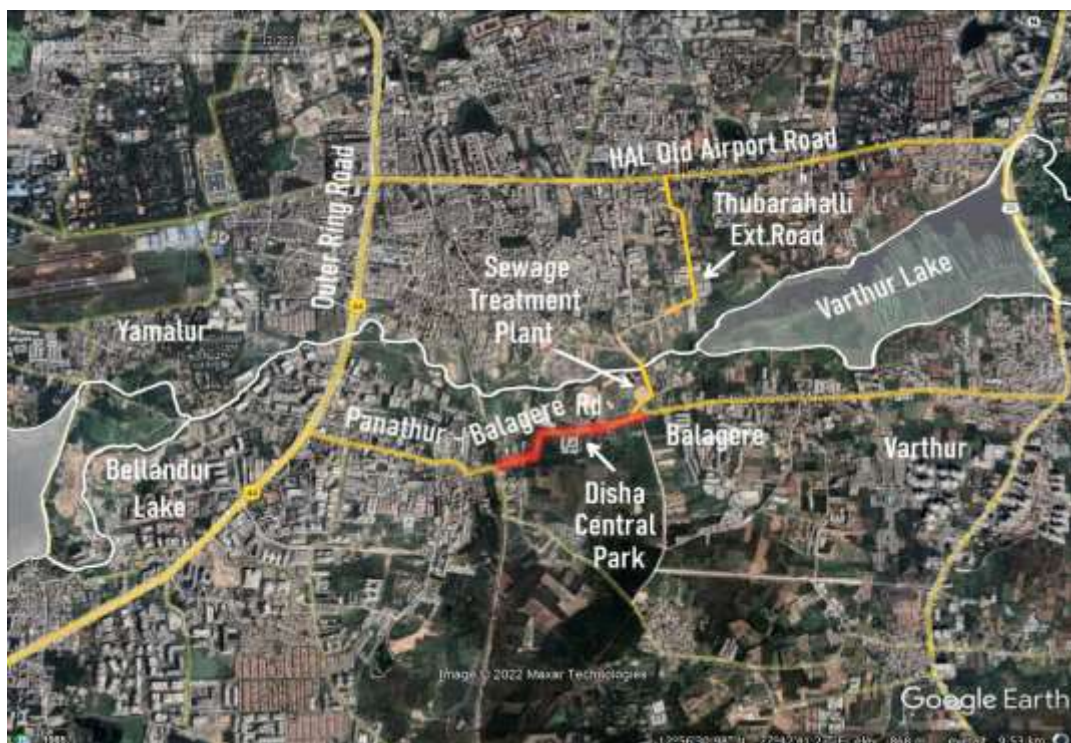


Figure 11.1: The location map of Panathur-Balagere road and the reach near Disha Central Park, which was severely affected during September, 2022 flood. The road runs parallel to the main stream of Bellandur lake and the severely affected reach is situated at the upstream of the newly constructed road near BWSSB Sewage Treatment Plant near Balagere.

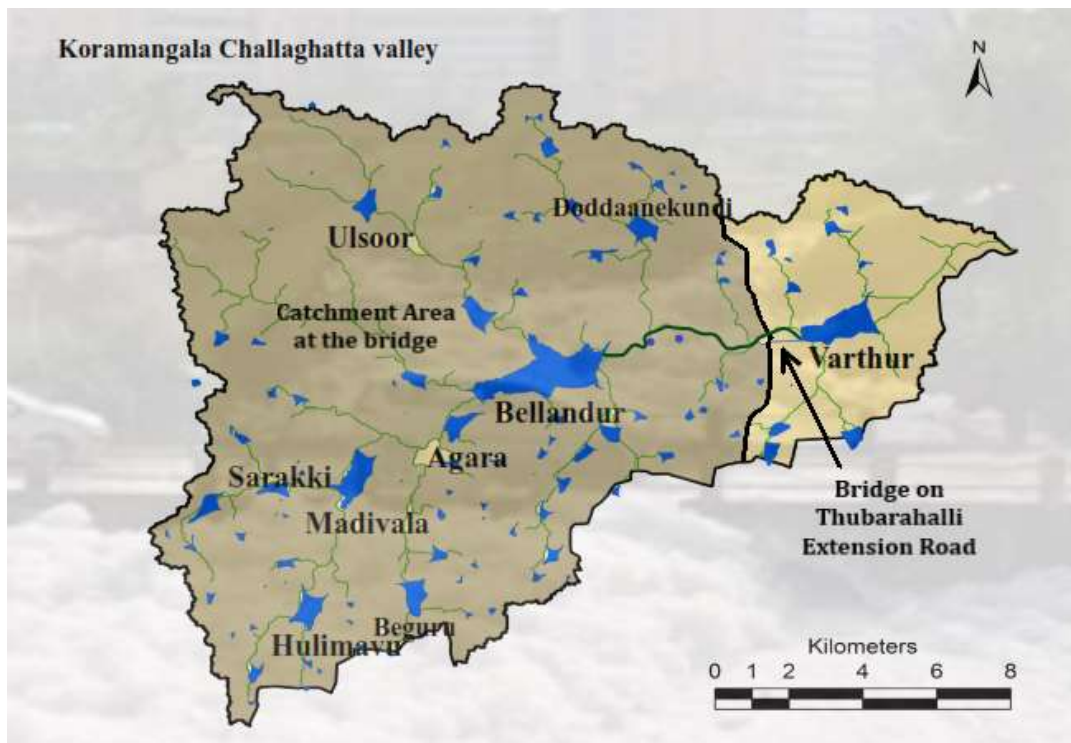
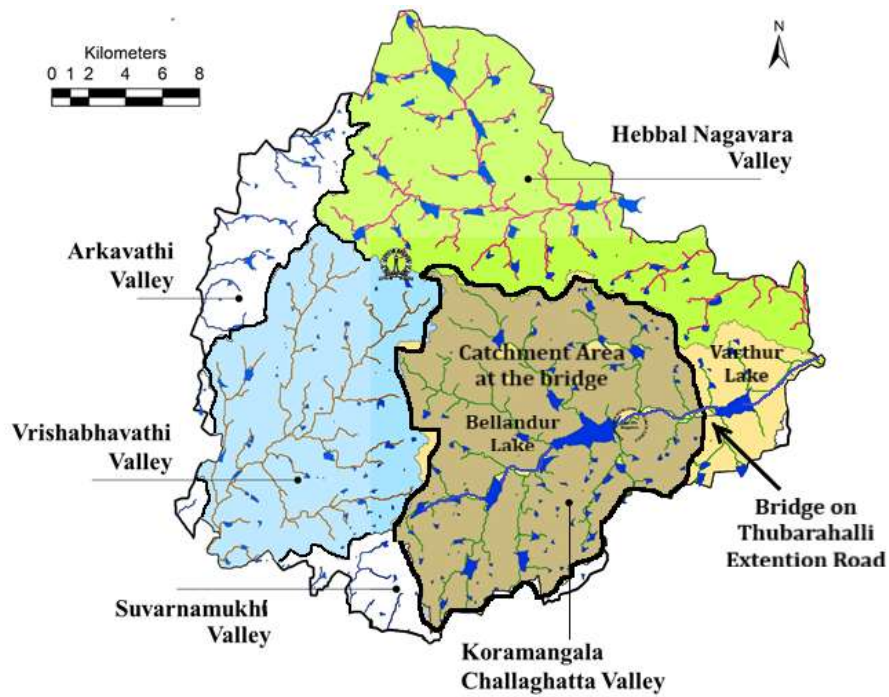


Figure 11.2: The catchment area of the main stream of Bellandur lake at the newly constructed road near the BWSSB Sewage Treatment Plant. It is about 209 sqkm or about 28% of the total area of the city.

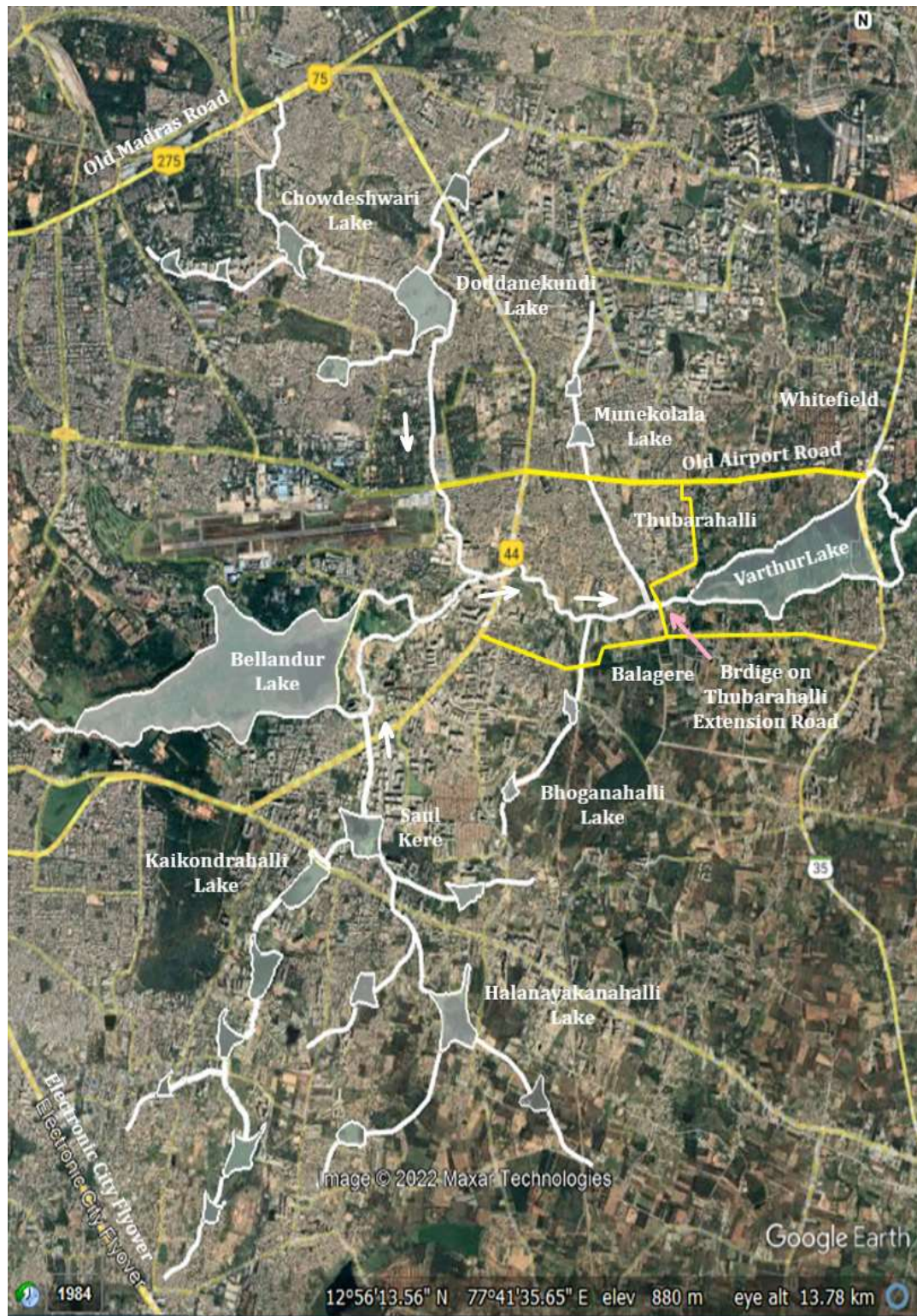


Figure 11.3: The view of the catchment area and lakes at immediate upstream of the newly constructed road near BWSSB Sewage Treatment Plant.



Figure 11.4: The Google image of the main stream of Bellandur lake at immediate upstream of Varthur lake in April 2015. It was completely a wet land. The sewage treatment plant was constructed in the above wet land, in the year 2014-15.



Figure 11.5: The Google image of the wet land in January 2018. The new road was proposed at the reach near the Sewage Treatment Plant connecting Thubarahalli Ext. and Panathur-Balagere roads.



Figure 11.6: The Google image of the newly constructed road near BWSSB Sewage Treatment Plant in March 2021. The construction of the bridge on the road was just commenced. The spillover from Bellandur lake and the runoff from its downstream catchment was still discharged freely at the reach.



Figure 11.7: The Google image of the road in March 2022. The construction of the bridge was completed.



Figure 11.8: The view of the bridge on the newly constructed road near BWSSB Sewage Treatment Plant. The runoff from about 28% of the total area of the city is directed at the bridge, besides sewage from Bellandur lake. But, it appears that it is just designed for discharge of sewage water. Its vent openings are very small and they get choked up by the weed during flood. The flow is invariably obstructed and held back.



Figure 11.9. The view of the newly constructed road near the bridge. A GI sheet barricade is put up all along its upstream edge. The free discharge over it is too obstructed by the barricade.

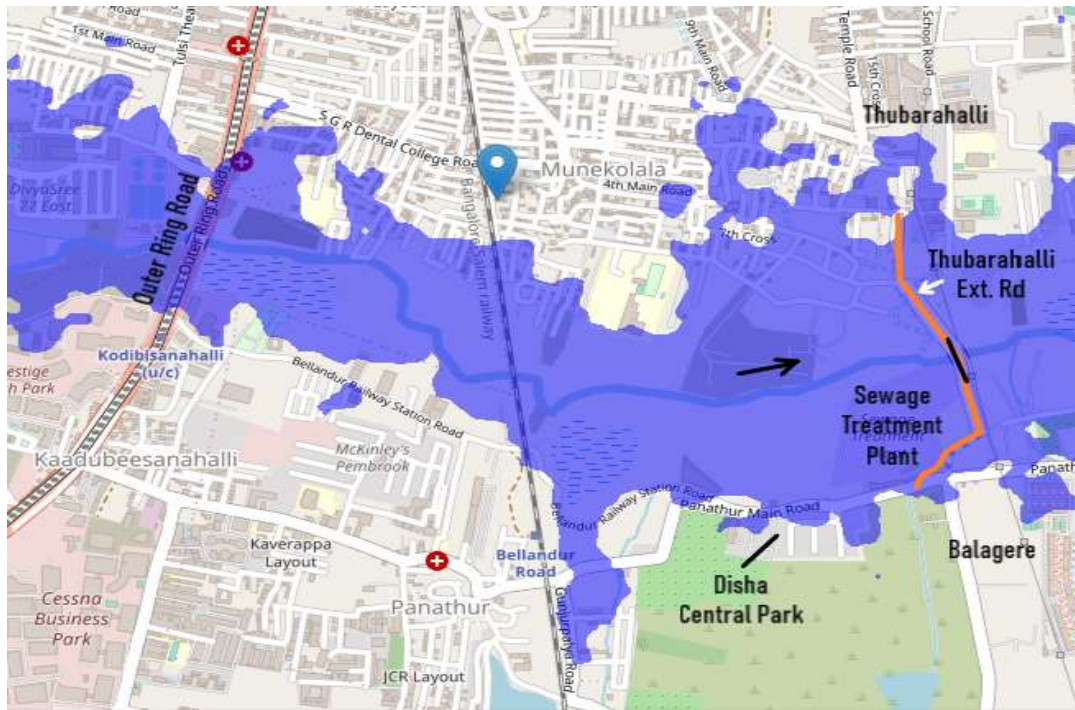


Figure 11.10: The inundation map at the upstream of the newly constructed road near BWSSB Sewage Treatment Plant. The obstructed discharge at the road is impounded at the valley on its upstream upto Outer Ring Road in Kadubeesanahalli. The Balagere-Panathur road and the apartments in its vicinity are inundated in the above water.



Figure 11.11: A 3D Google image of the valley at the upstream of the newly constructed road near BWSSB Sewage Treatment Plant, showing the extent of waterlogging at the valley.



Figure 11.12: A view of the waterlogging at Balagere-Panathur road near Disha Central Park on September 6, 2022. It is not the stormwater drained from its vicinity. It is actually the obstructed water at the newly constructed road near sewage treatment plant on the upstream of Varthur lake, impounded upto it. We could observe that there would not have been any waterlogging at the reach if there was a free discharge at the bridge on the newly constructed road. (Photo courtesy: Twitter)

Chapter 12

Stormwater Management in Silicon Valley of United States – A Case Study

An attempt is made in this chapter to understand the stormwater management at the Silicon Valley of United States.

12.1 The Silicon Valley

Silicon Valley is a region in Northern California that serves as a global center for high technology and innovation. Located in the southern part of the San Francisco Bay Area, it corresponds roughly to the geographical areas San Mateo County, Santa Clara County and Alameda County.

San Jose is Silicon Valley's largest city, the third-largest in California, and the tenth-largest in the United States; other major Silicon Valley cities include Sunnyvale, Santa Clara, Redwood City, Mountain View, Palo Alto, Menlo Park, Cupertino and Fremont. The San Jose Metropolitan Area has the third-highest GDP per capita in the world (after Zurich, Switzerland and Oslo, Norway), according to the Brookings Institution, and, as of June 2021, has the highest percentage of homes valued at \$1 million or more in the United States. (Ref: wikipedia.org)



USA on World map and California State
(Source: WorldAtlas.com)



California State and its major cities
(Source: www.reddit.com)



The Silicon Valley and city of San Jose
(<http://maps-san-francisco.com/silicon-valley-on-world-map>)

Figure 12.1: The location map of Silicon Valley and city of San Jose

As more high-tech companies were established across San Jose and the Santa Clara Valley, and then north towards the Bay Area's two other major cities, San Francisco and Oakland, the term "Silicon Valley" came to have two definitions: a narrower geographic one, referring to Santa Clara County and southeastern San Mateo County, and a metonymical definition referring to high-tech businesses in the entire Bay Area. The term Silicon Valley is often used as a synecdoche for the American high-technology economic sector. The name also became a global synonym for leading high-tech research and enterprises, and thus inspired similarly named locations, as well as research parks and technology centers with comparable structures all around the world. Many headquarters of tech companies in Silicon Valley have become hotspots for tourism.

12.2 Santa Clara Basin/Valley

The Santa Clara Basin is defined as the portion of the Bay south of Dumbarton Bridge and the 840-squaremile area of wetlands and uplands that drains to it. The basin is located at the southern end of the San Francisco Bay Area as shown in Figure 12.2. It is bounded by Dumbarton Bridge to the north, the crest of the Diablo Mountains to the east, and the crest of the Santa Cruz Mountains to the west and south.

The Basin includes about one-half of Santa Clara County and smaller portions of San Mateo and Alameda counties. Twenty cities lie within the Basin in whole or in part. The boundaries of the cities and counties in the Basin are shown in Figure 12.2. A number of special districts exist within the Basin. They include the Santa Clara Valley Water District (Water District), which is the primary water wholesaler in the Basin and is also responsible for flood protection.



Figure 12.2: Regional Location of Santa Clara Valley

12.3 Santa Clara Valley Water District

Santa Clara Valley Water District	
Formed	1968
Preceding agencies	Santa Clara Valley Water Conservation District Santa Clara County Flood Control and Water Conservation District
Headquarters	<u>San Jose, California</u>
Website	<u>www.valleywater.org</u>

The **Santa Clara Valley Water District** (also known as **Valley Water**) provides stream stewardship, wholesale water supply and flood protection for Santa Clara County, California, in the southern San Francisco Bay Area. The district encompasses all of the county's 1,300 square miles (3,400 km²) and serves the area's 15 cities, 2 million residents and more than 200,000 commuters. In terms of acres, the district includes 138,000 acres, and 120,700 of those acres are lands that people have built cities, roads or cultivate farms on. Almost 2,000 pumping wells supply the districts fields, houses and businesses with a clean reliable source of water. The water district has about 150 miles of pipelines, operates 10 dams and reservoirs, three treatment plants, many groundwater recharge basins, three pump stations and an advanced water purification plant.

Watersheds of Santa Clara Valley: There are 15 watersheds in the Santa Clara water district.

Watersheds Within the Santa Clara Basin ¹	
Watershed	Area (square miles)
Coyote Creek	321
Guadalupe River	170
Arroyo la Laguna	74
San Tomas Aquino/Saratoga Creeks	45
San Francisquito Creek	43
Baylands	33
Stevens Creek	29
Lower Penitencia Creek	29
Calabazas Creek	21
Permanente Creek	17
Matadero/Barron Creeks	17
Adobe Creek	11
Sunnyvale West Channel	8
Sunnyvale East Channel	7

¹ Eleven of the watersheds lie wholly within Santa Clara County. The Arroyo la Laguna and San Francisquito Creek watersheds lie primarily within Alameda and San Mateo counties, respectively. Watershed boundaries and areas were delineated by EOA, Inc.

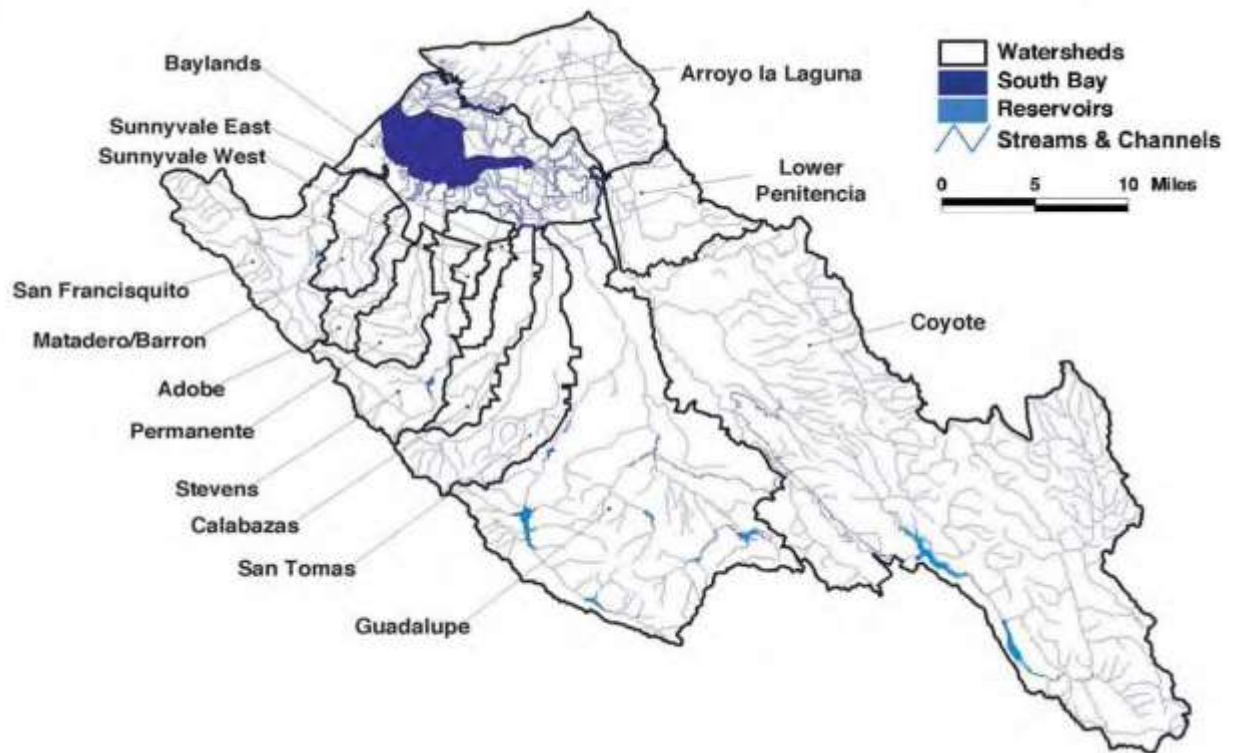


Figure 12.3: The watersheds of Santa Clara Basin

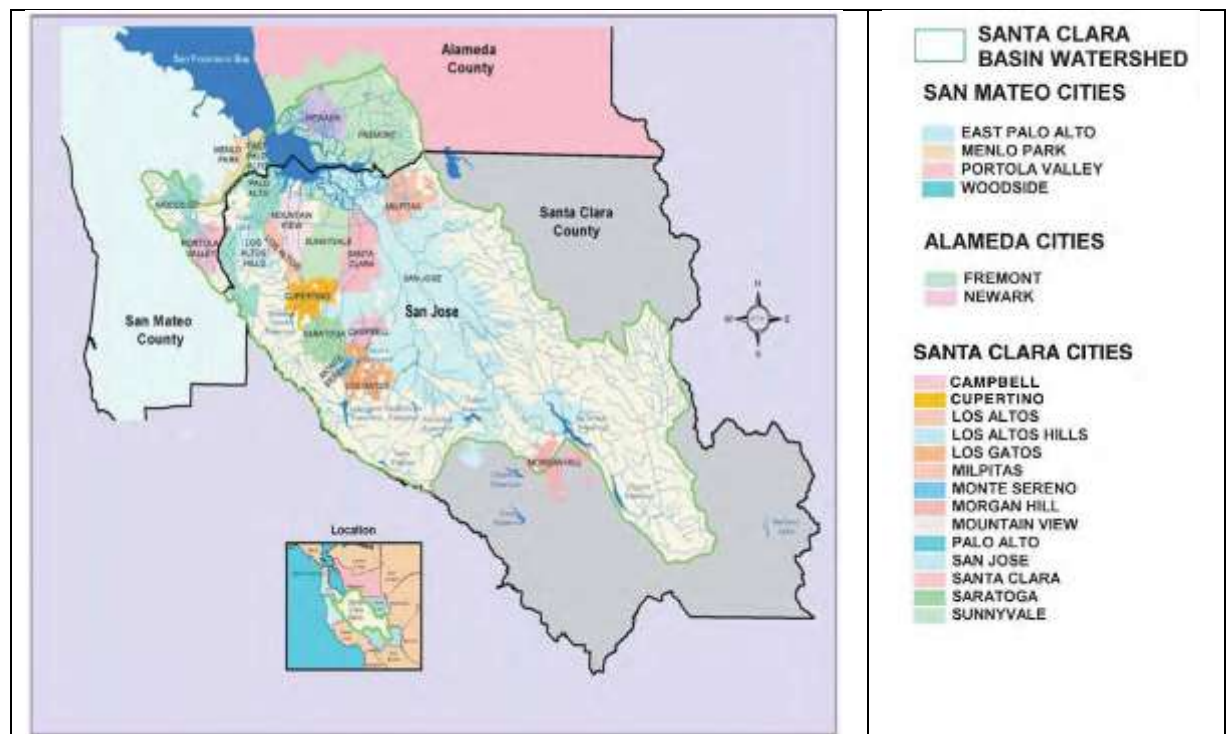


Figure 12.4: Counties and Municipalities in the Santa Clara Basin

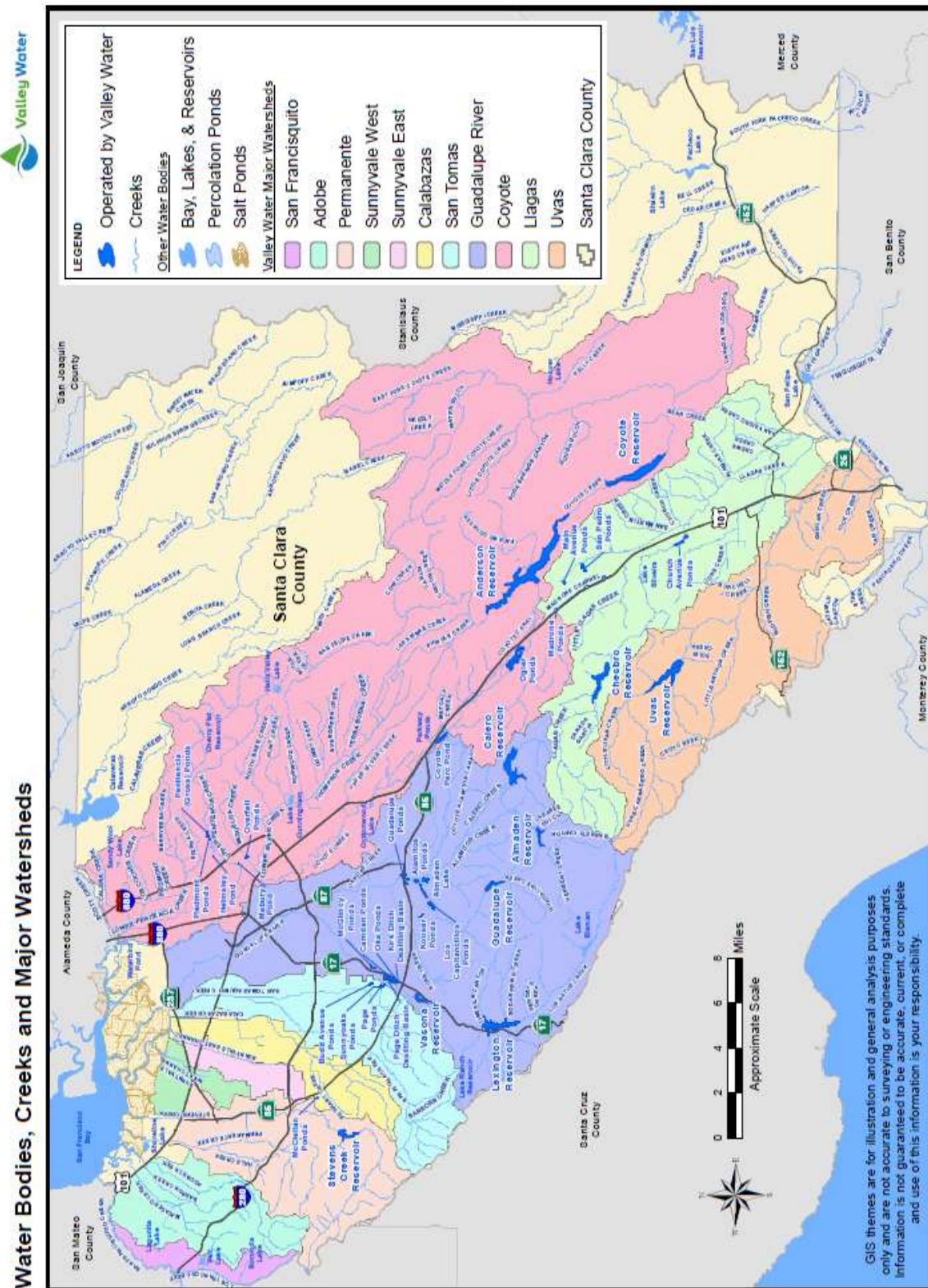


Figure 12.5: Water Bodies, Creeks and Major Water Watersheds of Santa Clara Basin

12.4 San Jose – GIS Plan

San José has a jurisdictional area of 180 square miles, and is located in the Santa Clara Basin, a sub-watershed of the larger San Francisco Bay watershed (the Bay) and a part of Santa Clara County, as shown in Figure 12.4. The Santa Clara Basin is bounded by the Diablo Mountains to the east and the Santa Cruz Mountains to the south and west. The City's six major watersheds (Coyote, Guadalupe, Lower Penitencia, San Tomas Aquino, Calabazas, and Baylands), all drain to southern San Francisco Bay. Within these watersheds, approximately 32,000 storm drain inlets receive runoff from surrounding urban areas into the storm drain system where it is carried and discharged to a creek, river, or the Bay.

The Coyote Creek watershed is the largest in the Santa Clara Basin watershed, covering 320 square miles, and includes two dammed reservoirs. The Guadalupe River watershed is approximately 171 square miles and discharges to the Lower South San Francisco Bay via Alviso Slough. Lower Penitencia Creek watershed, part of the Coyote Creek watershed, covers about 30 square miles, San Tomas Aquino Creek watershed encompasses about 45 square miles, and Calabazas Creek watershed covers about 20 square miles.

Federal & State Regulations & Initiatives: The U.S. Environmental Protection Agency (EPA) has authority under the Clean Water Act to promulgate and enforce stormwater related regulations. For the State of California, the EPA has delegated the regulatory authority to the State Water Resources Control Board (State Water Board), which, in turn, has delegated authority to the San Francisco Bay Regional Water Quality Control Board (Regional Water Board) to issue National Pollutant Discharge Elimination System (NPDES) permits in the San Francisco Bay Region. Stormwater NPDES permits allow stormwater discharges from municipal separate storm sewer systems (MS4s) to local creeks, San Francisco Bay, and other water bodies as long as they do not adversely affect the beneficial uses of or exceed any applicable water quality standards for those waters. Since the early 2000s, the EPA has recognized and promoted the benefits of using GSI in protecting drinking water supplies and public health, mitigating overflows from storm sewers, and reducing stormwater pollution. The EPA has encouraged the use of GSI by municipal agencies as a prominent component of their MS4 programs.

12.5 Stormwater & Wastewater Management at San Jose city

The website of the City of San Jose gives complete details of the Stormwater and Wastewater management at the city.

<https://www.sanjoseca.gov/your-government/environment/water-utilities/stormwater>

The City of San José is the lead agency for implementing stormwater and pretreatment management programs on behalf of the eight tributary jurisdictions whose sanitary sewer systems discharge to the San José-Santa Clara Regional Wastewater Facility. Stormwater and wastewater systems are separate systems with different functions.

STORMWATER

The **storm sewer system** refers to important infrastructure that prevents flooding when it rains. The system, also known as a Municipal Separate Storm

Sewer System or MS4, works to move water away from an area to a local water body, such as a creek or river. The storm sewer system includes the inlets (storm drains) and gutters on your street as well as pipes and outfalls, where the stormwater enters the water body. Because the MS4 flows directly into water bodies without any treatment it is critical to keep it pollutant free.

Stormwater Permits and Regulation

The work the City of San José does to prevent stormwater pollution is regulated under the jurisdiction of the San Francisco Bay Regional Water Quality Control Board, Region 2 (Regional Board), the enforcement arm of the State Water Resources Control Board (Water Board).

The Regional Board issues a National Pollutant Discharge Elimination (NPDES) Municipal Regional Permit (MRP) to the City of San José and 75 other co-permittees that covers stormwater activities for most of the Bay Area. Permittees in Santa Clara County, such as the City of San José and the Santa Clara Valley Water District, formed an association called the Santa Clara Valley Urban Runoff Pollution Prevention Program (SCVURPPP) to meet MRP permit regulations by sharing resources and collaborating on projects of mutual benefit. The MRP permit governs a variety of activities in the City of San José such as industrial and commercial businesses, new and redevelopment projects, construction sites, storm drain operation and maintenance, creek monitoring, pesticide applications, and illegal dumping of water and other pollution in the City's storm drain.

WASTEWATER

The sanitary sewer system refers to the important infrastructure that transports wastewater to the treatment facility. This system contains wastewater that requires treatment before it can be discharged. The sanitary sewer system includes the piping and pump stations that move the wastewater from households and businesses to the treatment facility.

Wastewater Regulations and Permits

The national pretreatment program, a component of the NPDES program, is a cooperative effort of federal, state, and local environmental regulatory agencies established to protect water quality. The EPA and authorized NPDES state pretreatment programs approve local municipalities to perform permitting, administrative, and enforcement tasks for discharges into the municipalities' publicly owned treatment works (POTWs).

The City of San José is responsible for administering pretreatment programs for eight tributary agencies: the cities of San José, Santa Clara and Milpitas; Cupertino Sanitary District; West Valley Sanitation District including Campbell, Los Gatos, Monte Sereno and Saratoga; County Sanitation Districts 2-3; and Burbank Sanitary District whose sanitary sewer systems discharge to the San José-Santa Clara Regional Wastewater Facility.

As the control authority, the City of San José regulates businesses including commercial and industrial facilities, construction activities, food service establishments, mobile businesses, septic haulers, and dental offices.

12.6 Creek and Watershed Maps

<https://www.sanjoseca.gov/your-government/environment/our-creeks-rivers-bay/watershed-maps>

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WATERSHED MAPS

WHAT'S A WATERSHED?

A watershed is a land area that drains water into a stream, lake, wetland, bay or estuary, and an area that percolates into groundwater. Watersheds come in different shapes and sizes. Local watersheds are parts of larger, regional basins. The Santa Clara Basin, for example, is a sub-basin of the larger San Francisco Bay Basin.

A watershed begins at the top of a hill or mountain ridge and is ultimately defined by gravity. The Santa Clara Basin is bounded by the Diablo Mountains to the east and the Santa Cruz Mountains to the south and west.

For more information about watersheds, including an interactive history map, visit [Valley Water's](#) watershed site.

WHICH WATERSHED ARE YOU LOCATED IN?

Storm drains collect runoff from both overflowing irrigation water and rain from streets and parking lots that flows to an underground storm drain system. There are approximately 35,000 inlets in the City of San Jose, and each flow to a creek or river, which then flows to the San Francisco Bay.

Find out which watershed you live in and where your inlets flow by viewing the maps linked below. You can also read the stencil on the sidewalk above or beside the inlet near your home or business that says "NO DUMPING", the name of the waterbody that the inlet drains to, and the stormwater hotline phone number, (408) 545-3000. Residents or visitors can call this hotline or [use this complaint form](#) to report illegal dumping if they have witnessed anything other than rain water entering an inlet or its surrounding area.

- **All Watersheds in San Jose**
- **Baylands Watershed:** Inlets at the northern tip of the City will be marked as *Flows to Bay*.
- **Calabazas Watershed:** Inlets will be marked as *Flows to Bay*.
- **Coyote Watershed:** Inlets will be marked as one of three subwatersheds:
 - *Flows to Coyote Creek*
 - *Flows to Silver Creek*

<http://explore.museumca.org/creeks/terms.html>

Creek & Watershed: Terms & Definitions

Base level	An elevation in a stream channel that is fixed over the long term by being relatively unerodable, such as a dam, or not subject to erosion, such as sea level.
Creek	A naturally flowing stream that is usually considered smaller than and often tributary to a river.
Floodplain	Flat lowland area that borders a stream and is covered by its water when the stream overflows. The floodplain is typically built up by alluvial sediments deposited by the stream.
Slough	Dead-end channels of slow-flowing or stagnant water in low marshy ground.
Storm drain	The opening into a culvert system that carries storm-water to a discharge point in a stream or ocean.
Tributary	A stream or river that flows into another.
Watershed	The entire area of land that drains into a particular channel or water body.
Wetlands	Areas such as swamps, marshes and bogs that contain permanent or temporary standing water.



Figure 12.6: Coyote Creek aerial view from the south in San Jose, from south of Hellyer County Park up to Interstate 280 (Ref: [https://www.wikiwand.com/en/Coyote_Creek_\(Santa_Clara_County\)](https://www.wikiwand.com/en/Coyote_Creek_(Santa_Clara_County)))



Figure 12.7: Confluence of Coyote Creek and San Gabriel River, Long Beach, California, on Approach to Long Beach Airport (Ref: [https://commons.wikimedia.org/wiki/File:Confluence_of_Coyote_Creek_and_San_Gabriel_River,_Long_Beach,_California,_on_Approach_to_Long_Beach_Airport_\(6013277245\).jpg](https://commons.wikimedia.org/wiki/File:Confluence_of_Coyote_Creek_and_San_Gabriel_River,_Long_Beach,_California,_on_Approach_to_Long_Beach_Airport_(6013277245).jpg))

12.7 Stormwater Master Planning

The Appendix-5-B of Virginia Stormwater Management Handbook gives an exhaustive description of general stormwater Management at the cities of US. It is reproduced below.

Virginia Stormwater Management Handbook, Chapter 5 July 2013

Appendix 5-B

Watershed-Based Stormwater Planning

5-B.1.0 STORMWATER MASTER PLANNING

5-B.1.1 Introduction

Stormwater master planning is an important tool with which communities can assess and prioritize both existing and potential future stormwater problems, as well as use to consider alternative stormwater management solutions. A stormwater master plan is prepared to consider, in detail, what stormwater management practices and measures are to be provided for an urban drainage area or a large development project.

Stormwater master plans are most often used to address specific single functions such as drainage provision, flood mitigation, cost/benefit analysis, or risk assessment. These plans prescribe specific management alternatives and practices. Multi-objective stormwater master planning broadens this traditional approach to potentially include land use planning and zoning, water quality, habitat, recreation, and aesthetic considerations. The broadest application of stormwater master planning is the comprehensive watershed plan which is described in detail in this Appendix.

....

5-B.1.2 Watershed Planning Flexibilities

Although site-by-site compliance with stormwater management requirements is much better than no stormwater management at all, evidence from across the nation indicates that individual controls on stormwater discharges are inadequate as the sole solution for stormwater in urban watersheds. Ideally, BMP implementation needs to be designed as a system, integrating structural and nonstructural BMPs and incorporating watershed goals, site characteristics, development land use, construction erosion and sediment controls, aesthetics, monitoring and maintenance. Stormwater cannot be adequately managed on a piecemeal basis due to the complexity of both the hydrologic and pollutant processes and their effect on habitat and stream quality.

Section 9 VAC 25-870-96 of the regulations allows local governments to develop comprehensive watershed-based stormwater management plans as an alternative way to comply with the water quality requirements, the water quantity requirements, or both. State and federal agencies intending to develop large tracts of land also may develop or participate in comprehensive watershed stormwater management plans where practicable. Section 9 VAC 25-870-76 also allows linear development projects, such as streets and highways, to achieve

compliance in accordance with such a watershed plan, as an alternative to strict on-site compliance.

Those who develop such plans must demonstrate to DEQ and the State Water Control Board (Board) that the results of implementing the plan will be at least as good as, if not better than, those that would be achieved from straightforward implementation of the regulation requirements on a site-by-site basis. The Board must approve local watershed plans before they may be implemented.

5-B.2.2 Scale of Watershed Management

Watersheds are typically defined according to the resource area or downstream water body of interest. Although there are no maximum size limits for defining a watershed, a manageable watershed for local planning efforts is usually no greater than 100,000 acres (~150 square miles). It is important to remember that larger watershed boundaries require the involvement of more jurisdictions and stakeholders.

Ideally, planning take place at both the watershed and smaller “subwatershed” scales. Typically, the broad, “big picture” planning takes place at the watershed level, and the more refined objectives and implementation plans are pursued at a subwatershed level (see **Table 5-B.1** and **Figure 5-B.1** below). Finally, individual projects and controls are carried out at the project or catchment level.

Table 5-B.1. Description of New USGS Hydrologic Mapping System Units

Order	No. Digits in New ID#	New Name	Unit Size	Sample Management Measures
1	2	Region	Avg. 177,560 sq. mi.	
2	4	Subregion	Avg. 16,800 sq. mi.	
3	6	(River) Basin	Avg. 10,596 sq. mi.	
4	8	Sub-basin	Avg. 703 sq. mi.	Basin-wide planning
5	10	Watershed ¹	Range: 40,000 acres (62.5 sq. mi.) to 250,000 acres (390+ sq. mi.)	Basin-wide planning combined with watershed-based development standards
6	12	Subwatershed	Range: 10,000 acres (15+ sq. mi.) to 40,000 acres (62.5 sq. mi.)	Watershed-based development standards combined with stream classification and management
		Catchment ²	Avg. Range: 5 - 15 sq. mi.	Stream classification and management
		Subcatchment ²	Avg. Range: 0 - 5 sq. mi.	Site design measures and structural controls
NOTES: ¹ 100,000 acres or 150 sq. mi. may be the upper practical local planning limit ² These terms are not part of the new USGS classification system, but they may help local planners develop more discrete planning units				

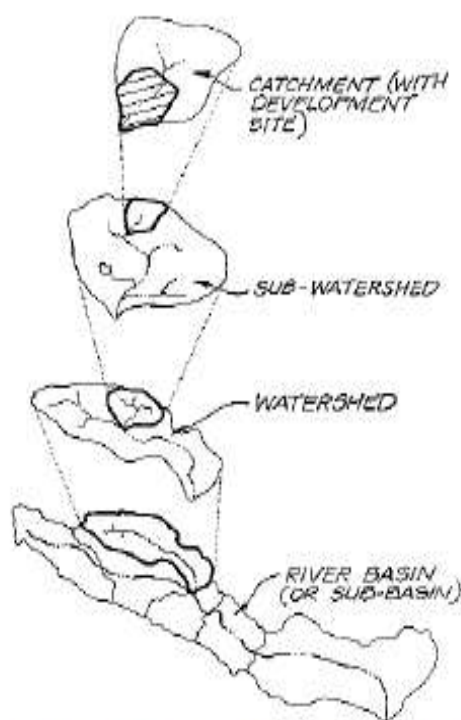


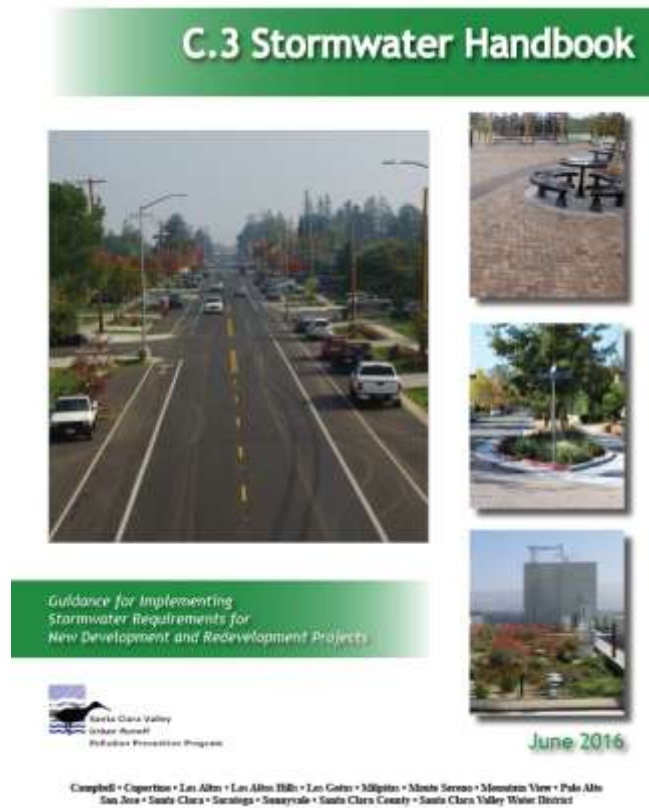
Figure 5-B.1. Watershed Management Units

The local program must document nutrient reductions achieved during the plan's implementation, in order to demonstrate the actual equivalence of compliance results. If the percent of impervious area upon which the plan was based changes or if any other amendments are deemed necessary by the local program, the local program must provide plan amendments to the Soil and Water Conservation Board for review and approval. For example, if the plan's target total nutrient removal for the watershed is based on an expected build-out resulting in a composite 53 percent impervious cover, and subsequently the locality approves comprehensive plan and zoning changes that will result in a composite 65 percent imperviousness at build-out, then the plan's original targets will no longer achieve results equivalent to those required in the regulations. The locality would need to amend the plan to achieve equivalence and submit the amendments to the Board for review and approval.

Section 9 VAC 25-870-63 of the regulations allows watershed plans to allow for compliance offsets (off-site mitigation, compliance trading, or fee-in-lieu options), where compliance is not feasible or cost-effective on the development site due to physical constraints, etc. In such cases, the chosen offset measure must ensure that the resulting stormwater control is equal to or greater than what would be required on each contributing land disturbing site. In fact, since the watershed planning process accounts for ultimate pollutant load reductions, such plans provide the best opportunity to optimize the most cost-effective strategy and mix of practices to achieve compliance. The regulations require that offsets must be achieved within the same Hydrologic Unit Code (HUC) watershed, or within HUCs established by the locality for this purpose. *Watershed plans also provide the best opportunity for communities to achieve an effective approach to encouraging and stimulating redevelopment and infill development and discourage continued sprawl into outlying areas.*

12.8 Stormwater Handbook

Another important aspect of the stormwater management at Santa Clara basin is a detailed reference Stormwater Handbook. It gives guidance for developers, builders and project applicants for implementing stormwater regulations in the development projects, at the valley. Its Chapter 1 which gives a synopsis of the handbook is reproduced below.



Chapter 1

Introduction and Guide to Using this Handbook

This Chapter describes the purpose of this handbook and gives an overview of its contents.

1.1 Purpose of this Handbook

This handbook was written to help developers, builders, and project applicants include appropriate post-construction stormwater controls in their projects, to meet local municipal requirements and requirements of the Bay Area Municipal Regional Stormwater Permit (MRP). Municipalities covered by the MRP must require postconstruction stormwater controls on development projects as part of their obligations under Provision C.3 of the MRP. This permit is a National Pollutant Discharge Elimination System (NPDES) permit issued by the San Francisco Bay Regional Water Quality Control Board (Regional Water Board), allowing municipal stormwater systems to discharge stormwater to local creeks, San Francisco Bay, and other water bodies if municipalities conduct prescribed

actions to control pollutants. In case of conflicting information between this handbook and the MRP, the MRP requirements prevail.

The term “post-construction stormwater control” refers to permanent features included in a development project to reduce pollutants in stormwater and/or erosive flows during the life of the project – after construction is completed. The term “post-construction stormwater control” encompasses low-impact development (LID) site design, source control, and treatment measures as well as hydromodification management measures. LID techniques reduce water quality impacts by preserving and recreating natural landscape features, minimizing imperviousness, maximizing opportunities for infiltration and evapotranspiration, and using stormwater as a resource construction is available on the Urban Runoff Program’s website www.scvurppp.org as well as the California Stormwater Quality Association’s website www.casqa.org.

Post-construction stormwater controls are required for both private and public projects. Although this handbook is written primarily for sponsors of private development projects, its technical guidance also applies to publicly-sponsored projects. Municipalities may also find the handbook useful for training municipal staff and consulting plan checkers.

1.2 Background on the Urban Runoff Program

The Santa Clara Valley Urban Runoff Pollution Prevention Program (Urban Runoff Program) is an association of thirteen cities and towns in Santa Clara Valley, the County of Santa Clara, and the Santa Clara Valley Water District that share a common NPDES permit to discharge stormwater to South San Francisco Bay. Member Agencies (also called Co-permittees) include Campbell, Cupertino, Los Altos, Los Altos Hills, Los Gatos, Milpitas, Monte Sereno, Mountain View, Palo Alto, San Jose, Santa Clara, Saratoga, Sunnyvale, the County of Santa Clara, and the Santa Clara Valley Water District.

The Urban Runoff Program’s member agencies, and other agencies throughout the region, are joint holders of the MRP. Each member agency is individually responsible for implementing the MRP requirements, but participating in the Urban Runoff Program helps it collaborate on Program-wide initiatives that benefit all members. More information on the Urban Runoff Program is available on its website, www.scvurppp.org.

1.3 How to Use this Handbook

When using this Program-wide handbook, please keep in mind that some requirements may vary from one local jurisdiction to the next. In the very early stages of project planning, contact the municipal planning staff to learn how the C.3 requirements – and other planning, zoning and building requirements – will apply to your project. Also, because regulatory requirements may change, be sure to ask the local municipal staff to provide any updates of information or requirements.

It’s important to note that post-construction stormwater design requirements are complex and technical: most projects will require the assistance of a qualified civil engineer, architect, landscape architect, and/or geotechnical engineer.

A synopsis of the handbook's chapters and appendices is provided below:

Chapters

- Chapter 2 explains how development affects stormwater quality and how postconstruction stormwater measures help reduce these impacts, and gives a detailed explanation of Provision C.3 requirements.
- Chapter 3 gives an overview of how the post-construction stormwater requirements fit into a typical development review process, and offers step-by step instructions on how to incorporate stormwater control/LID techniques into planning permit and building permit application submittals for your project.
- Chapter 4 presents information on site design measures, including guidance for self-treating and self-retaining areas, which can help reduce the requirements for stormwater treatment measures.
- Chapter 5 provides general technical guidance for stormwater treatment measures, including hydraulic sizing criteria, getting runoff into stormwater treatment measures, infiltration guidelines, underdrains, bypassing high flows, using "treatment trains", mosquito control, and plant selection and maintenance.
- Chapter 6 gives technical guidance for specific types of stormwater treatment measures, including bioretention area, flow-through planter, tree well filter, infiltration trench, subsurface infiltration system, rainwater harvesting and use, media filter, extended detention basin, green roof, and pervious paving.
- Chapter 7 explains the requirements for hydromodification management measures, which keep the flow rates, volumes, and durations of post-project stormwater flows at pre-project levels, in order to minimize development-induced erosion in susceptible creek channels.
- Chapter 8 explains the operation and maintenance requirements for stormwater treatment measures.
- Chapter 9 describes the MRP's Provision C.3.e., which allows projects to construct or contribute to off-site alternative compliance projects instead of constructing on-site stormwater treatment measures.

Appendices

- Appendix A presents guidelines for using stormwater controls that promote onsite infiltration of stormwater, and includes a depth to groundwater map.
- Appendix B provides worksheets for hydraulic sizing of stormwater treatment measures, including a map of Mean Annual Precipitation and hydraulic sizing curves for treatment measures, as well as examples showing how to use the worksheets and sizing approaches.
- Appendix C provides regional Biotreatment Soil Specifications for use in stormwater biotreatment measures.
- Appendix D includes a list of plants appropriate for use in landscape-based treatment measures. It also offers general guidance on plant selection and maintenance.

Chapter 13

Regulatory Setting

The main aspect of the stormwater management at Silicon Valley is that it is backed by specific Acts, Ordinances and Guidelines. The following are the three main among them; they are reproduced below.

- i. Cobey-Alquist Floodplain Management Act
- ii. Santa Clara County Floodplain Ordinance, and
- iii. Policy and Guidelines for Dedication and Reservation of lands for Flood Control Purposes

13.1 Cobey-Alquist Floodplain Management Act

(Ref: <https://www.fao.org/faolex/results/details/en/c/LEX-FAOC156031/> retrieved on 01.10.2022)

This Chapter of Division 5 concerns flood plain management and in particular the establishment of a local regulatory framework for areas designated by the Department of Water Resources or the State Reclamation Board as “designated floodway”. Within any area classified “designated floodway” the appropriate public agency shall establish the necessary flood plain regulations after completion of water projects. The Department or Board shall review the adequacy of such regulations. If a public agency fails to adopt flood plain regulations, they may be adopted by the local flood control agency having jurisdiction over the project area. The Department and the Board, within their respective jurisdictions shall establish general policy and may prescribe such rules and regulations as may be necessary to carry out this Chapter

WATER CODE - WAT

DIVISION 5. FLOOD CONTROL [8000 - 9651]

(Division 5 added by Stats. 1943, Ch. 369.)

PART 2. STATE FLOOD CONTROL [8300 - 8457]

(Part 2 added by Stats. 1943, Ch. 369.)

CHAPTER 4. Cobey-Alquist Flood Plain Management Act [8400 - 8415]

(Chapter 4 added by Stats. 1965, Ch. 506.)

ARTICLE 1. Declaration of Policy [8400 - 8401]

(Article 1 added by Stats. 1965, Ch. 506.)

8400.

This chapter may be cited as the Cobey-Alquist Flood Plain Management Act.
(Added by Stats. 1965, Ch. 506.)

8401.

The Legislature finds and declares that:

- (a) A large portion of the state’s land resources is subject to recurrent flooding by overflow of streams and watercourses causing loss

of life and property, disruption of commerce, and interruption of transportation and communications, all of which are detrimental to the health, safety, welfare, and property of the people of this state.

- (b) The public interest necessitates sound development of land use, as land is a limited, valuable, and irreplaceable resource, and the flood plains of this state are a land resource to be developed in a manner which, in conjunction with economically justified structural measures for flood control, will result in (1) prevention of loss of life, and (2) prevention of economic loss caused by excessive flooding.
- (c) The primary responsibility for planning, adoption, and enforcement of land use regulations to accomplish flood plain management rests with local levels of government.
- (d) It is the policy of this state to encourage local levels of government to plan land use regulations to accomplish flood plain management and to provide state assistance and guidance therefor as appropriate.

(Added by Stats. 1965, Ch. 506.)

ARTICLE 2. Definitions [8402- 8402.]

(Article 2 added by Stats. 1965, Ch. 506.)

8402.

Unless the context otherwise requires, the following definitions apply throughout this chapter:

- (a) "Department" means Department of Water Resources.
- (b) "Board" means the State Reclamation Board.
- (c) Wherever the words "department or board" or "department and board" are used together in this chapter they shall mean board as to any area within a project which by law is under the jurisdiction of the State Reclamation Board, and the department in all areas of the state outside these areas.
- (d) "Public agency" means any city, city and county, county, or district organized, existing, and acting pursuant to the laws of this state.
- (e) "Design flood" means the selected flood against which protection is provided, or eventually will be provided, by means of flood protective or control works. When a federal survey has been authorized the design flood will be determined by the appropriate federal agency and in all other cases it will be determined by the responsible local agency. It is the basis for design and operation of a particular project after full consideration of flood characteristics, frequencies, and potentials and economic and other practical considerations.
- (f) "Designated floodway" means the channel of a stream and that portion of the adjoining flood plain required to reasonably provide for the construction of a project for passage of the design flood including the lands necessary for construction of project levees.

- (g) “Restrictive zone” means the portion of the natural floodway between the limits of the designated floodway and the limits of the flood plain where inundation may occur but where depths and velocities are generally low.

(Added by Stats. 1965, Ch. 506.)

ARTICLE 3. Review of Flood Plain Management Plans [8403 - 8404]

(Article 3 added by Stats. 1965, Ch. 506.)

8403.

Prior to the establishment by a public agency of flood plain management regulations, the public agency may request and receive a review by the department or the board of the public agency’s flood plain management plans and shall furnish with the request necessary hydrologic studies, including flood routings or surveys adequate to properly delineate flood plain categories together with such additional supporting data as the department or the board may require.

(Added by Stats. 1965, Ch. 506.)

8404.

Where adequate personnel are not available to a public agency to prepare the necessary flood routing or other engineering data needed for support of its flood plain management plans, the department or board may, upon request of the public agency, make or cause to be made the necessary investigation and provide the data to the requesting public agency for its use in developing such plans. The cost of such investigations shall be borne by the public agency making the request.

(Added by Stats. 1965, Ch. 506.)

ARTICLE 4. Establishment of Flood Plain Regulations [8410 - 8414.2]

(Article 4 added by Stats. 1965, Ch. 506.)

8410.

Flood plain regulations established to comply with requirements of Section 8411 shall meet the following criteria:

- (a) Construction of structures in the designated floodway which may endanger life or significantly restrict the carrying capacity of the designated floodway shall be prohibited. For the purpose of this subdivision, the word “structures” does not include public utility electric, gas, or communication lines which may be located within the designated floodway; provided, that any permit or permits as may be required by law, other than this chapter, to so locate such lines have been granted.
- (b) Developments shall be permitted within the restrictive zone in accordance with local agency policy, giving full consideration to the protection of human life and the carrying capacity of the flood plain.

(Added by Stats. 1965, Ch. 506.)

8410.5.

All public agencies shall comply with flood plain regulations established to meet the requirements of Section 8411. For the purposes of this section, "public agencies" includes the state or any department or agency thereof.

(Added by Stats. 1971, Ch. 410.)

8411.

Within any area classified "designated floodway" by the department or board after completion of a federal project report the appropriate public agency shall establish the necessary flood plain regulations within one year following notification by the department or board. The state shall not pay any of the cost of lands, easements, and rights-of-way associated with the flood control project as set forth in the project report of the federal agency unless flood plain regulations for the designated floodway are adopted in accordance with the requirements of this chapter.

(Amended by Stats. 1972, Ch. 1304, Sec. 1.)

8412.

The department or board shall review the adequacy of flood plain regulations established by a public agency to meet the requirements of Section 8411, and shall notify the appropriate public agency in the event the flood plain regulations are determined by the department or board to be inadequate to comply with the requirements of Section 8411. The department or board may recommend any provisions necessary to provide adequate flood plain regulations. Any such public agency shall within 180 days of receipt of the notice from the department or board, adopt new flood plain regulations. Upon their adoption a copy of such flood plain regulations shall be sent to the department or board.

(Added by Stats. 1972, Ch. 1304.)

8413.

Flood plain regulations adopted pursuant to Section 8412 shall take effect 60 days from the date of adoption by the public agency and shall be deemed to meet the requirements of Section 8411 unless the department or board determines that such flood plain regulations are not adequate to comply with the requirements of Section 8411. If the department or board makes such a determination, it shall so report to the public agency and shall recommend such regulations or modifications of the regulations of the agency, which it determines are necessary.

(Added by Stats. 1972, Ch. 1304.)

8414.

If a public agency fails to adopt flood plain regulations pursuant to Section 8411, or fails to adopt new flood plain regulations within 180 days of receipt of the department's or board's notice pursuant to Section 8412, or fails to adopt or modify such regulations in the manner determined as necessary by the department or board pursuant to Section 8413 within 90 days of receipt of the department's or board's notice pursuant to Section 8413, the local flood control agency having jurisdiction over the project area may adopt flood plain regulations for the designated floodway necessary to comply with the requirements of Section 8411. Such flood plain regulations adopted by the local flood control agency shall take effect 30 days from the date of their adoption and shall be enforced by the public agency and have the same force and effect as if adopted as flood plain regulations by the public agency.

(Added by Stats. 1972, Ch. 1304.)

8414.2.

No public agency shall revise flood plain regulations established to meet the requirements of Section 8411 or shall grant a variance from such regulations without the consent of the department or board and of the local flood control agency having jurisdiction over the project area.

(Added by Stats. 1972, Ch. 1304.)

ARTICLE 5. Administrative Regulations [8415- 8415.]

(Article 5 added by Stats. 1965, Ch. 506.)

8415.

The department and the board, within their respective jurisdictions as provided in this chapter, shall establish general policy and may prescribe such rules and regulations as may be necessary to carry out this chapter.

(Added by Stats. 1965, Ch. 506.)

13.2 Santa Clara County Floodplain Ordinance

(Ref: <https://plandev.sccgov.org/ordinances-codes/floodplain-ordinance#1849274314-1383086610> retrieved on 18.09.2022)

County of Santa Clara

Department of Planning and Development

Santa Clara County Floodplain Ordinance

01 – Floodplain Development

What Is a Floodplain?

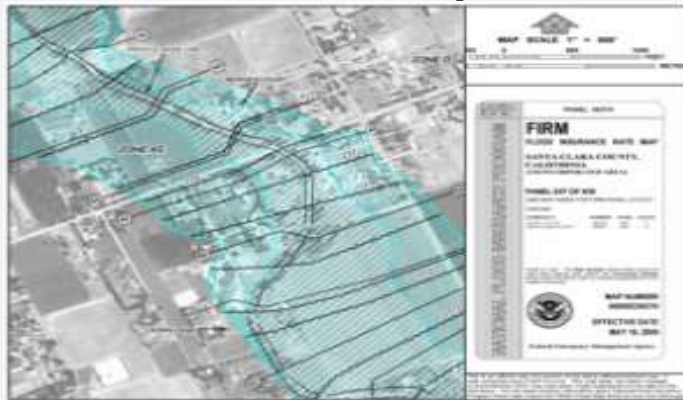
Floodplains are low-lying lands adjacent to waterways that are naturally subject to periodic flooding. Special development standards apply to those areas that have been designated as the “100-year floodplain.”

What Is the “100-Year Flood”?

Floodplain development regulations are based on the 100-year flood, which is more accurately described as the flood that has a 1% probability of being equaled or exceeded in any year. Engineering models are used to determine both the area subject to inundation and the height of floodwaters during this model flood event. The area that would be flooded is called the “Special Flood Hazard Area” and is subject to floodplain development regulations. The water height is called the “Base Flood Elevation” and is the basis for development standards within the regulated floodplain area.

The 100-year flood, or base flood, has a 1% probability of being equaled or exceeded in any given year. In some places, this flood has never occurred in recorded history. Other areas have experienced more than one “100-year flood” in a single year.

Where Do I Find a Flood Hazard Map?



Regulated Special Flood Hazard Areas are delineated on Flood Insurance Rate Maps (FIRMs) issued by the Federal Emergency Management Agency (FEMA). In some municipalities additional maps (Flood Boundary and Floodway Maps) and/or a Flood Insurance Study provide additional information. These maps and studies are available from the County Building Inspection Office. The FIRM maps can also be viewed at [FEMA's online Map Service Center](#). The "Make a FIRMette" feature enables you to save or print a selected portion of a map along with identifying information.

What Can I Learn from a Flood Hazard Map?

Areas subject to floodplain development regulations are shaded areas labeled with the letter "A," sometimes followed by a number or letter (or "V" in coastal areas). If Base Flood Elevations have been determined, they will be indicated on the map and in a Flood Insurance Study. "Floodway" areas may also be shown, either on the FIRM or on a separate Flood Boundary and Floodway Map. More stringent development controls apply within the regulatory floodway portion of the floodplain. The 500-year floodplain may be shown as a lightly shaded area labeled Zone B or Zone X. Although most municipalities do not impose special development standards within the 500-year floodplain, the flood hazard in these areas should be considered, particularly with regard to the location of critical facilities or storage of hazardous materials.

What Activities Are Regulated?

All development within the areas mapped as the 100-year floodplain (or Special Flood Hazard Area) is subject to floodplain development regulations. "Development" is defined as "any man-made change to improved or unimproved real estate" and includes (but is not limited to):

- the construction, reconstruction, renovation, repair, expansion, or alteration of buildings, accessory structures, or other structures;
- the placement of manufactured homes;
- mining;
- dredging;
- bridges, streets, and other paving;
- docks;
- utilities;
- filling, grading, and excavation;
- drilling operations;
- storage of equipment or materials; and
- installation of manufactured home, recreational vehicle, or trailer parks.

How Do I Get a Permit to develop in a floodplain?

Development in Special Flood Hazard Areas is regulated by the Santa Clara County Department of Planning and Development, which is responsible for issuing permits. In most cases, a Building or Grading Permit is required for all development in a floodplain.

What Are the Requirements for Floodplain Projects?

Minimum standards for floodplain development have been established by the federal government as a condition for the sale of federally subsidized flood insurance and eligibility for federal flood recovery assistance. Some states and communities have chosen to provide an extra margin of safety by exceeding the minimum federal standards.

The specific requirements for a floodplain development project depend on the flood zone and the type of development. The basic standards that must be met by any floodplain development are:

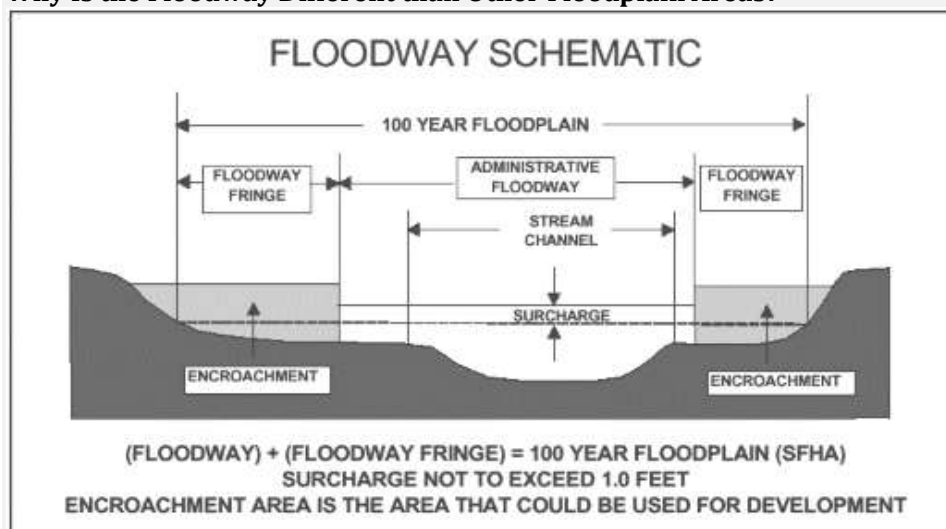
Proposed development must be **reasonably safe from flood damage**, which for most buildings means elevated above the height of floodwaters.

Proposed development **shall not result in physical damage to any other property**. Technical analysis may be required if there is potential for increased flood heights, diversion of flow, flotation of materials, or release of pollutants.

Applicant shall **obtain any other required permits**. Note that obtaining state or federal permits does not mean that floodplain requirements have been met – a project in the floodplain also requires a local floodplain development permit.

12 – Floodway Encroachments

Why Is the Floodway Different than Other Floodplain Areas?



A basic principle of floodplain management is that **development must not increase the flood hazard on other properties**. "Floodways" are areas where fill or other development is likely to divert flow and contribute to increased water

depths during a flood. Floodways may also be subject to high velocities, which can cause severe damage to structures and high risks for occupants and emergency responders. **Ideally, floodways should be undeveloped areas that can accommodate flood flows with minimal risk.** Any new development in the floodway generally requires an engineering analysis of the impact on flood hazards.

The floodway is the channel of a river or stream and the overbank areas that must remain open to carry the deeper, faster moving water during a flood. If the remainder of the floodplain, called the floodplain fringe, is completely obstructed, the 100-year flood elevation would not increase more than one foot. The regulatory floodway may be shown on the Flood Insurance Rate Map or on a separate Flood Boundary and Floodway Map. Because floodway boundaries are delineated using computer modeling, they often do not correspond to any features visible at the site.

What Is Meant by Encroachment?

An “encroachment” is any floodplain development that could obstruct flood flows, such as fill, a bridge, or a building. A driveway, road, or parking lot at grade (without any filling) would not cause an obstruction. Development of lakeshore floodplains, where there is no flow, is not considered an encroachment.

How Are Floodplain Encroachments Regulated?

The development standards for a floodplain encroachment depend on both the project location and the amount of information provided on flood hazard maps:

Floodplain fringe: The modeling used to establish floodway boundaries indicated that any encroachment or obstruction in the fringe area (outside the floodway) would not result in a “significant” increase in flood levels (i.e. no more than one foot), so no encroachment analysis is required.

Floodway: No new development is permitted within the regulatory floodway unless a licensed professional engineer demonstrates that the proposed encroachment shall not result in **any** rise in the 100-year flood elevation. This no-rise requirement is in addition to all other floodplain development standards applicable to the proposed project. A variance is a request to vary from the rules, not to ignore them. Any variance should allow only minimum deviation from the local requirements. The approved project should provide as much flood protection as is possible. Because the community should not be exposed to higher flood risks, federal standards prohibit variances from the requirements concerning floodway encroachments.

Riverine floodplain with base flood elevations, but no floodway: When the flood hazard map designates base flood elevations (100-year flood heights) but no floodway is delineated, the applicant must demonstrate that the cumulative effect of the proposed development, when combined with all other existing and anticipated floodplain development, would not increase the water surface elevation of the 100-year flood more than one foot at any location.

Approximate A Zones: When floodplain boundaries were established using approximate techniques (which produce neither floodways nor base flood elevations), the municipality may require an analysis to demonstrate that the project would not result in physical damage to any other property.

Are There Exceptions to the Encroachment Requirements?

Federal standards do not allow communities to issue variances for development within the floodway that would result in increased flood levels. However, there are some situations (such as dams, bridges, or roads) in which a project in the floodway may be justifiable even though it would cause a rise in the flood elevation. This necessitates that the flood hazard map be changed to reflect the new hazard. The applicant must apply to the Federal Emergency Management Agency (FEMA) for (1) a conditional map revision before the development occurs and (2) a final letter of map revision after the development has been completed.

No-Rise Certification for Floodway Encroachments

Any proposed encroachment in the floodway requires a technical evaluation by a licensed professional engineer to demonstrate that the project will not affect flood heights. The results of this analysis must be maintained in the municipality's permit file. This can be in the form of a No-Rise Certification supported by technical data and signed by a registered professional engineer. The supporting technical data should be based on the standard step-backwater computer model used to develop the floodway shown on the flood hazard map. Hydraulic modeling of the pre-project and post-project conditions should demonstrate that the change in the 100-year flood height is 0.00 feet.

12.3 Policy and Guidelines for Dedication and Reservation of lands for Flood Control Purposes

POLICY AND GUIDELINES FOR DEDICATION AND RESERVATION OF LANDS NEEDED FOR FLOOD CONTROL PURPOSES

STAFF COMMITTEE REPORT TO THE JOINT COMMITTEE

Board of Supervisors - County of Santa Clara

Board of Directors - Santa Clara County Flood Control and Water District

Planning Commission - County of Santa Clara

Selby Brown, II Jr.

March 21, 1973

RECITALS

County Land Development Regulations authorize the requirement of dedications for flood control channel facilities.

Present regulations do not set forth all the necessary criteria.

It is important for developing properties to receive flood protection commensurate with land use.

Where a change of land use adds to the need for, and benefits by, a public flood control improvement, it is deemed equitable to secure some reasonable measure of contribution to the public cost of such improvement.

Guidelines need to be established for the general public and county agencies and advisory bodies involved in regulating land development.

STATEMENT OF PURPOSE AND OBJECTIVES

The guidelines contained herein are for the purpose of fulfilling certain responsibilities of local government in helping to provide flood protection to developing properties in Santa Clara County. The primary objective is control of recurrent flooding and overflow commensurate with land uses permitted under county zoning and land development regulations. Flood control measures, where necessary, shall be so planned as to preserve and enhance environmental values of the community to the greatest extent feasible taking into consideration the public health, safety and welfare. Private loss and inconvenience created may also be taken into consideration.

Appropriate dedication in each case is a function of land use where variables can be related in terms of intensity of development and benefits derived from flood protection.

DEDICATION POLICY

As a condition of approval of division or development of land, the necessary right of way for the designated floodway (planned width) established by the Santa Clara County Flood Control and Water District may be required by dedication, reservation, or combination thereof, with exceptions and limitations as set forth herein. Where the dedicated portion is less than the right of way required for the full designated floodway, the reserved area not dedicated shall be purchased by the district at fair market value. The one-half channel requirements described below apply to each side of the channel in the land division or development which is traversed by or abuts a natural watercourse or proposed diversion facility.

DEDICATION REQUIREMENTS

Dedication on each side of the channel shall consist of such land as is needed for one-half of the total flood control right of way width for a floodway designed on the basis of a 100 year (1%) flood. The width so designated shall be measured from the center line of the planned final channel improvement to the outside line of the designated floodway.

EXCEPTIONS AND LIMITATIONS

- A. "A" Zoning Districts (Exclusive Agriculture) Property zoned "A", whether or not subject to a land conservation agreement or contract - No dedication will be required where the affected parcel after division or development is 10 acres or more.

B. Minor Land Divisions and Single Family One Parcel Developments in AI and RI Zoning Districts

- (1) Dedication requirements shall not exceed 15% of the gross land area before division or development.
- (2) No dedication will be required where the affected parcel after division or development is 10 acres or more.

C. Flood Plain or Modified Floodplain

Design In floodplain or modified floodplain situations, or in the case of very wide natural channels, the dedication requirement shall not exceed the amount of real property needed for one half of the total right of way on each side for a trapezoidal earth channel with levees and/or parallel maintenance roads, if required.

D. Existing Watercourses having adequate capacity

In the case of an existing watercourse having reasonably narrow width, and sufficient depth and capacity without enlargement to accommodate local drainage and design flood, dedication shall consist of the area between the tops of natural banks that is needed for flood control purposes (including necessary access and maintenance roads), in which case the 15% maximum dedication provision shall be inapplicable. For watercourses located in mountainous or hilly terrain, dedication where required by the district shall consist of that area subject to inundation by the design flood flow (high water line) plus necessary maintenance roads and access.

E. Dedication on Both Sides of Channel

Where there is developable land within the proposed division or development on both sides of the flood control facility after all dedications, the dedication requirements specified above shall apply to each side. If there is not developable land within the proposed division or development on both sides of the channel, the dedication requirement shall apply to only one side plus that part of the natural channel on the undevelopable side of the centerline.

F. Diversion and Baylands Channels

Where a diversion or other entirely new flood control facility is planned, or where a channel is subject to backwater effect from the San Francisco Bay, the owner may be required to reserve such land as is required for flood control purposes. Dedications will not be necessary in such cases unless the flood control facility provides a drainage benefit to the owner's land reasonably commensurate with the value of any required dedication.

OTHER GUIDELINES

1. Acquisitions by the District

In situations where land has been reserved for flood control purposes as a condition of approval for division or development, the district shall purchase the reserved area as soon as possible. Such purchase may be deferred where mutually agreeable to the owner and the district.

2. Type of Dedication

Fee dedication may be required except in the following situations where the dedication shall be in easement for flood control purposes only, unless otherwise agreed to by owner. Minor land divisions and one parcel single family development where the channel is either (a) floodplain or modified floodplain, or (b) unimproved earth channel that is to remain in its natural condition, except cleaning and erosion control and/or containment by levees. In all situations where fencing is required for public safety, fee dedication may be required irrespective of the type of development.

3. Dedication Forms

Dedications shall be prepared by the district on its deed forms based on right of way calculations by the district and reasonable survey information furnished by the applicant.

If the dedication pertains to a single building site proposed for development with a land use intensity of duplex or less, the district shall furnish any necessary field survey information.

4. Offsetting Density

In situations where the dedication is in easement or fee in minor land divisions for residential or agricultural purposes, density shall be computed on the basis of net area prior to deduction of flood control right of way, providing, however: (a) All parcels shall meet applicable land development regulations, and (b) the final allowed density after dedication shall be no greater than the density that would have been allowed prior to the dedication.

5. Water Rights

Dedication requirements and acquisition procedures for flood control rights of way shall to the extent reasonably possible be administered to preserve existing compatible rights to water use upon and of benefit to adjacent lands.

6. Tax Considerations

The County Assessor shall be notified by the district regarding dedications for flood control purposes in order that they may be considered in property valuations for tax purposes.

7. Floodplain Management

The Santa Clara County Flood Control and Water District shall prepare maps for all designated floodways within Santa Clara County, record the same with the County Recorder for official and public notice, and supply the same in appropriate form and scale to the County of Santa Clara as soon as possible. The county shall cooperate with the district in sound floodplain management by adopting necessary regulations and control procedures.

8. Agency Jurisdiction

The district shall have jurisdiction and responsibility for designated floodways and the watercourses tributary thereto which have a drainage area greater than 640 acres. The terminal point of jurisdiction in each case (main watercourse or tributary greater than 640 acres) shall be at the point on that watercourse where the contributing drainage area is diminished to 320 acres; or other size area when agreed upon by the Department of Public Works and the district. Natural

watercourses whose total tributary area upstream from the point of confluence with the designated floodway is 640 acres or less, and all other unincorporated land areas, shall be considered local drainage where the county shall have jurisdiction and responsibility.

9. Development Plans and Information

All proposals which affect existing or planned flood control and water utility facilities of the district including use permits, minor land divisions, subdivisions, planned developments and single site developments, shall be referred to the district for review and preparation of recommended conditions of approval pursuant to these guidelines for consideration at the first scheduled committee or advisory agency meeting after the referral.

The applicant for land division or development shall furnish on the tentative maps and development plans (if any) reasonably accurate information regarding essential land topography, existing and proposed property lines, the existing center line and high bank of the watercourse, designated floodway lines where maps have been prepared by the district, and the type, size and location of existing culverts, bridges, structures and other divided improvements existing or proposed on the land to be/or developed.

10. Flood Control Plans and Information

The proposed land division or development plans shall conform to the designated 100 year (1%) floodway plan. Where development is proposed in advance of the floodway map preparation, the district shall develop the one percent floodway information essential to such proposals. The district shall furnish all necessary information relating to right of way widths, design criteria, and the type and location of interim and ultimate flood control improvements.

11. Flood Protection Improvements

In situations where property presently is subject to flooding, the district shall specify reasonable interim measures necessary to reduce potential flood damage until the full improvements by the district are completed. The district also may require the property owner to enter into a deferred improvement agreement with the district.

12. Channel Design

Designated floodway lines and the design of flood control channels and facilities shall be based upon a level of protection adequate for the one percent flood. In locations where underground storm drainage facilities could normally be expected in the course of development, the flood control channels or watercourses shall be planned for sufficient depth to facilitate installation of underground local drainage facilities in accordance with good design practice and constructed to this design as soon as practicable.

13. Channel Alignment

As far as practicable, rights of way are to be established in conformance with the alignment of existing natural channels; i.e., the center line of the final right of way shall approximate the center line of the existing creek in order that properties burdened by the natural creek will accommodate the channel improvements. In the case of artificial channels, rights of way will normally be selected and positioned to follow property lines, where possible, to avoid or minimize property severance.

14. Setbacks

All structures or other improvements, including fencing, shall be set back an adequate distance to provide for the designated floodway (full planned width of right of way) except as may be otherwise allowed by permit from the district.

15. Joint Use of Right of Way

Use of flood control rights of way for other public or private purposes may be permitted to the extent that district property rights allow and that maintenance and operation by the district are not impaired. Where fencing of the right of way is necessary pursuant to these guidelines, joint use in conjunction with adjacent private property is not acceptable. Where fencing of the right of way line by the district or property owner is not necessary pursuant to these guidelines, private or public use for agriculture, landscaping, paving and other nonstructural purposes may be permitted where plans are acceptable and proper agreements and indemnifications are executed.

16. Fencing

Fencing in accordance with district standards, or other design approved by the district and county, may be required at the time of development in situations where the right of way is dedicated in fee title (see guideline 2) or where it is needed for public safety.

17. Wells and Septic Tanks

Underground facilities such as wells and septic tank drainfields normally shall be outside the flood control facility right of way line. If a proposed right of way line for a channel that is to remain as a floodplain or a modified floodplain would render the abutting parcel undevelopable because of Health Department standards, wells and septic tank drainfields may be permitted within the right of way providing they meet Health Department regulations with respect to sanitation and setback requirements from any proposed flood control improvement, top of high bank, high water line, road or other controlling factor.

18. Existing Fixed Improvements

Existing fixed improvements such as buildings, structures and agricultural wells, within an area to be dedicated that are of substantial value to the abutting property after division or development, shall be paid for by the district or excluded from the dedication.

19. Appeals

In situations where an appeal as provided in county ordinances is made pertaining to a flood control dedication, it shall be referred to the Board of Directors of the Flood Control and Water District for review and recommendation to the appeal body (whether Planning Commission or Board of Supervisors, or both) as provided in county ordinances. Such review shall be made at a meeting at which due notice has been given to all parties to the appeal.

APPLICATION OF GUIDELINES TO DRAINAGE EASEMENTS

The guidelines contained herein pertaining to dedications to the district for flood control purposes also shall apply to dedications to the county for drainage purposes with the following exceptions and modifications:

- (a) The word "flood" shall be construed to mean "drainage" and the word "District" shall be construed to mean "County".
- (b) The design of drainage facilities shall be in accordance with the adopted County Drainage Manual.
- (c) Appeals shall be processed in accordance with county regulations. Guideline 19 shall not apply.
- (d) Guideline 7 pertaining to floodplains shall not apply.
- (e) Dedications shall be taken in easement only. If the easement is for a pipe or interim watercourse improvement that benefits applicant's property, or a small watercourse, where there is to be no improvement or impairment of access, the one-half channel dedication requirements specified in the guidelines shall apply to both sides of the watercourse within the applicant's property.

NOTE: Guidelines are to be implemented by ordinance changes as necessary.

DEFINITIONS

Flood - This is the selected flood against which it is reasonable and practicable to provide protection. The design flood is often described in terms of frequency of occurrence. The generally accepted design flood and the one used by the District is the one percent flood, having a flow that has a one percent chance of being equaled or exceeded in any one year. This is commonly referred to as the 100-year flood.

Natural Water Course – This is a stream, creek, river, or even a long established man-made channel in which water does or can flow. It is not necessary that the flow be continuous throughout the year. A watercourse is usually defined in the law as visibly having a bed and banks in which water may flow. Generally speaking, property through which natural watercourses traverse is impressed with the obligation to receive and discharge all of the waters naturally originating within the upstream watershed.

Diversion Channel - It is an artificial channel which diverts runoff from a natural channel to a new point of outfall. The lands on which the diversion is constructed are not naturally impressed with any duty to transport the waters so diverted upon and across them.

Floodplain - It is that area adjacent to a stream which is subject to flooding by waters of that stream. The area is typically divided into two zones, namely, the floodway zone and the fringe or "restrictive" zones. Within the floodway which is considered to include the channel or main waterway, depths and velocities of flood flows will be so great that development should be prohibited or severely restricted. While flood conditions in the fringe areas are normally less severe, of saving lives and protecting property and is an ordinary exercise of police power.

Modified Floodplain – This is a design concept which uses limited structural measures to restrict the existing floodplain to a reasonable width. The width will be greater than required for channelization works, but by minimizing structural work the stream and adjacent floodplain areas may remain in a more natural state. Low levees are the structural measures usually required to contain the design flood flows which acceptable right of way widths.

Designed Floodway Map – For the purpose of assisting land use planning, maps depicting the location and extent of designated floodways are being prepared for all major streams in the District. Ordinances designating the floodway as shown on such maps are adopted by the District in accordance with provisions of the Cobey-Alquist Floodplain Management Act. In order to insure general availability of the map information, the maps are recorded with the County Recorder and districted to the County of Santa Clara and each affected City.

Chapter 14

Green Stormwater Initiative

In developed areas, building footprints and paved surfaces such as driveways, sidewalks, and streets prevent rain from soaking into the ground. As urban areas develop, the increase in impervious surface also results in increases in peak flows and volumes of stormwater runoff from rain events. Traditional “gray” stormwater infrastructure, like most of the city’s storm drain system, is designed to convey stormwater flows quickly away from urban areas.

However, the increased peak flows and volumes can cause erosion, flooding, and habitat degradation in downstream creeks to which stormwater is discharged, damaging habitat, property, and infrastructure. Integrating Green Stormwater Infrastructure (GSI) into public spaces typically involves construction of stormwater capture and treatment measures in public streets, parks, and parking lots or as part of public buildings. Types of GSI measures that can be constructed in public spaces include: (1) bioretention; (2) stormwater tree well filters; (3) pervious pavement, (4) infiltration facilities, (5) green roofs, and 6) rainwater harvesting and use facilities.

The Green Stormwater Infrastructure is **an essential component** of stormwater management at the cities of US. The GSI plan for the City of San Jose presented in the website of the city is reproduced here.

City of San Jose Green Stormwater Infrastructure Plan – September 2019

Executive Summary

The City of San José envisions a thriving watershed with healthy creeks and rivers and a healthy San Francisco Bay. Green stormwater infrastructure (GSI) is a tool that can help achieve this vision. The multidisciplinary approach of GSI uses soil, plants, and pervious surfaces to capture, treat, infiltrate, and/or use stormwater runoff. GSI can provide multiple benefits such as improved water quality, reduced localized flooding, potable water conservation, increased groundwater recharge, reduced urban heat island effect, and public space beautification. GSI can be integrated with building and roadway design, complete streets, drainage infrastructure, urban forestry, soil conservation, and landscaping. As California’s weather becomes increasingly unpredictable and extreme, GSI strategies can provide the city with enhanced climate resiliency, local water supplies, and energy savings, consistent with the City’s sustainability goals.

The City of San José has developed this Green Stormwater Infrastructure Plan (GSI Plan) to lay out the approach, strategies, targets, and tasks needed to transition traditional “gray” infrastructure to include green stormwater infrastructure over the long term and to implement and institutionalize the concepts of GSI into standard municipal engineering, construction, and maintenance practices.

The GSI Plan is intended to serve as an implementation guide for reducing the adverse water quality impacts of urbanization and urban runoff on receiving waters over the long term, and a reporting tool to provide reasonable assurance that specific pollutant reductions from discharges to local creeks and San Francisco

Bay will be met. The GSI Plan is required by the City's Municipal Regional Stormwater National Pollutant Discharge Elimination System (NPDES) Permit for the discharge of stormwater runoff from the City's storm drain system. The GSI Plan—along with the appended Reasonable Assurance Analysis (RAA) for bacteria—also describes how the City will reduce the storm drain system's contribution to bacteria loads entering the Guadalupe River, Coyote Creek, and other City watersheds per the Consent Decree with San Francisco Baykeeper.

Introduction

Green stormwater infrastructure (GSI) is a natural stormwater collection and treatment system that uses soil, plants, and pervious surfaces to capture and treat stormwater runoff. GSI facilities can also be designed to capture stormwater for uses such as irrigation and toilet flushing. The GSI Plan lays out the City's approach, strategies, targets, and tasks to transition traditional "gray" infrastructure to include green stormwater infrastructure over the long term, in a manner consistent with the City's planning and sustainability goals and regulatory requirements.

2.2 Types of GSI Facilities

Integrating GSI into public spaces typically involves construction of stormwater capture and treatment measures in public streets, parks, and parking lots or as part of public buildings. Types of GSI measures that can be constructed in public spaces include: (1) bioretention; (2) pervious pavement, (3) infiltration facilities, (4) green roofs, and (5) rainwater harvesting and use facilities. Descriptions of these facility types are provided as follows.

2.2.1 Bioretention Areas

Bioretention areas are depressed landscaped areas that consist of a ponding area, a mulch layer, plants, and a special biotreatment soil media composed of sand and compost, underlain by drain rock and an underdrain, if required. Bioretention is designed to retain and filter stormwater runoff through biotreatment soil media and plant roots, and either infiltrate stormwater runoff to underlying soils as allowed by site conditions, release treated stormwater runoff to the storm drain system, or both. They can be of any shape and size and are adaptable for use on a building or parking lot site or in the street right-of-way.

Bioretention systems in the streetscape have specific names: stormwater planters or stormwater curb extensions. A stormwater planter is a linear bioretention facility in the public right-of-way along the edge of the street, often in the planter strip between the street and sidewalk. They are typically designed with vertical (concrete) sides; however, they can also have sloped sides depending on the amount of space that is available.

A stormwater curb extension (or bulb-out) is a bioretention system that extends into the roadway and involves modification of the curb line and gutter. Stormwater curb extensions may be installed midblock or at an intersection. Curb bulb-outs and curb extensions installed for pedestrian safety, traffic calming, and other transportation benefits can also provide opportunities for siting bioretention facilities. Parking lots can accommodate bioretention areas of any shape in medians, corners, and pockets of space unavailable for parking.



Bioretention Curb Extension on Chynoweth Avenue



Linear Bioretention Facilities on Autumn Parkway

2.2.2 Pervious Pavement

Pervious pavement is hardscape that allows water to pass through its surface into a storage area filled with gravel prior to infiltrating into underlying soils. Types of pervious pavement include permeable interlocking concrete pavers, pervious concrete, porous asphalt, and grid pavement. Pervious pavement is often used in parking areas or on streets where bioretention is not feasible due to space constraints. Pervious pavement does not require a dedicated surface area for treatment and allows a site to maintain its existing hardscape.

There are two types of pervious pavers: Permeable Interlocking Concrete Pavers (PICP) and Permeable Pavers (PP). PICP allow water to pass through the joint spacing between solid pavers, and PP allow water to pass through the paver itself and therefore can have tighter joints. Porous asphalt and pervious concrete are similar to traditional asphalt and concrete, but do not include fine aggregates in the mixture, allowing water to pass through the surface. All types are supported by several layers of different sizes of gravel to provide structural support and water storage.



Permeable Pavement in Martha Gardens Alley Way

2.2.3 Infiltration Facilities

Where soil conditions permit, infiltration facilities can be used to capture stormwater and infiltrate it into native soils. The two primary types are infiltration trenches and subsurface infiltration systems. An infiltration trench is an excavated trench backfilled with a stone aggregate. Infiltration trenches collect and detain runoff, store it in the void spaces of the aggregate, and allow it to infiltrate into the underlying soil.

Subsurface infiltration systems are another type of GSI facility that may be used beneath parking lots or parks to infiltrate larger quantities of runoff. These systems, also known as infiltration galleries, are underground vaults or pipes that store and infiltrate stormwater while preserving the uses of the land surface above, such as parking lots, parks, and playing fields. Storage can take the form of large-diameter perforated metal or plastic pipe, concrete arches, concrete vaults, plastic chambers or crates with open bottoms.



Infiltration Trench



Subsurface Infiltration Gallery

2.2.4 Green Roofs

Green roofs are vegetated roof systems that filter, absorb, and retain or detain the rain that falls upon them. Green roof systems are comprised of a layer of planting media planted with vegetation, underlain by other structural components including waterproof membranes, synthetic insulation, geofabrics, and underdrains. A green roof can be either “extensive,” with 3 to 7 inches of lightweight planting media and low-profile, low-maintenance plants, or “intensive,” with a thicker (8 to 48 inches) of media, more varied plantings, and a more garden-like appearance. Green roofs can provide high rates of rainfall retention via plant uptake and evapotranspiration and can decrease peak flow rates in storm drain systems because of the storage that occurs in the planting media during rain events.



Fourth Street Apartments Green Roof

2.2.5 Rainwater Harvest & Use Facilities

Rainwater harvesting is the process of collecting rainwater from impervious surfaces and storing it for later use. Storage facilities that can be used to harvest rainwater include rain barrels, blue roofs, aboveground or belowground cisterns, open storage reservoirs (e.g., ponds), and various underground storage devices (tanks, vaults, pipes, and proprietary storage systems). The harvested water is then fed into irrigation systems or non-potable water plumbing systems, either by pumping or by gravity flow. Uses of captured water may include irrigation, vehicle washing, and indoor non-potable use such as toilet flushing, heating and cooling, or industrial processing. The award winning public artwork Watershed designed by artist Peter Richards is a good example of a rainwater harvest system. This public artwork captures rainwater from the San José Environmental Innovation Center roof and channels it into a 6,600-gallon water storage tank which is then used to irrigate a grove of 100-year-old olive trees.

The two most common applications of rainwater harvesting systems are:

1. Collection of roof runoff from buildings; and
2. Collection of runoff from at-grade surfaces or diversion of water from storm drains into large underground storage facilities below parking lots or parks.

Rooftop runoff usually contains lower quantities of pollutants than at-grade surface runoff and can be collected via gravity flow. Underground storage systems typically include pretreatment facilities to remove pollutants from stormwater prior to storage and use.

2.3 Types of Subprojects

Applications for GSI facilities as part of GSI projects can be categorized as green streets, parcel-based LID retrofits, and regional projects as shown in Figure 2-1. The types of GSI projects described in this GSI Plan are typically constructed on publicly owned property or within public right-of-ways (e.g., along streets) and will often involve retrofitting an existing impervious area such that the area will drain to a pervious area and/or a GSI facility.

2.3.1 Green Streets

GSI roadway projects are typically called “green streets.” Green street projects are located in the public right-of-way and capture runoff from the street and adjacent parcels. Green streets are distributed, small-scale GSI projects spread throughout an urban area that provide localized treatment and flood reduction for relatively small drainage areas. For example, green streets can include facilities such as bioretention bulb-outs, stormwater planters, or permeable pavers along street right-of-ways.

There are often opportunities to combine GSI projects with another type of street design known as “complete streets.” This latter term refers to streets that incorporate all modes of travel equally and are designed to increase safety and access for cyclists and pedestrians regardless of age or ability. The integration of the goals of both complete streets and green streets has coined several new terms such as “living streets,” “better streets” and “sustainable streets.” This movement recognizes the multiple benefits that environmentally and holistically designed streets can achieve.



Green Streets



LID Retrofits



Regional Projects

Figure: Green Streets, LID Retrofits, and Regional Projects

2.3.2 Parcel-Based Low Impact Development Retrofits

LID retrofit projects mitigate stormwater impacts by reducing runoff through capture and/or infiltration and treating stormwater on-site before it enters the storm drain system. LID retrofit projects may include bioretention facilities, infiltration trenches, detention and retention areas in landscaping, pervious pavement, green roofs, and systems for stormwater capture and use. These measures help to protect water quality by filtering stormwater through plants and soil and allowing stormwater to infiltrate into the ground, thus mimicking the pre-urbanized natural hydrology of the undeveloped site. For the purposes of this GSI Plan, LID retrofit projects are GSI facilities that are built on a parcel to treat runoff generated from impervious surfaces on that parcel. These projects may or may not be regulated projects (see Section 6.3).

2.3.3 Regional Projects

Regional GSI projects are large-scale stormwater capture and treatment measures that are intended to collect and treat runoff from a large drainage area, including runoff from on-site and off-site areas. Off-site surface runoff can come from diversions from storm drains, channels, culverts, and streams. These types of projects include aboveground or underground runoff capture facilities or subsurface infiltration galleries located in large open space areas or under existing uses (such as parking lots or parks) to which runoff from large areas of impervious surface can be directed. Benefits of regional stormwater capture projects include flood risk reduction, stormwater treatment and use, groundwater recharge, and the potential to augment alternative water supplies. They are often the most cost-effective projects due to the multiple benefits achieved and the economies of scale. The site characteristics and uses will determine what types of regional projects are feasible, e.g., how much flow the project can divert from the storm drain network, whether the project is aboveground or underground, and the size of the project. All projects proposed in City parks must conform to the City Charter Section 1700 Parks.

Chapter 15

Ideal Control Measures

15.1 The Stormwater management at Bengaluru city vis-a-vis at San Jose City

The stormwater management at the two cities may briefly be summarized as below,

San Jose City:

- i. San Jose city has a jurisdictional area of 180 square miles. It is located in the Santa Clara Basin or Valley, a sub-watershed of the larger San Francisco Bay watershed and a part of Santa Clara County.
- ii. The Santa Clara Valley encompasses parts of Santa Clara, Alameda and San Mateo counties. There are 15 watersheds in the valley. San Jose city spreads in 6 watersheds, viz., Coyote, Guadalupe, Lower Penitencia, San Tomas Aquino, Calabazas and Baylands. They all drain to southern San Francisco Bay.
- iii. Each watershed has a major watercourse or Creek at its centre.
- iv. The Storm sewer system, also known as **Municipal Separate Storm Sewer System or MS4**, receives runoff from the surrounding urban areas and discharges to a creek, river or the Bay. The storm sewer system includes the inlets (storm drains) and gutters on street as well as pipes and outfalls, where the stormwater enters the water body.
- v. The Santa Clara Flood Control and Water District has jurisdiction and responsibility for designated floodways and the watercourses tributary thereto which have a drainage area greater than 640 acres.
- vi. The terminal point of its jurisdiction in each case (main watercourse or tributary greater than 640 acres) would be at the point on that watercourse where the contributing drainage area is diminished in 320 acres.
- vii. All the main watercourses and tributary thereto greater than 640 acres within the district are designed for protection against **100-year return flood**, within their designated floodways. The floodway is the channel of a river or stream and the overbank areas that must remain open to carry the deeper, faster moving water during flood.
- viii. All proposals of land development on the watercourse and tributary are referred to Water District for review. No development is permitted within the designated floodway, unless a licensed professional engineer demonstrates that the proposed encroachment would not result in any rise in the 100-year flood elevation.
- ix. Any development on the floodplain that could obstruct flood flows, such as fill, a bridge or a building is considered as an encroachment.

Bengaluru city:

- i. It has three major drainage valleys.
- ii. The Bangalore Development Authority (BDA) prepares the master plan for the city including its sewage and stormwater drainage system.
- iii. Its Comprehensive Master Plan (CDP) of 1995 proposes **three categories** of drains in the stormwater drainage system, viz., primary, secondary and tertiary drains.
- iv. It stipulates buffer zones (no development area) of 50, 25 and 15 mtrs measured from the centre of the drain) on either side of primary, secondary and tertiary drains respectively, for providing utilities such as power, pipelines for water/oil/gas etc., and also to facilitate easy maintenance of drains.
- v. This buffer area was modified (May 2016) by the NGT to 50m, 35m and 25m (measured from the edge of the drain).
- vi. None of the drains have 'boundary markings' clearly specifying the 'no-development area' and the stipulation regarding buffer zone is not adhered to in respect of any of the drains. (CAG Report – 2021)
- vii. The BBMP has got a separate master plan of drains prepared during 2010-11. It **does not provide for any buffer zone**.
- viii. The land is developed on the either side of the drain upto its edges. The encroachment here is considered only for the extent of construction on the drain proper.
- ix. Most of the stormwater drains carry sewage water.
- x. The sections of the drains are arbitrarily decided.
- xi. The entire stormwater drainage system upto tertiary drain is maintained by BBMP.

There is a scientific approach in the stormwater management at the city of San Jose. The basin is divided into watersheds and sub-watersheds. The run-off from the sub-watersheds upto 640 acres is collected through storm sewer system. It is managed by the local authority. The watercourses or creeks are designed to discharge 100-year return flood. They are managed by the basin authority. No development is permitted in their floodplain.

The stormwater at Bengaluru city on the other hand is managed by adhoc planning. Though its CDP proposes three types of drains viz., Primary, Secondary and Tertiary, they are not actually specified on the field. Some primary drains are called Rajakaluves. No free overbank area is provided for any of the drains. The land is developed on them upto their edges. Most of them carry sewage.

15.3 The actual reasons for the floods

In the light of the discussion, the actual reasons for the floods at the city could be identified as below,

i. Land development right on the stream path between the lakes

The towns and villages at the erstwhile outskirts of the city were managed by the local authorities, before they were merged in BBMP. The spillover from the lakes was thinly discharged over the agriculture lands and there were no defined waterways. Layouts were developed right on the stream paths, before stormwater drainage system was developed or planned for them. The spill-over from the lakes is now obstructed by the above layouts and structures. The sewage drains at the layouts are connected together to regulate the discharge from one lake to another. This is the main reason for the floods in east and Mahadevapur zones of the city, which account for the majority of the floods at the city. For example, the RBD layout and Wipro Corporate Office campus on Sarjapur road are inundated since they are no defined stormwater drains connecting Halanayakanahalli and Junnasandra lakes and Saul Kere. The Outer Ring Road near Ecospace is inundated since there is no defined drain connecting Saul Kere to the main stream of Bellandur lake.

ii. Too arbitrary provision of waterway for the bridges and culverts

The bridges and culverts on the drains are designed without estimating the actual stormwater discharge at them. Such an exercise is not carried out for any of the bridges and culverts at the city. Even the bridges and culverts on the waterways between the lakes are designed with too small vents and are provided with too arbitrary sections. They choke up by the weed during floods. The large discharge from the lakes is invariably obstructed, held back and impounded at the valley on their upstream. This is the main reason for huge waterloggings at the city, particularly at its peripheral area. For example, the Outer Ring Road near Ecospace is inundated due to no provision of appropriate culvert on the stream path between Saul Kere and the main stream of Bellandur lake. Epsilon and other layouts and apartments in Yemalur are inundated due to too small waterway at the bridge on the Outer Ring Road in Kadubeesanahalli. The Panathur-Balagere road is inundated due to too small waterway at the bridge on the newly constructed road on the upstream of Varthur lake.

iii. No provision of floodway

All major stormwater drains/Rajakaluves at the city including those connecting two lakes are designed to have only drain section at their centre. No provision of floodway is made for them. They carry sewage. They invariably spill over during rain. When there is new development on their banks, the free discharge of water is obstructed. The waterloggings at most of the areas at the city are caused due to the above reason.

iv. No provision of park area at the layouts in the periphery of the city

The layouts at the periphery of the city are developed with no park area, though the planning rules stipulate for its provision for certain percentage of the layout. Wide roads are constructed. They are completely asphalted. The lakes are encroached upon. Layouts are developed on them. As a result, the entire catchment area is concretized and near entire rainfall is converted into run-off. The discharge from the lakes is increased. The capacity of the drains turn out to be too inadequate. They invariably spill over. It is the main reason for the large intensity of the floods at the periphery of the city.

v. Proposing Development works without their Drainage impact assessment study

Due to inadequate discharge capacity of the drains, the flood is discharged over the road at many locations. If its level is raised as a part of its development or the median wall of Metro railway line is constructed on it, it is obstructed by it and held back. The Rainbow Drive layout is inundated due to gradual raising of the level of Sarjapur road near its entrance gate, the Wipro Corporate Office on Sarjapur road was inundated due to construction of road divider on it and Outer Ring Road near Ecospace experienced more severe waterlogging during this year due to construction of median wall of Metro line at its centre. They are mainly affected since no drainage impact studies proposed prior to their construction.

The above five factors individually or combinedly explain the reasons for the most of the floods at the city.

18.2 Need for an immediate intervention

We observe that,

- i. The floods at the city have assumed new dimension in the recent years. They were only street flooding earlier. The entire localities are inundated now. The valleys are filled up. There is not a single instance where the flood at a location is effectively controlled at the city. Once it is affected, it continues to be affected during every subsequent rainy season. For example, the RBD layout is affected since last 10 years. The Outer Ring Road is affected since 2007. Only an adhoc measure is attempted for their control, viz., anti-encroachment drive. The encroachment of the drains may not be the actual reason or the only reason for the floods/waterlogging at all the places. Therefore, there is a need for taking up a detailed study of the waterlogging at each individual location and initiate specific measure for its immediate control.
- ii. Every year, they are affecting new locations. For example, Yemalur and Panathur-Balagere road were affected for the first time during this year. Therefore, there is a need for initiating certain measures in advance for preventing them affecting new areas.

- iii. The city is fast developing in all the direction, e.g., Sarjapur road, Devanahalli, Tumkur road, Bengaluru-Mysore Expressway and so on. They may too face floods in few years, as the periphery of the city is experiencing now. Therefore, there is a need for initiating certain measures at them too for preventing them from experiencing similar floods.

The stormwater management at the city actually demands certain pro-active measures in all the three above directions immediately for effective control of its floods.

15.3 Ideal measures

The ideal measures for the effective control of the floods at the city could be proposed as below,

i. Initiate specific flood control measures for each severely affected area

As the study reveals, all major waterloggings at the city could be prevented by small interventions. For example, the waterlogging at RBD layout could be prevented by providing an escape route for the discharge of Halanayakanahalli and Junnasandra lakes at Sarjapur road. The waterlogging at Outer Ring Road near EPZ Ecospace could be prevented by proposing a new drain and a culvert on the road near its foot bridge. The waterlogging at Yemalur road could be prevented by proposing a large bridge on Outer Ring Road in Kadubeesanahalli. Similarly, the waterlogging at Panathur-Balagere road could be prevented by proposing a large bridge at the newly constructed road on the upstream of Varthur lake and so on. Therefore, the actual reasons for the waterlogging at each severely affected area at the city may be studied and specific measures for its immediate control shall be initiated.

ii. Prepare Master Plan of creeks for each Drainage area

As the study reveals, the city demands scientific planning of its stormwater system similar to the city of San Jose at the Silicon Valley of US. It could be thought off in the following lines. The city has six lake systems. The drainage area of each lake system constitutes a major watershed. The drainage area of each individual lake constitutes a sub-watershed. All drains connecting one lake to another shall be designated as creeks. If a large catchment directly drains into a lake or to the stream between the lakes, it shall also be proposed as a creek. A master plan of creeks for each lake system shall be prepared. No land development shall be allowed on their floodplain.

iii. Propose separate Institution for management of creeks

The study of the stormwater management at the city of San Jose reveals that the creeks and storm drains there are considered as two distinct entities. Creeks are managed by basin authority, while storm drains are managed by local authorities. The creeks are referred as "Primary drains" or "Rajakaluves" at the city's stormwater system. "Secondary drains" and

“Tertiary drains” constitute the storm drains. But both are designed on the same concepts and provided with just a drain section at their centre. All drains are designed and managed by a single agency, viz., Stormwater Drainage Management Department of BBMP. Most of the floods at the city are created due to adhoc planning of the creeks and bridges and culverts on them. Therefore, a separate institution may be proposed for the management of the creeks at the city.

iv. Propose floodway for all creeks

The floods at the city could be prevented effectively only if the floodway is proposed for all its major drains/Rajakaluves. The present planning does not propose its provision. A new law may be proposed on the lines of “Policy and Guidelines for dedication and reservation of lands needed for Flood Control purpose” of Santa Clara County at Silicon Valley of US, which is reproduced in Chapter 13. Where there are already developments, they may be proposed to be acquired with consideration.

v. Propose bridges and culverts with large waterways

Another main reason for the floods at the city is the construction of bridges and culverts with too inadequate waterway and small vents. They invariably choke up by the weed during the flood. e.g., The Outer Ring Road near Ecospace is inundated due to provision of too small culvert on it. Yemalur road is affected due to too small vents at the bridge on Outer Ring Road in Kadubeesanahalli. The Panathur-Balagere road is affected due to too small vents at the bridge on the newly constructed road on the upstream of Varthur lake and so on. Therefore, they shall be scientifically designed and provided with large vents. The existing bridges with too inadequate waterway and causing major waterlogging shall be re-built.

vi. Prepare Master plan of creeks for all outskirts of the city

The city is fast expanding. There is huge development at the outskirts of the city. No stormwater drainage systems are planned for them. They may too face floods in due course as the peripheral area at the city is experiencing now. Therefore, the Master plans for creeks shall also be prepared for all the outskirts of the city before there is a large scale development at them.

vii. Propose Drainage Impact Study for all Infrastructure and Development works

A large number of infrastructure and development works are initiated at the city. New ring roads and peripheral roads are proposed. New Metro rail lines are proposed. They are causing floods at a number of locations. Therefore, their Detailed Project Report (DPR) may be proposed to include **Drainage Impact Assessment study** and requisite measures for free discharge of stormwater at their locations.

viii. Propose large public parks and gardens at outer zones and outskirts

Another main reason for too severe floods at the outer zones of the city is the total concretization of their catchment areas. It is causing too sudden and large discharges at the water bodies. Therefore, large public gardens and parks may be proposed similar to Cubbon Park and Lalbagh at the outer zones and outskirts, to increase the green space at them.

ix. Propose Green Stormwater Infrastructure (GSI) measures

Green Stormwater Infrastructure is a multi disciplinary approach proposed extensively and as an essential component of the stormwater management at the cities of US. It uses soil, plants and pervious surface to capture and infiltrate stormwater runoff in public streets, parks and parts of public buildings. Types of GSI measures include (i) bioretention (ii) stormwater tree well filters (iii) pervious pavement (iv) infiltration facilities (v) green roofs and (vi) rainwater harvesting. The measures may be proposed at the city too and a comprehensive plan for their implementation may be developed.

The groundwater at the city is presently recharged by percolation of water from large number of artificial lakes created during rain. It is meeting nearly fifty percentage of the total drinking water demand of the city. If effective measures are initiated and free discharge of stormwater is ensured at all the reaches, its entire run-off would drain either to Cauvery or Pennar river. The borewells may get dry. The city may face severe shortage of drinking water, an issue of different sort. It is confronted with two typical issues **at the same time**, viz., (i) the floods and (ii) limited availability of drinking water. If it attempts to control the floods and facilitate free discharge of stormwater, it may have to face the scarcity of drinking water. If it does not wish to face the scarcity of drinking water, it has to bear with the floods. Only the right mix of measures as discussed in the study could enable it to face both the issues effectively.

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From,
Dr. Sudhir Sajjan,
Deputy Chief Engineer,
o/o. the Managing Director,
Krishna Bhagya Jala Nigam Limited,
Almatti.

To,
The Secretary,
Water Resources Department,
Government of Karnataka,
4th Floor, Vikas Souda,
Bengaluru.

(Submitted through the **Managing Director**, KBJNL, Bengaluru)

Sub: Floods at Bengaluru City - a Technical Study.

Ref: Water Resources Department, GoK, Notification dated 09.07.2018
inviting suggestions from experts and public on the issues and
management of the water resource of the State.

Respected Sir,

The Bengaluru city is experiencing floods quite frequently in the recent years. Experts attribute them mainly to the encroachment of the drains and their shrinking widths. Attempts are being made by the State to clear the encroachments. However, a study of the stormwater management at the city of San Jose in Silicon Valley of US reveals that its stormwater drainage system has two distinct components, viz., (i) **stormwater drains** and (ii) **creeks**.

The entire basin is divided into watersheds and sub-watersheds. The creeks run at their centre. The stormwater drains collect runoff from the streets and drain it to the creeks. A large creek runs at the centre of the basin and all creeks from small watersheds drain to it. It finally drains to a river or tributary, reservoir or sea. A creek would have a natural channel at the centre and free overbank section or **floodway** to carry the deeper, faster moving water during flood. No construction activity is entertained on the floodway.

The Bengaluru city is situated at the ridge line and the stormwater here is collected in the lakes nearby, then it drains from lake to lake and finally to the tributaries of the Cauvery and Penner rivers. Logically, the stream paths from one lake to the another shall assume the role of the creeks. But, they are also designed as just drains with **no provision of free overbank area**. Most of them carry sewage and have the capacity just adequate for its discharge. They invariably spill over during the rain. The width of the bridges/culverts on them too are limited only for the width of the drains. When the spill-over from the lakes is obstructed by the development on the banks of the drains or at the bridges/culverts on them, it is held back and impounded.

A detailed study of the severely affected areas at the city reveals that they are all located on the stream paths between the lakes. For example, the RBD layout and Wipro complex on Sarjapur road are situated on the stream path between Halanayakanahalli lake and Saul Kere. The Outer Ring Road near Ecospace is situated on the stream path between Saul Kere and the main stream from Bellandur lake. The Borewell road in Whitefield is located on the stream path between Nallurahalli lake and Sheelvanthakere. When the run-off from the streets is obstructed, it finds its way through the gullies. It only creates local waterlogging. It would not create significant nuisance. But if the discharge from a lake is obstructed, it is **held back and accumulated at the valley** on its upstream for a long stretch.

Therefore, the issue here is not stormwater drainage management; it is the management of discharge from one lake to another. It demands two-pronged strategy to effectively control the floods at the city, viz., (i) identify all the severely waterlogged/affected locations and **initiate specific measures** for control of flood at them immediately and (ii) **propose floodway** for all major drains and for drains connecting one lake to another.

I have attempted to study the actual phenomena of the floods at the city and propose ideal measures for their control. I am submitting herewith my report for favour of your kind perusal.

With regards,

Almatti
05.12.2022

Yours faithfully,



(Dr. Sudhir Sajjan)
Deputy Chief Engineer,